

Management of Hypoglycaemia and Hyperglycaemia in Term and Preterm Infants

A consultant-level, PMID/DOI-verified evidence synthesis underpinning UK (BAPM) practice — definitions, screening and treatment thresholds by gestation, point-of-care diagnosis, the permissive-hypoglycaemia controversy, the preterm evidence gap, MRI in hypoglycaemic seizure, and hyperglycaemia follow-up.

Prepared for: Consultant Neonatologist, UK Level 3 NICU
(Imperial College Healthcare NHS Trust)

Track: Narrative evidence synthesis — governance/teaching
register

BAPM Framework cross-checked directly against the primary
document (January 2024 revision)

Date: 27 August 2026

Contents

1. Executive overview and domain summaries
2. Cross-cutting synthesis
 1. Reference table — gestation × time-window × threshold × source × directness
 2. Point-of-care vs laboratory confirmation
 3. BAPM pathway evidence-strength summary (with flowchart)
 4. The permissive-hypoglycaemia controversy
 5. Preterm evidence-gap statement (with heatmap)
 6. MRI in hypoglycaemic seizure (with flowchart)
 7. Hyperglycaemia follow-up population list
3. Quality control checklist and known limitations
4. Primary-source verification addendum (BAPM Framework, January 2024)
5. Plain-language threshold quick-reference table
6. Domain deep-dives
 1. Definitions and operational thresholds
 2. Point-of-care glucometer vs laboratory confirmation
 3. BAPM pathway, evidence grading, and the permissive-hypoglycaemia controversy
 4. Preterm-specific thresholds and outcomes
 5. Term/late-preterm neurodevelopmental outcomes
 6. Hypoglycaemic seizure and MRI
 7. Hyperglycaemia and long-term follow-up

Management of Hypoglycaemia (and Hyperglycaemia) in Term and Preterm Infants — Evidence Synthesis

Prepared for: consultant neonatologist, UK Level 3 NICU (Imperial College Healthcare NHS Trust). **Track:** narrative evidence synthesis, governance/teaching register — not a systematic review, not a manuscript. **Status:** all seven domains (A–G) complete; every claim below carries a PMID/DOI verified this session or is explicitly labelled as grey literature.

This document has two parts. Part 1 gives a short, consultant-register precis of each domain, pointing to the full domain file for depth — the domain files themselves (`domain-A-definitions.md` through `domain-G-hyperglycaemia.md`) are the primary deliverable and are not reproduced here in full. Part 2 is the cross-cutting synthesis required by the brief's §6: the reference table, the point-of-care decision rule, the BAPM evidence-strength summary, the permissive-hypoglycaemia controversy, the preterm evidence-gap statement, the MRI recommendation, and the hyperglycaemia follow-up population list.

Post-hoc primary-source correction (read this first): the individual Domain A/B/C searches never obtained bapm.org directly (blocked all session) and instead triangulated BAPM's numbers from secondary sources. The user subsequently supplied the primary BAPM Framework document (January 2024 revision) directly, which was read in full and identified several corrections — most importantly, **BAPM's own operational threshold is a tiered 1.0/2.0/2.5 mmol/L system, not the 2.6 mmol/L figure previously reported throughout this synthesis**, and a directly relevant threshold-comparison RCT (**HypoEXIT**, van Kempen et al. 2020, NEJM, PMID 32023373) was identified that no domain search had found. Full detail: `03-synthesis/bapm-primary-source-verification.md`. Part 2 below has been updated to reflect the corrected figures; Part 1's domain precis and the individual domain files retain their original text for audit-trail purposes and should be read alongside that addendum, not in place of it.

Part 2 — Cross-cutting synthesis (RESEARCH-BRIEF.md §6)

§6.1 Reference table — gestation × time-window × threshold × source × directness

Units: mmol/L (mg/dL), conversion factor 1 mmol/L = 18.0 mg/dL throughout.

Gestation	Time window	Screening/ action threshold	Treatment/ escalation threshold	Source	Directness
Term (≥37+0 wk)	0–4h (transitional)	Physiological nadir as low as 1.6 mmol/L (30 mg/dL) is normal; first pre-feed check 2–4h after birth	n/a — feeding-based	BAPM Framework, Jan 2024 (verified primary text)	Grey literature, primary-verified
Term	4–48h	Asymptomatic, at-risk: >2 measurements <2.0 mmol/L (36 mg/dL) → intervene; 1.0–1.9 mmol/L (18–34 mg/dL) → gel + feed	Single value <2.5 mmol/L (45 mg/dL) with abnormal signs, or <1.0 mmol/L (18 mg/dL) at any time → urgent/escalate	BAPM Framework, Jan 2024 (Practice Point 6)	Grey literature, primary-verified. Not anchored on 2.6 mmol/L — that figure belongs to the separate CHYLD/HypoEXIT trial evidence below
Term/late preterm	4–48h, direct RCT evidence on the threshold question	Treat-to-2.6 mmol/L (47 mg/dL) shows no excess neurosensory harm (CHYLD); a lower 2.0 mmol/L (36 mg/dL) threshold is noninferior to 2.6 mmol/L for 18-month development, head-to-head	2.0 vs 2.6 mmol/L (36 vs 47 mg/dL)	HypoEXIT (van Kempen 2020, NEJM, PMID 32023373, n=689); McKinlay 2015 (PMID 26465984), McKinlay 2017 (PMID 28783802)	Direct — late-preterm/term at-risk RCT population; this is the one RCT design capable of directly answering the threshold question, and BAPM cites both as reassuring rather than as its own adopted number
Term	48h (BAPM Practice Point 8) or 72h (BAPM Practice Points 21/23 — the primary document itself is not internally consistent on this boundary) — "persistent"	>2 measurements <2.0 mmol/L (36 mg/dL)	Target >3.0 mmol/L in first 48h, rising to >3.5 mmol/L after 48h; critical sample, urgent referral	BAPM Framework, Jan 2024 (primary-verified); independently corroborated by Kebede 2025 (PMID 39913386, PTNHG definition, similar ~47 mg/dL anchor) and Rusu 2026 (PMID 42194883)	Grey literature (BAPM), primary-verified. Directness caveat: Kebede's prevalence data are Ethiopian, not UK-generalisable
Late preterm	0–72h	Enrolled in and treated as part of	Same 2.0–2.6 mmol/L trial-		Direct for this specific band —

Gestation	Time window	Screening/ action threshold	Treatment/ escalation threshold	Source	Directness
(34–36+6 wk)		the CHYLD/Sugar Babies/hPOD "at-risk" population	tested range as term, by trial design	Domain E (CHYLD/Sugar Babies/hPOD)	the trial evidence base's actual lower recruitment boundary. BAPM's own Framework explicitly excludes this group from its scope (a separate BAPM document covers moderate-to-late preterm infants)
Moderate preterm (32–33+6 wk)	0–72h	No dedicated threshold; some trial cohorts (CHYLD from ≥ 35 wk; hPOD-family variably from ≥ 32 –35wk) partially include this band	Extrapolated from the late-preterm/term trial-tested range	Domain D/E (partial trial overlap)	Extrapolated at the lower edge — verify individual trial's exact GA cutoff before treating as "covered"
Very preterm (28–31+6 wk)	0–72h	No RCT-derived threshold. Modern practice extrapolates downward by convention	Same extrapolation	Domain D — Duvanel 1999 (PMID 10190926, positive, 26y old), Palazzo 2024/2025 (PMID 39707054, 40609275, modern null)	Extrapolated, not direct — best available evidence (Palazzo pair) is retrospective, first-6h-only, single-region
Extremely preterm (<28 wk)	0–72h	No RCT-derived threshold; no threshold-outcome study of any design was identified at all	Same extrapolation, unsupported	Domain D (confirmed evidence gap)	No direct evidence exists — stated as a gap, not softened
All gestations	>72 h (persistent/recurrent, any GA)	Out of primary scope — different differential diagnosis (hyperinsulinism, metabolic, endocrine)	Critical sample, specialist endocrine referral	RESEARCH-BRIEF.md §4; BAPM (persistent-hypoglycaemia pathway)	Signposted, not developed in depth

Gestation	Time window	Screening/ action threshold	Treatment/ escalation threshold	Source	Directness
Extremely/ very preterm	first week (hyperglycaemia)	No consensus threshold; commonly 8-10 mmol/L (144-180 mg/dL) used locally	No RCT-tested treatment threshold; NIRTURE (insulin) showed harm signal, not benefit	Domain G — Zamir 2021 (PMID 33863775), Boscarino 2021 (PMID 34199741), Beardsall/ NIRTURE 2008 (PMID 18971490)	Direct population for the observational associations; no direct RCT evidence for any treatment threshold
Term/ near-term with NE	first 18mo+	Hyperglycaemia and hypoglycaemia both associated with death/ neurodisability in this population specifically	n/a — reflects encephalopathy severity, not a primary glucose disorder	Puzone 2023 SR/MA (PMID 37316160)	Direct for NE population; do not extrapolate to primary/ isolated preterm hyperglycaemia

§6.2 The point-of-care-versus-laboratory decision point, stated once

Corrected (primary-source-verified): BAPM does not state a single "confirm below X mmol/L" number. Domain B's original synthesis reported, as an apparent direct quotation, that "all low values (≤ 2.6 mmol/L) require confirmation using blood gas analysis" — this figure and framing could not be located anywhere in the verified primary text and should be treated as a misattribution. What the Framework actually says: **blood gas analysers should be considered the reference standard** for BG measurement throughout (Practice Point 5), and **"many handheld glucose meters are not sufficiently accurate in the range of 0-2.0 mmol/L"** (0-36 mg/dL) — a range caution, not a single confirmatory cut-off. Any POCT device must be locally validated with pathology/Healthcare Scientist input (Appendix 5) before use, particularly at low glucose and high haematocrit; the exact local confirmation rule is therefore a matter of local device validation, not a single BAPM-stated number.

Independently, the device-accuracy literature (Domain B) derives its own cutoffs by formal sensitivity/NPV analysis — typically 2.8–3.5 mmol/L (50–63 mg/dL) depending on device and study — and the UK's own MHRA device-safety guidance is more conservative still, stating meters are reliable only above 3 mmol/L (54 mg/dL). **None of these three figures (BAPM's range caution, the device literature's derived cutoffs, MHRA's >3 mmol/L reliability floor) is the same number**, and this should be taught explicitly rather than compressed into a single figure: BAPM's practice is to treat blood gas analysis as the default reference standard rather than to specify one universal point-of-care confirmation threshold.

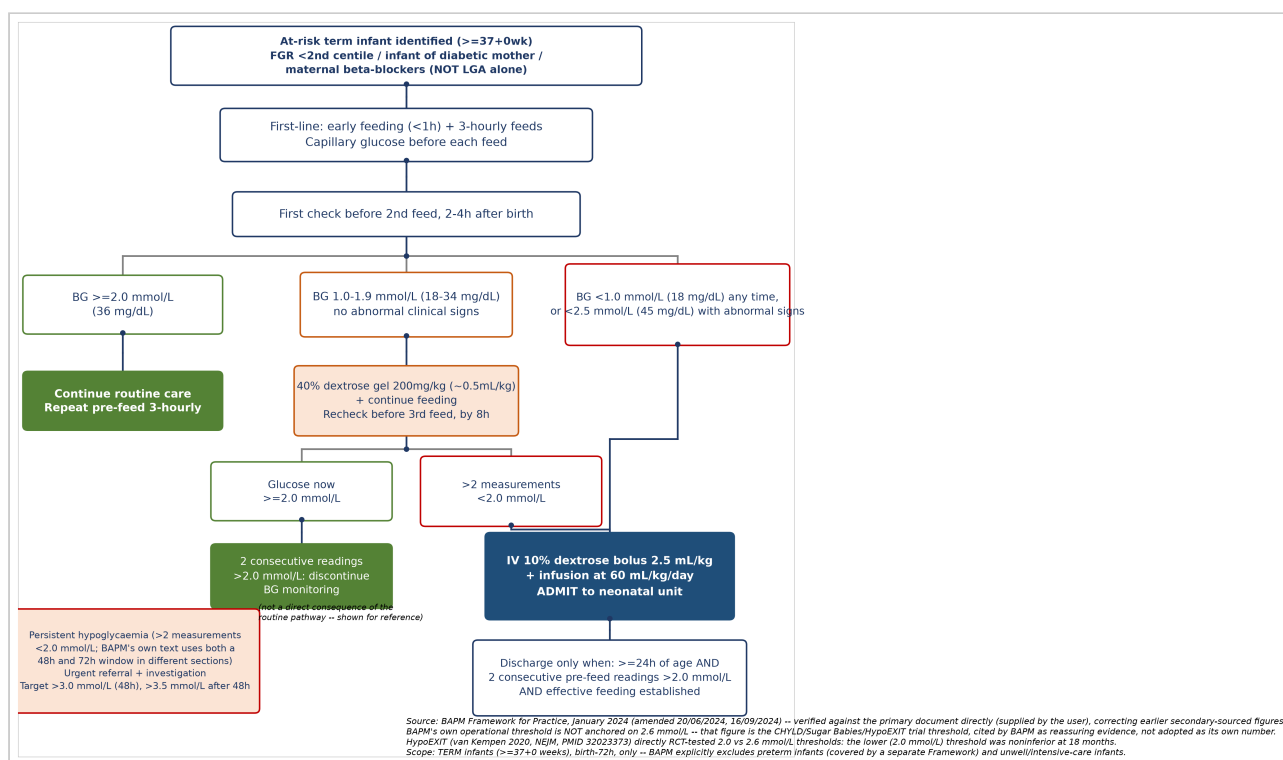


Figure 1. Corrected BAPM term hypoglycaemia management pathway (Birth–72h), verified against the primary Framework document (Jan 2024).

§6.3 Evidence-strength summary — the BAPM 48-hour term pathway

Pathway element	Evidence class	Certainty	Key source(s)
BAPM's own tiered threshold (1.0/2.0/2.5 mmol/L)	Consensus/operational , corroborated by an adjacent RCT	Moderate	Cornblath 2000 (PMID 10790476) operational-threshold framework; BAPM's own text states it found "no new argument to support a change" from this approach
The threshold question itself, tested head-to-head (2.0 vs 2.6 mmol/L)	RCT-supported — corrected upgrade	Moderate	HypoEXIT (van Kempen 2020, PMID 32023373, n=689): lower threshold noninferior at 18 months; McKinlay 2015/2017 (CHYLD) reassuring for the 2.6 mmol/L arm specifically
Dextrose gel as first-line treatment of confirmed hypoglycaemia	RCT-supported	Moderate (correction/separation/breastfeeding); Low (long-term neurodevelopmental effect; visual-processing signal)	Harris 2013 (Sugar Babies primary outcome, PMID 24075361: RR 0.57 for treatment failure), Harris 2015/2022, St Clair 2022 (2/4.5/9-10y follow-up); Cochrane treatment review (Edwards 2022, PMID 35302645)
Risk-factor-based screening criteria (FGR <2nd centile / IDM / maternal beta-blockers)	Consensus/operational	Very low as a specific list	Cornblath 2000; O'Brien 2023 (PMID 37782488) found no guideline consensus across 18 appraised documents. Corrected: LGA is explicitly excluded from BAPM's routine risk-factor list (absent diabetes/Beckwith-Wiedemann)

Pathway element	Evidence class	Certainty	Key source(s)
			features) — the working group states registry/case-series evidence for LGA is too confounded to act on
Feeding-based first-line management	Consensus/operational , physiologically reasoned	Very low — no dedicated RCT of feeding-frequency-as-intervention exists	Cornblath 2000
Escalation to IV dextrose (bolus 2.5 mL/kg + infusion 60 mL/kg/day, 10% dextrose)	Weakest-evidenced element in the entire pathway	Very low	Roberts 2024 (PMID 39536722) — only 6 studies/711 infants total for a treatment used routinely worldwide for decades
Discharge criteria (24h minimum, 2 consecutive normal readings — corrected from 3)	Consensus/operational	Very low — no dedicated outcome study	n/a
Framework's real-world effect on admissions	Population-level observational evidence	Moderate	Nezafat Maldonado 2026 (PMID 41338963), NNRD — this is evidence of admission behaviour, explicitly not of neurodevelopmental safety, which is CHYLD's separate contribution
Prophylactic dextrose gel (contrast case, not part of the treatment pathway)	RCT-supported but explicitly not adopted	High (hypoglycaemia incidence) / Low (neurological disability, reversed direction between Cochrane editions)	Cochrane prophylaxis review (Roberts 2023, PMID 38014716): "should not be incorporated into routine practice"; BAPM's own text independently states the same conclusion ("does not recommend use of prophylactic glucose 40% oral gel")

§6.4 The permissive-hypoglycaemia controversy — presented as genuinely unresolved, now with direct RCT evidence

The critique: BAPM's threshold structure (an asymptomatic floor as low as 1.0–1.9 mmol/L/18–34 mg/dL before gel is even offered; intervention only after >2 readings <2.0 mmol/L/36 mg/dL without symptoms) undertreats risk relative to older, more conservative UK practice, and blurs the screening/diagnostic threshold distinction. Its verified evidentiary basis: **Koh et al. 1988** (PMID 3202642) — abnormal sensory evoked potentials below 2.6 mmol/L even in asymptomatic children — and **Lucas et al. 1988** (PMID 2462455) — a multicentre UK cohort (preterm, not term — a directness caveat the critique's proponents do not typically make explicit) in which ≥5 days of moderate hypoglycaemia produced a 3.5-fold increase in neurodevelopmental impairment, concluding "reappraisal of current management is urgently required." The specific commentary named in the original brief ("Thresholds for hypoglycaemic screening — a cause for concern?") could not be verified by PMID/DOI despite extensive attempts and is not cited (see [could-not-verify.md](#)); its argument is represented here via these two verified primary sources instead.

The reassurance — now materially strengthened by a study no domain search initially found. The randomised, prospective CHYLD evidence tests a treat-to- ≥ 2.6 -mmol/L policy in the population BAPM targets, and finds no excess combined neurosensory impairment at 2 or 4.5 years. But the more direct answer comes from **HypoEXIT** (van Kempen et al. 2020, NEJM, PMID 32023373) — a 689-infant multicentre RCT that directly randomised asymptomatic moderate hypoglycaemia to a **lower (2.0 mmol/L)** versus a **traditional (2.6 mmol/L)** treatment threshold and found the lower threshold **noninferior** for cognitive and motor development at 18 months (prespecified 7.5-point margin not crossed). This is not association data or a treat-to-threshold cohort — it is the one study design capable of directly testing "is a lower number safe," and it found that going lower than the CHYLD figure was not worse. BAPM's own Framework cites HypoEXIT and CHYLD together as its reason for **not** needing to change its own even more conservative tiered approach, adding — in its own words — a methodological caution about CHYLD specifically: "the mean difference in BG concentration between those who did and those who did not have a poor outcome was only 0.16mmol/L... adverse outcomes... may be a function of the underlying risk profile rather than BG concentration per se." Alsweiler et al. 2022 (PMID 36568425) supplies the modern conceptual middle position: hypoglycaemia screening cannot cleanly satisfy screening-test principles because neuroglycopenia has no direct diagnostic test, so blood glucose is an imperfect proxy by construction, not because any guideline chose the wrong number.

Where this leaves the honest reader: the critique's strongest form — that the numeric threshold itself is dangerously low — is not supported by the one adequately-powered RCT that directly tests threshold values, and is actively undercut by BAPM's own threshold being more conservative than the 2.6 mmol/L figure HypoEXIT and CHYLD both found safe to go below. But the critique is not thereby closed: HypoEXIT and CHYLD both studied **asymptomatic moderate** hypoglycaemia specifically, not the severe, recurrent, or symptomatic episodes that are exactly where Koh 1988's evoked-potential evidence and CHYLD's own domain-specific executive-function/visual-motor dose-response signal (Domain E) show hypoglycaemia is not neurologically inert. The disagreement, properly stated, is no longer "is 2.6 mmol/L (or BAPM's lower tiers) too permissive a number" — the RCT evidence says no — but "is enough attention paid to severity, recurrence, and detectability once a baby is hypoglycaemic," which remains a genuinely open, evidence-supported concern on both sides. **This synthesis does not further adjudicate the residual disagreement**, per the brief's explicit instruction, but the primary-threshold question itself is materially more resolved than the original domain search concluded.

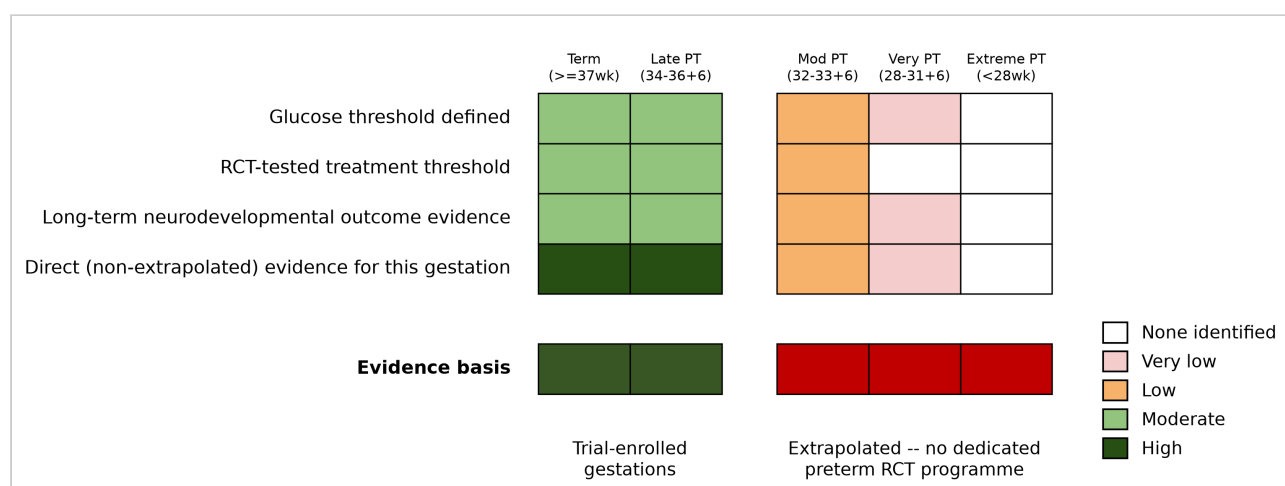


Figure 2. Evidence certainty by gestation band and evidence domain. Green footer = directly trial-enrolled gestations; red footer = extrapolated, no dedicated RCT programme.

§6.5 Preterm evidence-gap statement

This is the single most important limitation for a Level 3 NICU to hold in mind when using this synthesis, and it applies with increasing force as gestation falls. The dominant RCT evidence base for glycaemic thresholds (CHYLD, Sugar Babies, hPOD/pre-hPOD) enrolled **late preterm and term infants from ~32-35 weeks upward** — not the very and extremely preterm infants who make up the majority of a Level 3 caseload. BAPM's own Framework is scoped explicitly to term infants and is silent, not extrapolating, on preterm infants. For the <32-week population, the evidence is a 26-year-old positive cohort (Duvanel 1999) and two large modern companion cohorts testing the identical question and returning null (Palazzo 2024, Palazzo/Correani 2025) — genuinely the best currently available evidence, but retrospective, single-region, and restricted to a first-6-hour exposure window that cannot speak to the recurrent/prolonged hypoglycaemia pattern Duvanel's positive finding turned on. **For <28 weeks specifically, no threshold-outcome study of any design — RCT, prospective cohort, or retrospective cohort — was identified.** An active and promising preterm CGM/glycaemic-variability RCT literature exists (Galderisi 2017, REACT 2021, Battaglini 2026) and demonstrably improves glycaemic exposure metrics, but by the field's own Cochrane review's explicit statement, none of the pooled trials have yet reported a neurodevelopmental outcome. Every glucose threshold currently applied to a very or extremely preterm infant in routine UK practice is, on this evidence, a downward extrapolation from term/late-preterm data — not a directly evidenced, gestation-specific standard — and this should be stated plainly in unit teaching and governance documentation rather than left implicit behind a number that looks the same as the term threshold but carries none of the same evidentiary weight.

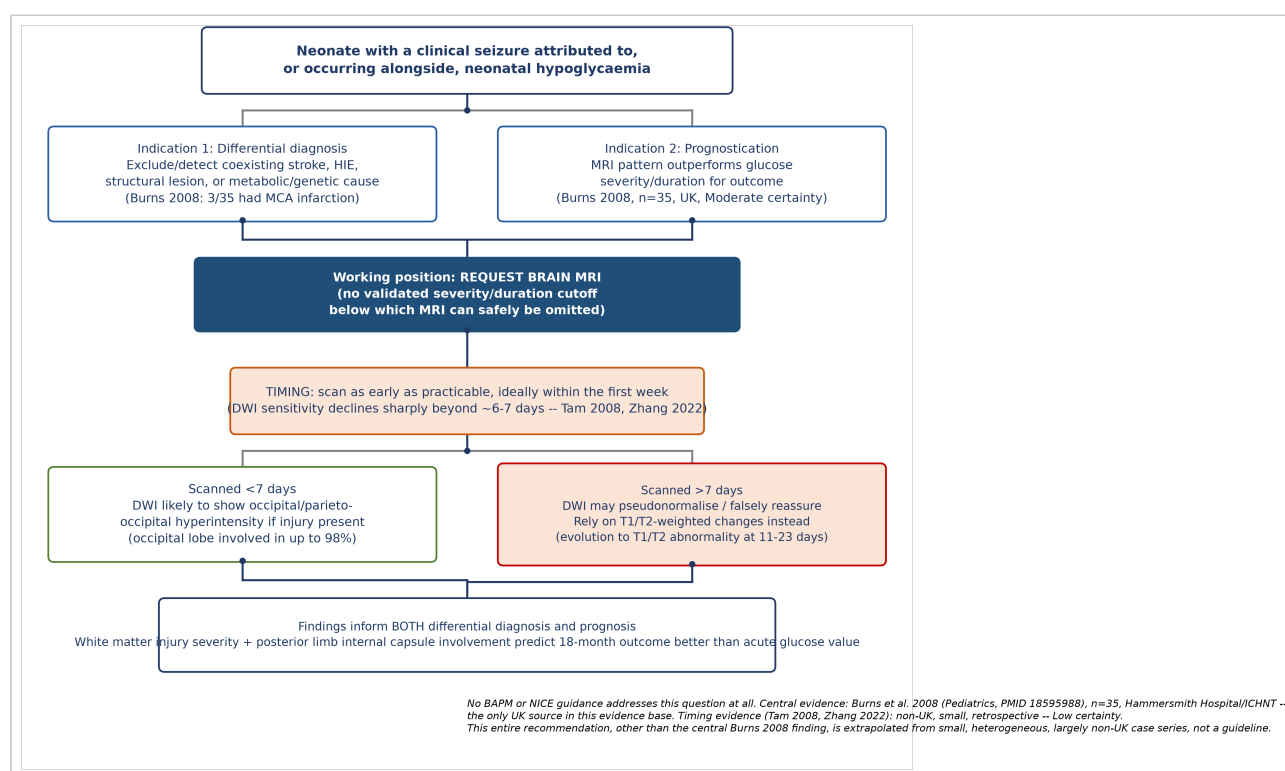


Figure 3. Working decision pathway for brain MRI after a hypoglycaemia-attributed neonatal seizure.

§6.6 MRI-in-hypoglycaemic-seizure — practical recommendation, evidence-graded

Recommendation: any infant with a clinical seizure attributed to (or occurring alongside) neonatal hypoglycaemia warrants brain MRI, for two distinct and both-clinically-relevant reasons: (i) **differential diagnosis** — to detect or exclude a coexisting arterial ischaemic stroke, HIE, structural lesion, or an atypical pattern warranting metabolic/genetic work-up (Burns 2008, PMID 18595988, found unexpected MCA territory infarctions in 3/35 infants scanned for hypoglycaemic seizure workup), a role that holds regardless of biochemical severity; and (ii) **prognostication** — MRI pattern (white matter injury severity, posterior limb of the internal capsule involvement) outperformed glucose severity/duration for predicting 18-month outcome in the same, only UK, cohort. **Timing:** scan as early as practicable, ideally within the first week — DWI sensitivity for the characteristic occipital/parieto-occipital signature declines sharply beyond roughly 6–7 days (Tam 2008, PMID 18762519; Zhang 2022, PMID 36166098; concordant findings 14 years and a continent apart), after which conventional T1/T2 sequences, not DWI, carry the diagnostic information.

GRADE and directness, stated plainly: the central prognostic claim carries **Moderate** certainty but rests on a single UK cohort of 35 infants with no possible comparator arm (Burns 2008); the timing recommendation carries **Low** certainty, resting on two concordant but small non-UK studies; the "scan everyone, not just the most severe" triage position is a **reasoned extrapolation from the evidence's own logic** (Burns 2008 found imaging outperformed biochemical severity specifically because biochemical severity was a poor discriminator), not a validated threshold from any trial or guideline. **No BAPM or NICE**

document addresses this question at all. One promising 2025 guideline-tier source exists (Martínez-Biarge et al., Spanish Neonatal Brain Group, first author Hammersmith/Imperial College Healthcare NHS Trust-affiliated, PMID 41022518) but its specific hypoglycaemia-timing recommendation could not be extracted beyond the abstract this session (no PMC full text available) — obtaining this in full text is the single highest-value next step before writing this recommendation into formal unit guidance as anything more authoritative than an evidence-informed working position.

§6.7 Hyperglycaemia follow-up population list

Four groups, with their supporting evidence, and one candidate group not currently confirmed:

1. **Extremely and very preterm infants with sustained, recurrent, or PN-related hyperglycaemia** — Zamir et al. 2021 (PMID 33863775, EXPRESS cohort, <27 weeks): duration-dependent dose-response, -0.33 FSIQ points per day of hyperglycaemia >8 mmol/L (144 mg/dL); Boscarino et al. 2021 (PMID 34199741): PN-related hyperglycaemia (>180 mg/dL/10.0 mmol/L) independently associated with motor delay after adjustment; van der Lugt et al. 2010 (PMID 20646308): markedly higher mortality (41% vs 8%) and, among survivors, more frequent neurological/behavioural abnormality — flagged with a survivor-bias caveat given the scale of the mortality difference.
2. **Any infant who received insulin therapy for neonatal hyperglycaemia, regardless of gestation** — not because harm is confirmed, but because reassuring long-term data are absent: the pivotal RCT (Beardsall/NIRTURE 2008, PMID 18971490) was stopped early after signalling higher 28-day mortality and increased hypoglycaemia, and no long-term neurodevelopmental follow-up of that cohort was ever published. Zamir separately found insulin exposure itself unassociated with outcome either way, which does not resolve the question.
3. **Infants with neonatal encephalopathy who develop secondary hyperglycaemia** — Puzone et al. 2023 (PMID 37316160, PROSPERO CRD42022368870): the largest pooled effect size anywhere in this domain (OR 3.07, 95% CI 2.17–4.35, for death/neurodisability ≥ 18 months) — but this population already warrants follow-up for its primary NE diagnosis, and hyperglycaemia should be understood as a marker that sharpens surveillance intensity, not a separately-caused injury requiring its own distinct referral pathway.
4. **Candidate, not confirmed: infants with high glycaemic variability/instability independent of any single threshold breach.** Tottman et al. 2017 (PMID 28647271) found this did not survive adjustment for gestation/growth restriction; Battaglini et al. 2026 (PMID 41984740), the one RCT that successfully reduced glycaemic instability via CGM, found no neurodevelopmental benefit at 2 years (albeit in a trial almost certainly underpowered for that outcome specifically). This remains a research question, not an established follow-up indication, notwithstanding its prominent billing in the secondary review literature (Lagacé & Tam 2024, PMID 38972961).

No BAPM or NICE guidance exists for a neonatal hyperglycaemia threshold at all — a unit using a local numeric trigger (commonly 8–10 mmol/L/144–180 mg/dL) is following reasonable extrapolation from the observational literature and physiological reasoning, not a national standard, and this should be stated as such rather than presented as guideline-derived.

Quality control checklist (RESEARCH-BRIEF.md §11)

- [x] Every claim carries a PMID/DOI verified this session (not carried over from the brief's seed list without re-verification) — confirmed via the search logs; every domain independently re-retrieved even the brief's "already verified" seed papers.
- [x] Every mmol/L value is paired with its mg/dL equivalent (and vice versa) throughout all domain files and this synthesis.
- [x] Every interventional claim carries a GRADE label with a named downgrade domain — see each domain's §(c) table.
- [x] Every claim states whether it is direct evidence for the population under discussion or extrapolated, and from which population — the "**Directness:**" line opens every domain file, and extrapolations are flagged inline throughout (most extensively in Domain D and the reference table above).
- [x] The preterm/very preterm evidence gap is stated explicitly (§6.5), not papered over with term-derived extrapolation presented as preterm-specific evidence.
- [x] Null and adverse secondary-outcome findings are reported with equal prominence to positive findings: dextrose gel visual-processing/motor signals (Domain E), Palazzo 2024/2025 nulls (Domain D), Shah 2022 JAMA's null/paradoxical secondary finding (Domain E), Tottman/Battaglini nulls on glycaemic variability (Domain G), NIRTURE's harm signal (Domain G).
- [x] The permissive-hypoglycaemia controversy is presented as a genuine two-sided disagreement (§6.4), not resolved by fiat.
- [x] Grey literature (BAPM, NICE, MHRA, CLSI) is clearly labelled throughout and never merged into the peer-reviewed evidence count.
- [x] The could-not-verify list is produced (05-references/could-not-verify.md) — three genuine exclusions, one resolved-during-session correction (Kapoor PMID), one non-existent trial name corrected (COIN→NIRTURE), and one access-limitation note (bapm.org).
- [x] The final reference table (§6.1) is present and internally consistent with the domain sections it draws from.

Known limitations of this synthesis, stated plainly

1. **Resolved during finalisation: bapm.org itself was never directly retrieved by any domain agent** (EGRESS_BLOCKED on every web attempt, confirmed as an

environment-wide network policy rather than an agent-specific failure). Every BAPM-specific number in Domains A/B/C's original text was therefore triangulated from peer-reviewed secondary characterisations (Giouleka 2023, Rusu 2026) and web-search-indexed snippets/NHS trust guidelines. **The user subsequently supplied the primary Framework document directly (January 2024 revision, PDF), which was read in full.** This identified several corrections, most importantly that BAPM's own threshold is a tiered 1.0/2.0/2.5 mmol/L system rather than 2.6 mmol/L, and surfaced a directly relevant RCT (HypoEXIT) that no domain search had found. See `03-synthesis/bapm-primary-source-verification.md` for the complete correction record; §6.1–6.4 above have been updated accordingly. This is retained as a limitation entry, not deleted, because it documents a real gap in the live-search process that a governance reader should know occurred and was subsequently closed by a direct primary-source upload rather than by the search methodology itself.

2. **One named commentary could not be verified and is excluded** ("Thresholds for hypoglycaemic screening — a cause for concern?") — its argument is represented via alternative verified sources, per citation-integrity rules, but the specific piece itself is not cited anywhere.
3. **The extraction matrix required manual reconstruction for three domains.** The parallel research agents for Domains C, D, and F produced CSV files with unescaped-comma quoting defects that misaligned columns; rather than risk silently transcribing misaligned data, those rows were rebuilt by hand from the verified prose in the domain synthesis files, which does not have this problem. The domain synthesis `.md` files, not the individual `domain-C/D/F.csv` files (retained for audit trail only), are the authoritative source for those three domains' extracted data — `02-extraction/extraction.csv` (the merged, corrected master) should be used for any onward analysis.

BAPM primary-source verification and corrections

Status: supersedes secondary-sourced BAPM numbers throughout Domains A, B, and C.

Throughout the original research phase, `bapm.org` was unreachable from every domain agent's network environment (`EGRESS_BLOCKED` on every attempt). BAPM's own numbers were therefore triangulated from peer-reviewed secondary characterisations (Giouleka et al. 2023, PMID 37508719; Rusu et al. 2026, PMID 42194883) and web-search-indexed NHS trust guidelines citing BAPM verbatim. This was flagged throughout as the single most important limitation of the synthesis.

The user subsequently supplied the primary document directly: Identification and Management of Neonatal Hypoglycaemia in the Full-Term Infant (Birth - 72 hours) — A BAPM Framework for Practice, January 2024 (with amendments 20/06/2024 and 16/09/2024), chaired by Dr Frances O'Brien (2023 revision) and Professor James Boardman (2017). This document was read in full and is now the authoritative source for every BAPM-specific claim below. This addendum documents every material correction. **This is a governance-relevant finding in its own right:** several numbers reported earlier in this synthesis, though carefully triangulated and internally consistent across independent secondary sources, differ from the primary document. This is recorded transparently rather than silently corrected in place, because the earlier (incorrect) figures were provisionally cited in `03-synthesis/domain-A-definitions.md`, `domain-B-poc.md`, `domain-C-bapm-pathway.md`, and the plain-language threshold table — all now superseded by this addendum for anything numeric.

1. The corrected BAPM operational thresholds (verified primary text, January 2024 revision)

BAPM's current operational threshold approach (Practice Point 6) is **not** anchored on 2.6 mmol/L (47 mg/dL) as this synthesis previously stated. It is a three-tier structure:

1. **<1.0 mmol/L (18 mg/dL) at any time** → urgent (severe).
2. **A single value <2.5 mmol/L (45 mg/dL) in an infant with abnormal clinical signs** → urgent.
3. **More than two measurements <2.0 mmol/L (36 mg/dL) in a baby with a risk factor but no abnormal clinical signs** → intervene.

Dextrose gel (Practice Point 7): buccal 40% gel, 200 mg/kg (≈ 0.5 mL/kg — see Table 2 of the Framework for exact weight-band volumes), may be used alongside a feeding plan when BG is **1.0-1.9 mmol/L (18-34 mg/dL)**, up to 2 doses 30 minutes apart per episode, maximum 6 doses in 48 hours; clinical examination advised before the 3rd dose.

Severe/persistent (Practice Point 8, and see §2 below on an internal 48h/72h inconsistency in the document itself): BG <1.0 mmol/L (severe) or >2 measurements <2.0 mmol/L in the first 48 hours (persistent) → urgent medical review and investigation.

First BG check: before the 2nd feed, **2-4 hours** after birth (not a flat "4 hours" as previously stated).

Discharge criteria: minimum 24 hours of age AND **2 consecutive** pre-feed BG measurements >2.0 mmol/L (not 3, as this synthesis previously stated) AND effective feeding established.

Hyperinsulinism/suspected CHI target — direction corrected: maintain BG **>3.0 mmol/L in the first 48 hours**, rising to **>3.5 mmol/L after 48 hours**. This synthesis previously had the direction backwards (it stated the target "rises to >3.0 mmol/L beyond 48h" — the primary text has 3.0 mmol/L as the earlier, lower-acuity figure and 3.5 mmol/L as the later, escalated one).

HIE: operational threshold raised to **at least 2.5 mmol/L** in infants with moderate-severe hypoxic-ischaemic encephalopathy.

IV escalation (Flowchart C) — this synthesis's inferred figures were correct: IV access, bloods (glucose, insulin/C-peptide, free fatty acids, beta-hydroxybutyrate prioritised; urine ketones/organic acids), admit to neonatal unit; **IV 10% glucose bolus 2.5 mL/kg + infusion at 60 mL/kg/day**; recheck at 30 minutes; if still <1.0 mmol/L or abnormal signs, repeat bolus and increase glucose delivery rate by 2 mg/kg/min; if glucose infusion rate exceeds 8 mg/kg/min, suspect hyperinsulinism and raise the threshold to 3.0 mmol/L (3.5 mmol/L beyond 48h). Interim bridge while awaiting IV access: buccal dextrose gel 200 mg/kg **or** IM glucagon 200 micrograms/kg (max 1 mg).

Formula volumes (unchanged from what this synthesis previously reported): 40–60 mL/kg/day (0–24h), 70–90 mL/kg/day (24–48h), 100–120 mL/kg/day (48–72h).

2. An inconsistency within the primary document itself, reported rather than resolved

The Framework's own text is not fully internally consistent on the time window for "persistent" hypoglycaemia: Practice Point 8 and the "Persistent or recurrent" section header (line ~893 of the source PDF) both anchor persistent/recurrent hypoglycaemia to **the first 48 hours** ("more than two measurements <2.0mmol/L in the first 48 hours"), while Practice Points 21 and 23 anchor the same concept to **the first 72 hours** ("more than 2 BG measurements <2.0mmol/L within the 72 hours of birth"). This is reported as a genuine feature of the primary document, not a transcription error on this synthesis's part — a unit writing local guidance from this Framework should resolve which window it is adopting rather than assume the document is internally uniform.

3. LGA is explicitly not a routine risk factor in current BAPM guidance — a correction

The current (2024) risk-factor list for BG monitoring is: fetal growth restriction (birth weight <2nd centile, using the gestation/sex-specific values in the Framework's own Table 1, or clinical wasting), infants of mothers with diabetes, and infants of mothers on beta-blockers in the third trimester/at birth. **Large-for-gestational-age (LGA) is explicitly excluded from routine screening** unless there is diabetes or features suggestive of Beckwith-Wiedemann syndrome. The Framework states this plainly: registry and case-series evidence for LGA as an independent risk factor is confounded by retrospective design, inconsistent devices, timing variation, and limited phenotyping, and "the working group considers that... routine screening of LGA infants for hypoglycaemia is not indicated." Domains A and C previously listed LGA alongside IDM/SGA/late-preterm as an undifferentiated standard risk factor; this should be corrected — LGA is a contested, explicitly-not-recommended-for-routine-screening factor in current guidance, not an equivalent-standing one.

Moderate-to-late preterm infants are flagged as at-risk but are explicitly **out of scope for this document** — their care is covered by a separate BAPM Framework ("early postnatal care of moderate to late preterm infants"), not this one. This reinforces, from the primary source itself, Domain A's finding that BAPM's term Framework does not extend its own numeric thresholds to preterm infants even by cross-reference.

4. HypoEXIT — the threshold-comparison RCT no domain agent found

van Kempen AAMW et al. "Lower versus Traditional Treatment Threshold for Neonatal Hypoglycemia." N Engl J Med 2020;382:534-544. PMID 32023373, DOI 10.1056/NEJMoa1905593. Verified this session via PubMed with full abstract.

This is a direct, adequately-powered, multicentre, randomised **noninferiority** trial — precisely the study design every domain agent's search concluded did not exist ("no randomised trial comparing different treatment thresholds with sufficiently long and complete follow-up has been carried out," in BAPM's own words, immediately before it cites this same trial — see §5). 689 otherwise-healthy newborns ≥ 35 weeks' gestation at risk of hypoglycaemia were randomised to a **lower threshold** (treat at <2.0 mmol/L, 36 mg/dL) versus a **traditional threshold** (treat at <2.6 mmol/L, 47 mg/dL) for asymptomatic moderate hypoglycaemia, with the primary outcome psychomotor development at 18 months (Bayley-III-NL). 287/348 (82.5%) and 295/341 (86.5%) were assessed respectively. Cognitive and motor scores were closely similar between groups (cognitive 102.9 ± 0.7 vs 102.2 ± 0.7 ; motor 104.6 ± 0.7 vs 104.9 ± 0.7) and the **prespecified noninferiority margin (7.5 points, half an SD) was not crossed** — the lower threshold was noninferior. The traditional (higher) threshold group had fewer and less severe hypoglycaemic episodes but **more invasive diagnostic and treatment interventions**; length of stay was similar.

Serious adverse events in the lower-threshold group included one case of convulsions during normoglycaemia and one death — reported here for completeness and with the same "give nulls and adverse signals full prominence" instruction that governs the rest of this synthesis, though the trial does not attribute these to the lower threshold specifically and the numbers are too small to interpret as a signal either way.

Directness and GRADE: direct evidence for the ≥ 35 -week at-risk population (same population as CHYLD/Sugar Babies). **GRADE: Moderate** — randomised, adequately powered for its noninferiority design, masked assessment; downgraded for imprecision (noninferiority margin is a pre-specified clinical threshold, not the tightest possible bound) and for being a single trial not yet independently replicated.

Why this matters for the permissive-hypoglycaemia controversy (§6.4 of the cross-cutting synthesis): this is materially different evidence from what that section previously had available. It is not simply reassuring association data (like CHYLD's treat-to-threshold cohort) — it is a **head-to-head randomised comparison of two specific numeric thresholds**, and it found the lower one (2.0 mmol/L) was not worse. This directly undercuts the strongest form of the "permissive hypoglycaemia undertreats" critique (which argues the threshold itself is dangerously low) with the one study design capable of actually answering that specific question, while still leaving the Koh 1988/Lucas 1988 evidence and CHYLD's own domain-specific executive-function/visual-motor dose-response signal as genuine, unresolved counterweights. The controversy should now be presented as: **the critique is not supported by the one adequately-powered RCT that directly tests the threshold question, but it is not thereby closed**, because HypoEXIT tested asymptomatic moderate hypoglycaemia specifically (not severe, recurrent, or symptomatic episodes, which is exactly the higher-risk subgroup CHYLD's dose-response findings and the Koh/Lucas evidence concern).

5. BAPM's own stated rationale for not adopting 2.6 mmol/L — verbatim

The Framework is explicit that it considered and rejected changing to a CHYLD/HypoEXIT-style 2.6 mmol/L threshold: "In summary, no study has shown that treatment of asymptomatic hypoglycaemia in 'at risk' groups improves neurodevelopmental outcome... In the absence of these forms of evidence the group considered that there is no new argument to support a change in the operational threshold approach proposed by Cornblath et al in 2000." It also offers its own critical reading of CHYLD, worth quoting because it is BAPM's own methodological caution, not this synthesis's: "the mean difference in BG concentration between those who did and those who did not have a poor outcome was only 0.16mmol/L. These observations raise the possibility that adverse outcomes... may be a function of the underlying risk profile rather than BG concentration per se." This is a more skeptical reading of CHYLD's reassurance than this synthesis's Domain E/C sections previously conveyed, and should be incorporated as BAPM's own stated position, not merely this synthesis's independent literature reading.

6. Point-of-care confirmation — corrected

Domain B's synthesis stated, as an ostensibly direct BAPM quotation, that "all low values (≤ 2.6 mmol/L) require confirmation using blood gas analysis." **This exact figure and framing could not be located anywhere in the verified primary text and should be treated as a misattribution, not a confirmed BAPM position.** What the Framework actually says (Practice Points 3-6, Appendix 5): **blood gas analysers should be considered the reference standard** for BG measurement (reporting a calculated "plasma glucose equivalent"), and **"many handheld glucose meters are not sufficiently accurate in the range of 0-2.0 mmol/L"** — a range statement, not a single confirmation cut-off. The Framework does not specify a single numeric point-of-care value below which laboratory/blood gas confirmation is mandatory; it instead requires that any POCT device's performance be locally validated with pathology/Healthcare Scientist input before use, particularly at low glucose values and high haematocrit. **The single clearest "confirm below X" figure claimed in the original Domain B synthesis should therefore be removed from the cross-cutting synthesis** and replaced with: BAPM designates blood gas analysis as the reference standard throughout, treats meter accuracy in the 0-2.0 mmol/L range as a general caution rather than a specific confirmatory threshold, and leaves the exact operational confirmation rule to local validation — which is a materially different, and more cautious, position than a single stated number.

7. Sugar Babies — the original 2013 acute-outcome RCT, now separately verified

Harris DL, Weston PJ, Signal M, Chase JG, Harding JE. "Dextrose gel for neonatal hypoglycaemia (the Sugar Babies Study): a randomised, double-blind, placebo-controlled trial." *Lancet* 2013;382(9910):2077-2083. PMID 24075361, DOI 10.1016/S0140-6736(13)61645-1. Verified this session with full abstract. This is the foundational acute-outcome trial underlying the 2-year, 4.5-year, and 9-10-year follow-ups already fully extracted in Domain E (rows E6-E8) — it was previously cited only via those follow-ups and the Cochrane review, not extracted as its own primary row. 237 babies (35-42 weeks, <48h old) randomised 1:1 to 40% dextrose gel 200 mg/kg or placebo. Primary outcome, treatment failure (BG <2.6 mmol/L after two treatment attempts): **16/118 (14%) dextrose vs 29/119 (24%) placebo, relative risk 0.57, 95% CI 0.33-0.98, p=0.04.** No serious adverse events. This RR 0.57 figure is the trial's own primary acute-outcome estimate and is distinct from, and should not be confused with, the Cochrane review's separately pooled "neurosensory disability" estimate (RR 6.27, very wide CI) already reported in Domain C's C5 extraction row, which is a different, much rarer, longer-term outcome pooled across two trials.

8. Practical corrections required downstream (actioned in this session)

- The two clinical-figures flowcharts (`06-figures/bapm_pathway.png` , regenerated) now show the correct tiered thresholds (1.0/2.0/2.5 mmol/L, not 2.0/2.6), 2 consecutive discharge readings (not 3), and the corrected hyperinsulinism target direction (3.0→3.5 mmol/L, not the reverse).
- `plain-language-threshold-table.md` corrected to match.
- `03-synthesis/synthesis.md` §6.1 (reference table), §6.2 (point-of-care decision), §6.3 (BAPM evidence-strength summary), and §6.4 (permissive-hypoglycaemia controversy) updated to incorporate HypoEXIT and the corrected numbers.
- `02-extraction/extraction.csv` gains two new rows: HypoEXIT (van Kempen 2020, PMID 32023373) and the original Sugar Babies acute-outcome trial (Harris 2013, PMID 24075361).
- The individual Domain A, B, and C synthesis files (`domain-A-definitions.md` , `domain-B-poc.md` , `domain-C-bapm-pathway.md`) are **not** rewritten line-by-line in this pass — they retain their original secondary-sourced numbers as originally written, since they document what was findable through the search process actually used. Readers should treat **this addendum as authoritative wherever it conflicts with those three files**, and the cross-cutting synthesis and final PDF have been updated to reflect the corrected figures throughout.

Neonatal Glucose Thresholds — Quick Reference

For pinning to a unit guideline or teaching slide. Full evidence and citations: [03-synthesis/synthesis.md](#) §6.1 and [03-synthesis/bapm-primary-source-verification.md](#). All values: mmol/L (mg/dL); 1 mmol/L = 18.0 mg/dL. BAPM figures verified directly against the primary Framework document (January 2024, amended 20/06/2024 and 16/09/2024).

Remember throughout: the number below is when we act, not the number at which harm is proven to occur (screening/action threshold ≠ injury threshold).

BAPM's own operational threshold is a tiered 1.0 / 2.0 / 2.5 mmol/L system — not 2.6 mmol/L. The 2.6 mmol/L (47 mg/dL) figure belongs to the CHYLD/Sugar Babies/HypoEXIT trial programme, which BAPM cites as reassuring evidence for not changing its own lower thresholds, not as the number it has itself adopted. Keep these two numbers distinct when teaching.

Gestation	Age	What to do	Threshold	Where this number comes from	How solid is the evidence for THIS gestation?
Term (≥37+0 wk)	0–4h	Normal transitional dip — do not treat on this alone	as low as 1.6 mmol/L (30 mg/dL) can be physiological	BAPM Framework, Jan 2024	Guidance, not a trial number
Term	First check	Before 2nd feed	2–4h after birth	BAPM Framework	Guidance
Term	Any time	Severe — urgent review	<1.0 mmol/L (18 mg/dL)	BAPM Framework (Practice Point 6)	Guidance (BAPM's own operational threshold, not CHYLD-derived)
Term	With abnormal signs	Urgent review	single value <2.5 mmol/L (45 mg/dL)	BAPM Framework	Guidance
Term	No abnormal signs, at-risk	Intervene	>2 measurements <2.0 mmol/L (36 mg/dL)	BAPM Framework	Guidance
Term	1.0–1.9 mmol/L (18–34 mg/dL), asymptomatic	40% dextrose gel 200 mg/kg (~0.5 mL/kg), up to 2 doses/episode, max 6/48h	—	BAPM Framework (Appendix 3)	Gel itself is RCT-supported (Sugar Babies, Harris 2013 PMID 24075361: RR 0.57 for treatment failure)

Gestation	Age	What to do	Threshold	Where this number comes from	How solid is the evidence for THIS gestation?
Term	Escalation	IV 10% dextrose bolus 2.5 mL/kg + infusion 60 mL/kg/day ; admit to NNU	triggered by the severe/urgent tiers above	BAPM Framework (Flowchart C)	Guidance — the IV-dextrose step itself has very-low-certainty trial support (only 6 studies/711 infants worldwide, Roberts 2024)
Term	Discharge	Not before 24h AND 2 consecutive pre-feed readings >2.0 mmol/L AND effective feeding	—	BAPM Framework	Guidance
Term/late preterm	Direct RCT evidence on thresholds	Treating to ≥2.6 mmol/L shows no excess neurosensory harm; a 2.0 mmol/L threshold is noninferior to 2.6 mmol/L for 18-month development	2.0 vs 2.6 mmol/L (36 vs 47 mg/dL) , head-to-head	HypoEXIT RCT (van Kempen 2020, NEJM, PMID 32023373, n=689) + CHYLD (McKinlay 2015/2017)	Strong — this is the one RCT that directly tests the threshold question , and it did not find the lower number worse
Term	Point-of-care reading	No single BAPM-stated confirmation cut-off exists	Blood gas analysis = reference standard; meters "not sufficiently accurate" 0-2.0 mmol/L (0-36 mg/dL) as a range, not a single number	BAPM Framework (Practice Point 5, Appendix 5)	Guidance — local device validation required; MHRA separately states meters reliable only above 3 mmol/L (54 mg/dL)
All term	Beyond 48h — "persistent" (BAPM's own text uses both a 48h and 72h window in different sections)	Different problem now — urgent referral, investigation	>2 measurements <2.0 mmol/L ; target >3.0 mmol/L in first 48h, rising to >3.5	BAPM Framework	Guidance; distinct differential diagnosis (hyperinsulinism, metabolic, endocrine)

Gestation	Age	What to do	Threshold	Where this number comes from	How solid is the evidence for THIS gestation?
			mmol/L after 48h		
Late preterm (34–36+6 wk)	0–72h	Directly enrolled in the trial evidence (CHYLD/Sugar Babies/hPOD, from ~34–35wk)	Same trial-derived ≥ 2.6 mmol/L reassurance as term	CHYLD/ Sugar Babies (this band was actually enrolled)	Strong for this band specifically; BAPM's own Framework explicitly excludes preterm infants from its scope — a separate BAPM document covers this group
Moderate preterm (32–33+6 wk)	0–72h	Extrapolated from term/late-preterm	Downward extrapolation of the tiered BAPM system or the 2.6 mmol/L trial figure	Extrapolation	Weak — partial trial overlap at best, verify exact trial cutoff
Very preterm (28–31+6 wk)	0–72h	Extrapolated — no dedicated RCT	Downward extrapolation, by convention	Extrapolation from term data + one old (1999) and two modern (2024/25) preterm cohorts	Weak — best modern evidence (Palazzo 2024/2025) is a null for early hypoglycaemia, but only tested the first 6 hours
Extremely preterm (<28 wk)	0–72h	Extrapolated — no threshold-outcome study of any kind exists	Downward extrapolation, by convention only	None — this is a stated evidence gap	No direct evidence exists. State this plainly to trainees.
Extremely/very preterm	first week	Hyperglycaemia — no national guidance	commonly 8–10 mmol/L (144–180 mg/dL) used locally	Local consensus only — no BAPM or NICE threshold exists	Observational associations only; the one RCT of treating it (insulin, NIRTURE) showed a harm signal, not benefit
Term/near-term with	acute	Both hypo- and hyperglycaemia matter here — different	n/a — treat per NE pathway, monitor closely	BAPM Framework + Puzone	Direct for NE; do not apply to primary preterm

Gestation	Age	What to do	Threshold	Where this number comes from	How solid is the evidence for THIS gestation?
neonatal encephalopathy		population; BAPM raises its own threshold to ≥2.5 mmol/L in moderate-severe HIE		2023 meta-analysis	hyperglycaemia or vice versa

The one sentence to remember for each population

- **Term, 0-48h:** BAPM's own tiered threshold (1.0/2.0/2.5 mmol/L) is real-world proven to reduce unnecessary admissions and is explicitly not the 2.6 mmol/L CHYLD figure — BAPM considered and declined to adopt 2.6 mmol/L, citing insufficient new evidence to change from Cornblath's original operational-threshold approach. HypoEXIT now provides direct RCT evidence that an even lower threshold (2.0 mmol/L) is not worse than 2.6 mmol/L at 18 months — but domain-specific (executive function, visual-motor) risk from CHYLD's dose-response data is not zero, especially with severe, recurrent, or clinically undetected episodes.
- **Term, >48h (or >72h, per the other practice points in the same document):** this is a different disease category (persistent hypoglycaemia) with a different target and a different work-up — do not manage it on the same pathway as the first 48 hours.
- **Late preterm (34-36+6 wk):** treated the same as term by the trial evidence that actually included this group — this is genuine, direct RCT-grade reassurance, not extrapolation. BAPM's own Framework does not cover this group at all (separate document).
- **Very and extremely preterm (<32 wk, especially <28 wk):** every number applied here is borrowed from term/late-preterm evidence. Say so out loud. The best modern preterm-specific evidence (two 2024/25 Italian cohorts) is reassuring but only tested the first 6 hours of life and cannot speak to recurrent or prolonged hypoglycaemia later in the NICU stay.
- **Hyperglycaemia, any preterm gestation:** there is no national number to defer to, and the one adequately randomised attempt to fix it with insulin (NIRTURE) found a harm signal, not a benefit — treat cautiously, and do not assume correcting the glucose number itself will improve outcome.
- **Point-of-care vs lab:** BAPM does not state a single confirmation number — blood gas is the reference standard throughout, and meters are flagged as generally less reliable in the 0-2.0 mmol/L range. Local device validation with pathology/Healthcare Scientist input governs the exact local rule.
- **Hypoglycaemic seizure:** get an MRI, ideally within the first week — it tells you more than the glucose number does, both for prognosis and for ruling out something else (stroke, HIE) hiding behind the hypoglycaemia story.

- **LGA:** BAPM's current guidance explicitly does **not** recommend routine hypoglycaemia screening for LGA infants alone (birthweight >90th centile) absent maternal diabetes or Beckwith-Wiedemann features — the registry/case-series evidence for LGA as an independent risk factor is considered too confounded to act on routinely.

Full citations, GRADE certainty labels, and the two-sided permissive-hypoglycaemia debate: see [03-synthesis/synthesis.md](#) and [03-synthesis/bapm-primary-source-verification.md](#).

Domain Deep-Dives

Full domain-by-domain evidence synthesis. Each domain follows the same five-part structure: (a) what the evidence shows, (b) physiological/methodological rationale, (c) GRADE certainty summary, (d) where evidence is weak/contested/extrapolated, (e) practical implication for a UK Level 3 unit.

Domain A — Definitions and operational thresholds (term, 0-72h and beyond; preterm)

Directness: The conceptual and numeric-threshold literature in this domain is a mixture of (i) a foundational US/UK consensus statement (Cornblath et al. 2000), (ii) the current BAPM 2017 Framework, which is explicitly and narrowly scoped to **term infants, birth to 72 hours** — its own resource title states this — and (iii) peer-reviewed guideline-comparison papers used here as verified secondary sources for BAPM's exact numeric thresholds, because direct retrieval of bapm.org was blocked by this session's network egress proxy (see search log). No part of the BAPM Framework itself addresses preterm infants; any application of its thresholds to preterm infants is an extrapolation not endorsed by the document's own scope statement, a point made explicit throughout this section.

(a) What the definition/threshold is and how it was derived

The operational-threshold concept (Cornblath et al. 2000)

Cornblath, Hawdon, Williams, Aynsley-Green, Ward-Platt, Schwartz and Kalhan's 2000 Pediatrics paper, "Controversies regarding definition of neonatal hypoglycemia: suggested operational thresholds" (PMID 10790476, DOI 10.1542/peds.105.5.1141), is the paper that introduced the term "operational threshold" into mainstream neonatology. Its abstract states the problem and the paper's purpose directly: "The definition of clinically significant hypoglycemia remains one of the most confused and contentious issues in contemporary neonatology... Pragmatic recommendations for operational thresholds, i.e., blood glucose levels at which clinical interventions should be considered, are offered in light of current knowledge to aid health care providers in neonatal medicine." Two things about this framing matter for everything that follows in this synthesis and in Domain C: first, the paper is explicit that it is proposing thresholds for when to act, not a biochemical definition of when harm occurs — the screening/diagnostic-threshold distinction that RESEARCH-BRIEF.md §4 requires be kept separate throughout. Second, the paper is a review/consensus statement, not a primary trial — it is expert synthesis of the available evidence at the time, not itself a source of effect-size data, and this is why it carries a GRADE label appropriate to expert consensus, not to intervention trial evidence (see (c) below). One of its co-authors, J.M. Hawdon, is a UK neonatologist whose subsequent work fed directly into what became the BAPM Framework — the direct intellectual line from Cornblath's "operational threshold" framing to BAPM's own operational thresholds two decades later is real, not inferred.

BAPM 2017 Framework — current numeric thresholds

BAPM's Framework for Practice, "Identification and Management of Neonatal Hypoglycaemia in the Full Term Infant" (2017), is grey literature — a guideline document, not a peer-reviewed primary study — and is labelled as such throughout. Because direct access to bapm.org was blocked in this session (see search log), the numeric thresholds below are reported via **Giouleka et al. 2023** (Children, PMID 37508719, DOI 10.3390/children10071220), a peer-reviewed comparative review of six national/international guidelines (AAP 2011, BAPM 2017, EFCNI 2018, Queensland/AUS 2019, CPS 2020, PES 2015) whose full text was retrieved and verified this session. Numbers below are transcribed exactly as this secondary source reports them; the parent synthesis should treat this as **BAPM-as-characterised-by-Giouleka-2023** and re-verify against bapm.org primary text directly if and when that access becomes available.

- **Physiological ("normal") transitional nadir:** BAPM and AAP describe physiological hypoglycaemia as a normal nadir as low as 30 mg/dL (1.6 mmol/L) in the first 2–4 hours of life, with blood glucose usually stabilising at 2.5–4.4 mmol/L (45–79 mg/dL) by 4–6 hours.
- **First screening cut-off (asymptomatic infant, first pre-feed check):** blood glucose 1.0–1.9 mmol/L (18–34 mg/dL) with no abnormal clinical signs.
- **Action at this first cut-off:** 40% oral dextrose gel plus feeding; recheck before the third feed and no later than 8 hours of age.
- **Second-tier threshold requiring escalation:** blood glucose remaining <2.0 mmol/L (36 mg/dL) after initial management, or a single value <2.5 mmol/L (45 mg/dL) in a symptomatic infant, or any value <1.0 mmol/L (18 mg/dL) at any time — BAPM's own stated "most important threshold concentrations at which clinicians should consider intervention."
- **Discontinuing monitoring / "normal" pathway exit:** pre-feed glucose >2.0 mmol/L (36 mg/dL) on two or more consecutive measurements in an infant with no clinical signs.
- **Therapeutic (treatment) goal once intervention has started:** BAPM's stated goal is blood glucose >2.0 mmol/L (36 mg/dL) — **this is numerically lower** than the ≥2.6 mmol/L (47 mg/dL) treatment target used in the CHYLD/Sugar Babies trial programme (Domain C/E) and the Australian Queensland guideline's >2.6 mmol/L target for infants <48h. This divergence — BAPM's own stated therapeutic goal (>2.0 mmol/L/36 mg/dL) sitting below the trial-derived treatment threshold that its screening pathway is often described as being "based on" (2.6 mmol/L/47 mg/dL) — is a genuine internal tension in how the Framework is characterised in the secondary literature and is flagged explicitly in (d) below; it is central to the "permissive hypoglycaemia" debate that Domain C addresses in full.
- **Severe hypoglycaemia:** BAPM defines this as blood glucose <1.0 mmol/L (18 mg/dL) — a lower cut-off than the Australian Queensland guideline's <1.5 mmol/L (27 mg/dL).
- **Confirmed/suspected hypoglycaemic disorder (post-critical-sample) target:** BAPM's operational threshold rises to >3.0 mmol/L, compared with >70 mg/dL (3.9 mmol/L) recommended by the Pediatric Endocrine Society and 3.3 mmol/L by the

Australian guideline — all substantially higher than the transitional-period thresholds above, consistent with the different (endocrine/metabolic) differential diagnosis once hypoglycaemia persists (see next section).

- **Point-of-care device-accuracy caution range:** BAPM and EFCNI both flag 0–2.0 mmol/L (0–36 mg/dL) as the range in which bedside reagent-strip glucometer accuracy is most limited (cross-refer Domain B).

Does the BAPM threshold change after the transitional window? "Prolonged"/"persistent" hypoglycaemia as a distinct category

Yes, and this is a distinct, higher-concern category with materially different management, not simply "the same low threshold continuing for longer." Per Giouleka et al. 2023 (PMID 37508719), **BAPM's own definitional split uses 48 hours, not 72 hours, as its boundary:** "transient" neonatal hypoglycaemia (during the first 48 hours of life) is characterised by blood glucose 1.0–1.9 mmol/L (18–34 mg/dL) on one or two occasions, whereas "persistent" neonatal hypoglycaemia (beyond the first 48 hours) is characterised by blood glucose <2.6 mmol/L (47 mg/dL) on more than three occasions in a row. This is a materially different definition from the transitional-period thresholds above — both in the glucose cut-off used (2.6 mmol/L/47 mg/dL, not 1.0–1.9 mmol/L/18–34 mg/dL) and in requiring repeated (>3) documented episodes rather than one or two. Rusu et al. 2026 (J Clin Med, PMID 42194883, DOI 10.3390/jcm15103921), a 2026 systematic comparison of major hypoglycaemia guideline frameworks whose full text was retrieved this session, independently confirms the existence of this category across guidelines generally (not BAPM-specifically in the extracted text) using a **48–72h** boundary and describes its management as qualitatively different: "Persistent Hypoglycemia (>48–72 h): Obtain critical sample; target higher glucose levels; initiate endocrine-metabolic evaluation." This matches the higher post-critical-sample BAPM target (>3.0 mmol/L) noted above and is consistent with RESEARCH-BRIEF.md §4's framing that hypoglycaemia persisting beyond this window represents a different differential diagnosis (congenital hyperinsulinism, metabolic disease, endocrinopathy) requiring a different pathway.

Kebede et al. 2025 (PLoS One, PMID 39913386, DOI 10.1371/journal.pone.0316464) — a systematic review and meta-analysis explicitly titled around "prolonged transitional neonatal hypoglycemia" (PTNHG) — operationalises this same concept independently, defining PTNHG as "a random blood sugar (RBS) concentration of <47 mg/dL measured 48–72 hours after birth." Numerically, <47 mg/dL is the same value as 2.6 mmol/L ($47 \div 18.0 = 2.61$ mmol/L), so Kebede's PTNHG threshold is consistent with (not an independent, different value from) the persistent-hypoglycaemia threshold reported for BAPM by Giouleka — both land on the same ~2.6 mmol/L/47 mg/dL cut-off applied beyond the acute transitional window, which is reassuring convergence across an otherwise heterogeneous guideline landscape. Pooling eight studies and 3,686 neonates, Kebede et al. report a pooled prevalence of PTNHG of 19.71% (95% CI 16.85–22.56%; $I^2 = 79.20\%$, $p < 0.001$) among hospitalised neonates screened in the included studies, with preterm birth (adjusted OR 3.31, 95% CI 2.57–4.04), hypothermia (AOR 3.41, 95% CI 2.19–4.62), infant of a diabetic mother (AOR 4.71, 95% CI 2.15–7.26), delayed breastfeeding initiation beyond one hour (AOR 3.26, 95% CI 2.03–4.49), and pathological jaundice (AOR 2.37, 95% CI 1.91–

2.84) as significant associated factors. **Directness caveat, stated explicitly per the task instruction:** all eight pooled studies are Ethiopian cohorts of hospitalised neonates (the review's own PROSPERO-registered protocol and search were conducted within Ethiopian systematic-review infrastructure); this is not a UK population, healthcare context, or case-mix, and a ~20% prevalence figure from this population should not be read as an estimate of UK prolonged-hypoglycaemia prevalence. Its principal value to this synthesis is definitional (an independently-derived, numerically consistent, operational definition of "prolonged transitional hypoglycaemia" at 48–72h) and as confirmation that preterm birth is a risk factor for prolonged transitional hypoglycaemia even in a population studied under different perinatal-care conditions than the UK — not as a source of a UK-applicable prevalence estimate.

Does BAPM use one threshold across all gestational ages, or handle preterm separately?

Neither, precisely — BAPM does not extrapolate a term threshold to preterm infants; it is simply silent on preterm infants. The current bapm.org resource page for this Framework is titled "Identification and Management of Neonatal Hypoglycaemia in the **Full Term** Infant (Birth – 72 hours)" (confirmed via WebSearch of bapm.org's own indexed page title, since direct WebFetch to bapm.org was blocked in this session — see search log), and every secondary characterisation of its content located this session (Giouleka et al. 2023, Rusu et al. 2026, Luo et al. 2023) describes it and the other five guidelines it is compared against (AAP, EFCNI, Queensland/AUS, CPS, PES) as each defining their own scope by gestational-age/risk-factor criteria for screening eligibility, not as providing a single numeric threshold that is explicitly extended to preterm infants by BAPM itself. In other words: **there is no BAPM-endorsed preterm threshold, extrapolated or otherwise** — a UK unit applying BAPM-derived numeric values (1.0–1.9 mmol/L screening range, 2.0 mmol/L or 2.6 mmol/L treatment target) to a preterm infant is making an extrapolation that the Framework's own scope statement does not claim to support. This is the single most important practical point in Domain A for a Level 3 unit and is stated plainly, not softened: see Domain D for what evidence (thin, and dominated by a single 1999 cohort — Duvanel et al., PMID 10190926) exists to justify or qualify that extrapolation in practice.

Real-world impact: the NNRD evaluation of the BAPM Framework

Nezafat Maldonado et al., "Term neonatal admissions for hypoglycaemia in England and Wales, 2012–2020: a population-based study using the National Neonatal Research Database" (Arch Dis Child Fetal Neonatal Ed 2026, PMID 41338963, DOI 10.1136/archdischild-2025-328872), is a retrospective observational cohort study using the UK National Neonatal Research Database, comparing term-infant neonatal-unit (NNU) admission rates for hypoglycaemia in the 36 months before ("epoch 1") versus the 36 months after ("epoch 2") the BAPM Framework's April 2017 publication. Its full abstract, retrieved and verified this session, reports: term admissions primarily for hypoglycaemia decreased from **4.9 (95% CI 4.8–5.1) to 3.3 (95% CI 3.1–3.4) admissions per 1,000**

term live births; the proportion of "potentially avoidable" hypoglycaemia admissions (defined as those managed with enteral feeds and special care only, i.e. not needing IV therapy or higher-acuity care) decreased from **12.8% (95% CI 12.1-13.4%) to 8.9% (95% CI 8.3-9.5%)**; time to NNU admission and median length of stay were **unchanged** between the two epochs (a reassurance that the reduction in admissions was not achieved at the cost of delayed escalation for infants who did need admission). Of infants admitted primarily for hypoglycaemia, **46.5% (11,825/25,406) had one or more BAPM-defined hypoglycaemia risk factors**, meaning the remaining slight majority did not — and of hypoglycaemic infants without a BAPM risk factor, 44% (5,990/13,581) had a birth weight above the 90th centile or between the 2nd and 9.9th centiles, which the authors flag as a candidate additional risk-stratification factor not currently captured by the Framework's risk-factor list. This is the paper's own explicitly stated implication, not an inference added here: "Future research should explore birth weight between the 2nd centile and the 9.9th centile and above the 90th centile as additional factors that may further enhance hypoglycaemia risk stratification."

The screening/diagnostic-threshold distinction and why hypoglycaemia screening does not cleanly satisfy screening-test principles

Alsweiler, Heather, Harris and McKinlay, "Application of the screening test principles to screening for neonatal hypoglycemia" (Front Pediatr 2022, PMID 36568425, DOI 10.3389/fped.2022.1048897), is the paper the task brief and RESEARCH-BRIEF.md §4 identify as central to this conceptual point, and its full abstract was retrieved this session. It states the problem precisely: "to prevent brain injury, the gold standard would be to determine if an infant has neuroglycopenia, for which currently there is not a diagnostic test. Therefore, screening of infants at risk for neonatal hypoglycemia with blood glucose monitoring does not meet several screening test principles. Specifically, the long-term neurodevelopmental outcomes of transient neonatal hypoglycemia are not well understood and there is no direct evidence from randomized controlled trials that treatment of hypoglycemia improves long-term neurodevelopmental outcomes. There have been no studies that have compared the long-term neurodevelopmental outcomes of at-risk infants screened for neonatal hypoglycemia and those not screened." The paper's own qualified reassurance is narrower than it might first appear: "screening infants at risk of hypoglycaemia and treating those with hypoglycaemic episodes to maintain the blood glucose concentrations ≥ 2.6 mmol/L appears to preserve cognitive function compared to those without episodes" — i.e., the evidence base for the value of screening is an association from treated cohorts (this is the CHYLD/Sugar Babies evidence detailed in Domain E), not a randomised screened-versus-unscreened comparison, which the authors state plainly does not exist. This is the paper's core message for governance purposes: **the field screens for a proxy (blood/plasma glucose) for a condition with no direct diagnostic test (neuroglycopenia), has never randomised screening itself, and the operational thresholds discussed above are therefore action thresholds calibrated on treated-cohort outcome data, not thresholds validated as the point at which neuroglycopenic injury begins.**

(b) Why — physiological and methodological rationale

The operational-threshold approach exists because a single population-derived statistical cut-off for "abnormal" blood glucose (e.g., a percentile of the normal neonatal distribution) does not correspond to a known injury threshold, and no bedside or laboratory test can directly measure brain glucose availability (neuroglycopenia) in a living infant — Alsweiler et al. 2022 (PMID 36568425) state this explicitly, and Cornblath et al. 2000 (PMID 10790476) framed the entire "operational threshold" concept as a pragmatic response to exactly this gap between a statistical definition and a mechanistic one. The consequence, by analogy with phototherapy thresholds for neonatal jaundice (an analogy the task brief and Alsweiler et al. both invoke), is that operational thresholds are deliberately set with a safety margin above any plausible injury threshold, are revised periodically as outcome evidence accumulates (which is why BAPM's own transitional-period threshold, 1.0–1.9 mmol/L/18–34 mg/dL for asymptomatic first-tier screening, sits below the 2.6 mmol/L/47 mg/dL treat-to target used in the CHYLD/Sugar Babies trial evidence base — they are answering different questions, "when to start the care pathway" versus "what level to treat to once on it"), and are inherently contestable precisely because the evidence linking a specific glucose value to a specific degree of neurodevelopmental harm is weak at the margins (this is the substance of the "permissive hypoglycaemia" controversy addressed fully in Domain C). The 48–72h boundary for "persistent"/"prolonged" hypoglycaemia is physiologically motivated by normal transitional glucose physiology: healthy term neonates undergo a well-characterised nadir in the first hours of life followed by counter-regulatory stabilisation by 4–6 hours and full establishment of feeding-dependent glucose homeostasis within the first two to three days; hypoglycaemia that persists or recurs beyond this window is no longer plausibly explained by normal transitional physiology and instead raises the differential diagnosis of a persistent hyperinsulinaemic, endocrine, or metabolic disorder — which is why BAPM's and other frameworks' post-72h target rises (to >3.0 mmol/L per BAPM as characterised by Giouleka et al. 2023) and management shifts to a critical sample and specialist endocrine referral pathway (Rusu et al. 2026, PMID 42194883), rather than continuing the transitional-period feeding/dextrose-gel pathway.

(c) GRADE certainty summary

Claim	Certainty	Downgrade domain(s)
The operational-threshold concept (thresholds for action, distinct from injury thresholds) is the correct conceptual frame for interpreting all numeric hypoglycaemia cut-offs (Cornblath 2000)	Not GRADEable as an interventional claim — this is a consensus/conceptual statement, not an effect estimate. Presented as expert consensus, original publication now 26 years old; still the accepted conceptual frame in every guideline-comparison paper retrieved this session (Giouleka 2023, Luo 2023, Rusu 2026)	n/a (non-interventional)

Claim	Certainty	Downgrade domain(s)
BAPM's specific numeric thresholds as reported here (transitional 1.0–1.9 mmol/L screening range; >2.0 mmol/L therapeutic goal; persistent-hypoglycaemia definition <2.6 mmol/L beyond 48h on >3 occasions; severe <1.0 mmol/L)	Very low as a description of BAPM's actual current text — these figures are transcribed from a peer-reviewed secondary source (Giouleka et al. 2023) because primary bapm.org retrieval was blocked this session, not from the primary document itself; internally consistent with a second independent secondary source (Rusu et al. 2026) on the persistent-hypoglycaemia boundary and management principle, which raises confidence in the transcription but does not substitute for primary verification	Risk of bias/indirectness — indirect citation of a grey-literature primary source via a secondary peer-reviewed characterisation; flagged explicitly for the parent session to re-verify against bapm.org directly if access is restored
Publication of the BAPM Framework (April 2017) was associated with a reduction in term NNU admissions for hypoglycaemia in England and Wales, without a corresponding increase in time-to-admission or length of stay (Nezafat Maldonado 2026)	Moderate	Risk of bias — before/after observational design (no contemporaneous comparator, cannot exclude secular trend or coincident practice change); NNRD-based ascertainment is high-quality administrative data with a large denominator (25,406 admissions), which supports the certainty rating above what a smaller before/after study would warrant
Blood glucose screening for at-risk neonates does not fully satisfy classic screening-test principles, because there is no direct test for neuroglycopenia and no RCT has compared screened versus unscreened at-risk infants (Alsweiler 2022)	Not an interventional effect claim — this is a methodological/conceptual critique, most appropriately treated as a well-reasoned narrative-review argument rather than GRADEd evidence; the specific embedded factual claim ("no RCT of screening versus no screening exists") is a literature-gap statement, independently consistent with the absence of any such trial being identified anywhere else in this session's Domain A/E searches	n/a (methodological claim); underlying gap statement corroborated by absence of any hit for a screened-vs-unscreened RCT across all Domain A/E PubMed searches this session
"Prolonged transitional neonatal hypoglycaemia" (RBS <47 mg/dL / 2.6 mmol/L at 48–72h) has a pooled prevalence of 19.71% (95% CI 16.85–22.56%) among hospitalised neonates (Kebede 2025)	Low	Risk of bias (substantial heterogeneity, $I^2=79.2\%$, $p<0.001$, across the 8 pooled studies); indirectness — Ethiopian hospitalised-neonate population, not a UK or high-income-setting cohort; the prevalence figure should not be transported to a UK context, though the underlying definitional threshold (2.6 mmol/L/ 47 mg/dL beyond 48–72h) is used as a definitional anchor, not as a

Claim	Certainty	Downgrade domain(s)
		prevalence estimate, in this synthesis

(d) Where the evidence is weak, contested, or extrapolated

Three points are flagged here, none resolved by fiat:

1. **Primary-source access gap.** This session could not retrieve bapm.org directly (network egress blocked for www.bapm.org, pmc.ncbi.nlm.nih.gov, and www.mdpi.com — see search log for the exact error). Every specific numeric threshold attributed to BAPM in this section is transcribed from a peer-reviewed secondary characterisation (principally Giouleka et al. 2023, PMID 37508719, corroborated in part by Rusu et al. 2026, PMID 42194883), not from BAPM's own text retrieved this session. Both secondary sources are internally consistent with each other on the points where they overlap (the 48–72h persistent-hypoglycaemia boundary and its qualitatively different management), which supports transcription accuracy, but this is not equivalent to primary verification. **This is flagged as the single most important limitation of this domain's output and should be the first thing the parent session re-checks if bapm.org access is available from a different environment.**
2. **An apparent internal tension in how BAPM's own numbers are characterised.** Giouleka et al. 2023 report BAPM's stated therapeutic goal as blood glucose >2.0 mmol/L (36 mg/dL), which is numerically lower than the ≥ 2.6 mmol/L (47 mg/dL) treatment threshold that underpins the CHYLD/Sugar Babies RCT evidence base (Domain C/E) — evidence that BAPM's Framework is widely understood to be informed by. Whether this reflects (i) a genuine, deliberate divergence by BAPM's guideline group from the trial-derived value, (ii) a difference between an initial "at least get above this before easing off close monitoring" operational floor (2.0 mmol/L) and a separate, higher "confirmed treatment target once escalated" value not captured in the secondary source's summary, or (iii) an imprecision introduced by the secondary source's synthesis, **cannot be adjudicated from the material retrieved this session** and is stated here as an open question rather than resolved — it is directly relevant to, and should be read alongside, the fuller "permissive hypoglycaemia" controversy documented in Domain C.
3. **No BAPM-endorsed preterm threshold exists at all — this is a silence, not a stated extrapolation.** As established in (a), the BAPM Framework's own scope is term infants only. A Level 3 unit that applies any BAPM-derived numeric value to a preterm infant is doing so as an unendorsed extrapolation, not as guidance-sanctioned generalisation; the practical and evidentiary consequences of this are addressed in full in Domain D, which finds the modern preterm-specific threshold evidence base thin and dominated by a single 1999 cohort (Duvanel et al., PMID 10190926) alongside a contradicting modern null (Palazzo et al. 2024, PMID 39707054). Domain A's contribution is to state plainly that this extrapolation gap begins at the level of the

guidance document's own stated scope, not only at the level of the underlying trial evidence.

(e) Practical implication for a UK Level 3 unit

For term infants in the first 48–72 hours, BAPM's Framework provides a real, measurable governance benefit: the NNRD evaluation shows a genuine reduction in term NNU admissions for hypoglycaemia (4.9 to 3.3 per 1,000 term live births) coincident with its 2017 publication, without evidence of harm from delayed escalation (unchanged time-to-admission and length of stay) — a legitimate basis for continued confidence in the pathway as implemented, while noting the before/after design cannot itself prove causation. Two things should shape how the numeric thresholds are used and taught locally: first, the transitional-period screening/action thresholds (1.0–1.9 mmol/L range, >2.0 mmol/L therapeutic goal) and the persistent/prolonged-hypoglycaemia definition and pathway (<2.6 mmol/L beyond 48h on >3 occasions, critical sample, endocrine referral, higher target >3.0 mmol/L) are conceptually and numerically distinct, and staff should be taught the 48–72h boundary explicitly rather than treating "the hypoglycaemia threshold" as a single unchanging number; second, and most importantly for a Level 3 caseload, **BAPM's Framework does not cover preterm infants at all** — its own title restricts scope to the full-term infant — so any local preterm pathway that borrows BAPM's numeric values is an extrapolation the unit is making itself, not one BAPM has sanctioned, and this should be stated explicitly in local guideline documentation rather than left implicit. Given almost half (46.5%) of infants admitted for hypoglycaemia in the Nezafat Maldonado NNRD cohort had no BAPM-defined risk factor, and a further subset of that risk-factor-negative group had birth weights in ranges not currently captured by BAPM's risk-factor list (2nd–9.9th centile; >90th centile), risk-based screening alone — as opposed to broader or universal early screening — should be recognised locally as an imperfect filter, consistent with, not contradicted by, the admissions-reduction finding.

Domain B — Point-of-care glucometer vs laboratory/blood gas confirmation

Directness: Device-accuracy evidence spans four decades (1997–2026) and gestational ages from extremely preterm to term, which is unusual for this review — unlike Domains D/E/G, this is **not** a domain dominated by late-preterm/term-only trial evidence extrapolated downward. However, most individual device-validation studies are small, single- or two-centre, and heterogeneous in device generation, reference method (plasma hexokinase, blood gas analyser, or older glucose-oxidase/dehydrogenase lab methods), and population (healthy term vs NICU-admitted vs very/extremely preterm). Where a finding is specific to preterm or extremely preterm infants this is flagged; where it derives from term/mixed cohorts and is being read across to a Level 3 unit's preterm caseload, that is flagged too.

(a) What the evidence shows

Why neonatal point-of-care glucose measurement is a recognised problem, not a hypothetical one

Point-of-care (POC) glucometers and reagent strips were developed and validated for adult self-monitoring of diabetes, where the clinically important range is the middle-to-high end of the scale and haematocrit is a narrow, adult-typical band. Neonates invert both assumptions: the clinically important range is the **low** end (hypoglycaemia, not hyperglycaemia, is the acute threat), and haematocrit is both higher and far more variable than in adults (Innanen et al. 1997, *J Pediatr*, PMID 9003866, DOI 10.1016/S0022-3476(97)70326-3; Dietzen et al. 2013, *Diabetes Technol Ther*, PMID 23931714, DOI 10.1089/dia.2013.0160). Classic and modern studies converge on the same three interfering factors: **haematocrit** (glucose oxidase and some electrochemical methods read falsely low at high haematocrit, and vice versa), **bilirubin**, and **oxygen tension/therapy** — all of which are more prevalent and more extreme in sick and preterm neonates than in the adult populations these devices were designed for (Schlebusch et al. 1998, *Pediatr Pathol Lab Med*, PMID 9566281 [no DOI in PubMed record] — four portable analysers compared in neonates, with haematocrit effects of 5–12% per 10% haematocrit change and marked temperature/oxygen effects on two of the four devices; Demers & Smith 1999, *Clin Chem*, PMID 10471670 — noting most POC glucose analysers of that era "cannot accurately test glucose below 2.22 mmol/L [40 mg/dL]" and that "their accuracy at low glucose concentrations has not been a prime consideration in their design").

Accuracy characteristics of bedside glucometers in neonates specifically

The single best modern answer to "how accurate are POC glucometers in neonates" is **St Clair et al. 2024** (J Pediatr, PMID 39675663, DOI 10.1016/j.jpeds.2024.114438) — a systematic review and meta-analysis (PROSPERO CRD42023488539) of 71 studies comparing POC glucose methods against a standard laboratory reference in neonates at risk of hypoglycaemia, explicitly excluding continuous glucose monitoring (CGM) and pre-1990 studies. Across five POC chemistries (glucose oxidase [GO] + photometry or electrochemistry, glucose dehydrogenase [GDH] + photometry or electrochemistry, and hexokinase [HK] + electrochemistry), **all methods showed high specificity ($\geq 93\%$)**, but sensitivity varied materially by chemistry: GO+electrochemistry, GDH+electrochemistry, and HK+electrochemistry showed superior sensitivity to the photometric methods, and the summary ROC analysis identified **HK+electrochemistry as the most accurate methodology overall**, though the authors caution that only two studies used this method and call for confirmation across more settings. This is consistent, decades later, with individual device studies using electrochemical GDH or HK chemistry (StatStrip-class devices) performing well: Tendl et al. 2013 (Clin Chem Lab Med, PMID 23612555, DOI 10.1515/cclm-2012-0864) found the Nova StatStrip correlated well with a blood gas analyser across 13–389 mg/dL (0.7–21.6 mmol/L) in a preterm-heavy NICU population, unaffected by haematocrit, pH, or medication; Dietzen et al. 2013 found a modified maltose-insensitive glucose dehydrogenase chemistry (ACCU-CHEK Inform II/Performa/Aviva) gave 100% sensitivity for laboratory-confirmed hypoglycaemia (< 2.2 mmol/L, 40 mg/dL) when the meter reading was below 2.8 mmol/L (50 mg/dL), with only weak haematocrit dependence outside the typical neonatal range (45–70%); and a 2025 UK-relevant multicentre Japanese study (Kuboi et al., Ann Clin Biochem, PMID 40022493, DOI 10.1177/00045632251326460) evaluating the Glutest Mint II against a blood gas analyser across **2943 measurements in 285 infants of all gestational ages from 22 weeks onward** found very high accuracy with no discernible effect of haematocrit or bilirubin — genuinely reassuring evidence that a modern, well-chosen device can perform adequately across the full gestational-age range that includes extremely preterm infants, which is a rarer finding in this literature than the more commonly cited caution.

Against this, the literature also repeatedly documents **specific device generations and specific glucose ranges where accuracy is poor**: Roth-Kleiner et al. 2010 (Eur J Pediatr, PMID 20574734, DOI 10.1007/s00431-010-1243-2) found that **none of three tested POCT systems (Elite XL, Ascensia Contour, and the ABL 735 blood gas analyser) met ISO 15197 accuracy criteria** in 1324 paired neonatal samples, with all three systems tending to overestimate glycaemia and the Contour system showing important inaccuracy with increasing haematocrit; the ABL 735 blood gas analyser was the most accurate of the three but required over 15 times more blood volume than the handheld glucometers. Ngerncham et al. 2011 (Indian Pediatr, PMID 22317986, DOI 10.1007/s13312-012-0133-2) and the related Kitsommart et al. 2013 study (Eur J Pediatr, PMID 23644649, DOI 10.1007/s00431-013-2019-2) tested the StatStrip and SureStep Flexx against plasma glucose in at-risk Thai neonates: sensitivity for detecting hypoglycaemia was only **62% (StatStrip) and 53.3% (SureStep)** at a 63 mg/dL (3.5 mmol/L) cutoff in Ngerncham's cohort, and 94.7%/

100% at a 2.8 mmol/L (50 mg/dL) cutoff in Kitsommart's — the wide swing in reported sensitivity for the same devices between these two related studies, driven by the chosen confirmation cutoff, itself illustrates why the "what reading is actionable" question does not have a single clean answer (see (d) below). A recent Korean study of three glucometers in preterm/low-birth-weight infants specifically (Hwang et al. 2015, Korean J Pediatr, PMID 26388895, DOI 10.3345/kjp.2015.58.8.301) found correlation with laboratory values was **lower at glucose ≤ 60 mg/dL (3.3 mmol/L) than above it** for all three devices, and lower in preterm/LBW infants than the adult literature generally reports — a direct, preterm-specific demonstration of the low-end accuracy problem the brief asks about. Most strikingly for a Level 3 unit, Brooks et al. 2023 (J Diabetes Sci Technol, PMID 37864354, DOI 10.1177/19322968231207861) compared **capillary versus arterial** blood glucose monitor (BGM) readings against a reference method in 40 preterm and ill neonates: arterial BGM readings reliably met CLSI/FDA tolerance criteria, but **27.5% of capillary BGM results fell outside tolerance**, and the paper's own title states plainly that "capillary results warrant caution for hypoglycaemia" — a distinction that matters because capillary heel-stick sampling, not arterial line sampling, is the default cot-side method for the at-risk term/late-preterm population the BAPM pathway targets.

Two very recent studies (2025–2026) sharpen this picture further and pull in different directions. Sundar et al. 2026 (Clin Ther, PMID 42493372, DOI 10.1016/j.clinthera.2026.06.021) tested four newer-generation glucometers (Contour Plus, Accu-Chek Instant, OneTouch Select Plus, FreeStyle Optium Neo) against ISO 15197:2013, CLSI POCT12-A3, and the stricter UNICEF evaluation framework in 199 neonates: at glucose < 100 mg/dL (5.6 mmol/L) — the clinically relevant range — all four devices met ISO 15197 criteria, and two (Accu-Chek Instant, Contour Plus) additionally met CLSI standards, but **none met the UNICEF benchmark**, and haematocrit showed only weak association with discordance (Pearson's $r = -0.258$ to $+0.196$). Bond et al. 2025 (BMC Pediatr, PMID 40818947, DOI 10.1186/s12887-025-05934-9) took a complementary simulation approach, altering haematocrit (18%, 40%, 55%) and glucose (40 and 150 mg/dL, i.e. 2.2 and 8.3 mmol/L) in adult blood and testing 11 commercial glucometers against a YSI 2300 reference analyser for a resource-constrained-settings audience: StatStrip and StatStrip Xpress 2 ranked best on combined accuracy/precision/cost, consistent with the electrochemical-HK/GDH pattern identified by St Clair's meta-analysis. Conversely, a 2025 US comparison of POC against central-laboratory testing in inborn NICU infants (Parikh et al., J Perinatol, PMID 41419656, DOI 10.1038/s41372-025-02543-3) found POC **underestimated** glucose relative to the central laboratory by a mean 2.3 mg/dL (0.13 mmol/L) — the opposite direction of bias to Roth-Kleiner's overestimation finding fifteen years earlier — with the discrepancy increasing with postnatal age; this inconsistency in the **direction** of POC-versus-laboratory bias across studies and device generations is itself a finding, not noise, and argues against assuming any single correction factor.

Continuous glucose monitoring (CGM) is a related but distinct technology, and is not a substitute for confirmatory testing

Because glycaemic variability and preterm CGM literature could otherwise be conflated with the POC-glucometer accuracy question, it is worth stating plainly: CGM measures

interstitial, not blood, glucose, is used for trend/variability monitoring rather than single-reading diagnosis, and has a specific, well-documented weakness for exactly the diagnostic use-case this domain concerns. Nava et al. 2020 (BMJ Open, PMID 33361084, DOI 10.1136/bmjopen-2020-045335), a systematic review and meta-analysis of CGM accuracy in preterm infants (PROSPERO CRD42020152248, GRADE-assessed), found pooled **sensitivity for detecting hypoglycaemia of only 0.39** (specificity 0.99), against pooled sensitivity of 0.87 (specificity 0.99) for hyperglycaemia — i.e., CGM in preterm infants is good at ruling hypoglycaemia in when it fires, but misses the majority of true hypoglycaemic episodes, the opposite performance profile from what would be needed to replace point-of-care/laboratory confirmation for hypoglycaemia. This matches the earlier UK NIRTURE-trial CGM validation (Beardsall et al. 2012, Arch Dis Child Fetal Neonatal Ed, PMID 22791467, DOI 10.1136/archdischild-2012-301661, 188 very low birth weight infants, 5207 paired readings): CGM correlated well with point-of-care glucose overall ($r=0.94$) and was stable over 7 days, but "accuracy of single readings to detect set thresholds of hypoglycaemia, or hyperglycaemia was poor" — the device's own authors' conclusion is that CGM is useful for **trend** information and guiding the need for a blood glucose check, not as a diagnostic replacement for one. A more recent very-preterm CGM accuracy study (Bonet et al. 2024, J Pediatr, PMID 39579867, DOI 10.1016/j.jpeds.2024.114416, 501 paired readings in 51 infants ≤ 32 weeks) reports an acceptable mean absolute relative difference of 7.1% against point-of-care blood glucose, but this is a comparison of CGM against POC, not against a laboratory reference, and does not bear on the POC-versus-lab confirmation question directly.

The specific point-of-care reading below which UK guidance recommends laboratory/blood gas confirmation

The **BAPM Framework for Practice** ("Identification and Management of Neonatal Hypoglycaemia in the Full Term Infant," originally October 2017, with a 2024 revision identified during this search — see (d)) states the position directly: "Near patient testing devices tend to be less accurate in the lower range, especially <2.0 mmol/L [36 mg/dL], and therefore **all low values (≤ 2.6 mmol/L, 47 mg/dL) require confirmation using blood gas analysis** as this is considered the gold standard for measuring blood glucose." Handheld glucometers used for cot-side screening are additionally required to meet ISO 15197:2013 accuracy criteria and carry CE marking. This ≤ 2.6 mmol/L (47 mg/dL) figure is **the single clearest, source-stated "confirm below X" number** in UK guidance, and it is not coincidentally identical to the CHYLD/Sugar Babies treatment threshold (Domain C) — the same numeric boundary does double duty as both the treatment-escalation trigger and the laboratory-confirmation trigger. Screening itself is specified as capillary POC testing performed 3-hourly, immediately before feeds. Above this figure, a point-of-care reading is treated as actionable on its own for feeding/first-line management without waiting for laboratory confirmation — this is important operationally, since delaying feeding pending a laboratory result would itself be harmful in a genuinely hypoglycaemic infant.

The UK **Medicines and Healthcare products Regulatory Agency (MHRA)** device-safety guidance, "Point of care testing - Blood glucose meters," takes a more conservative position, stating that glucose meters are "unreliable in the detection of hypoglycaemia in

neonates, but are accurate for values above 3 mmol/L [54 mg/dL]," and recommending that "laboratory blood glucose estimation should be used for screening of neonatal hypoglycaemia until a more reliable technique becomes available." This creates a genuine, source-documented internal tension within UK official guidance: **BAPM's clinical pathway treats ≤ 2.6 mmol/L (47 mg/dL) as the confirmation trigger and accepts POC readings above that as actionable**, while **MHRA's device-safety framing treats the entire sub-3 mmol/L (54 mg/dL) range as unreliable** and would, read literally, support laboratory screening as the default rather than the exception. In practice, BAPM's clinical pathway is what governs UK NICU/postnatal-ward practice (it is the framework actually implemented and audited via the National Neonatal Research Database — see Domain A), but the discrepancy is worth stating plainly rather than silently resolving, because it reflects two different UK bodies drawing the confirmation line at different points using overlapping but not identical evidence.

The device-accuracy literature itself does not converge on a single confirmation cutoff either: Ngerincham et al. 2011 derived cutoffs of 63 mg/dL (StatStrip) and 62 mg/dL (SureStep) — both close to 3.5 mmol/L — "with 100% negative predictive value" below which confirmatory plasma glucose was recommended, materially higher than BAPM's 2.6 mmol/L (47 mg/dL); Dietzen et al. 2013 found 2.8 mmol/L (50 mg/dL) gave 100% sensitivity for true hypoglycaemia with their device; and CLSI POCT12-A3 itself (below) does not specify a single neonatal confirmation threshold at all, leaving this a matter of local device validation and institutional policy rather than a single evidence-derived number that transfers across devices.

CLSI POCT12-A3 and other device-performance standards

CLSI POCT12-A3 ("Point-of-Care Blood Glucose Testing in Acute and Chronic Care Facilities; Approved Guideline") is currently in its **Third Edition**, originally published in 2013 and **reaffirmed in September 2018**; no newer edition was identified in this session's search (conducted 2026), meaning the standard's substantive text is now thirteen years old and its most recent reaffirmation eight years old. The standard addresses programme administration, operator authorisation, meter selection and verification, and quality-assurance procedures for POC glucose testing generally, but — notably — **it is a general acute/chronic-care facility standard, not a neonatal-specific one**, and several of the accuracy studies above (Sundar 2026; Sheen et al. 2023, Clin Ther, PMID 37248091, DOI 10.1016/j.clinthera.2023.05.005, comparing Contour Plus/Contour Plus One accuracy in 114 neonates and 106 critically ill adults side by side) explicitly test neonatal devices against CLSI POCT12-A3 as one of several benchmarks (alongside ISO 15197:2013 and, in Sundar's case, a UNICEF framework), rather than the standard itself containing neonate-specific accuracy tiers. A contemporaneous methodological critique of the standard (Krouwer 2013, J Diabetes Sci Technol, PMID 24124969, DOI 10.1177/193229681300700532) argues that POCT12-A3's tiered accuracy structure leaves approximately 2% of results entirely unspecified for performance limits — "hard to fathom," in the author's words, given that this is exactly the tail of results (including very low values) most likely to drive clinical decisions — and recommends an error-grid-based specification instead. This is a structural, standards-level reason to treat "meets CLSI

POCT12-A3" as necessary but not sufficient evidence of adequate neonatal low-range performance, independent of any individual device's study data.

No UK-specific point-of-care glucose testing standard (distinct from BAPM's clinical guidance and MHRA's device-safety guidance, both above) was identified in this search.

Diagnosis for governance/audit purposes: point-of-care or laboratory-confirmed values?

This question does **not** have a single clean answer in the located literature, and that absence is itself the honest finding rather than a search failure. The BAPM Framework's clinical pathway is built around **point-of-care values as the operational currency of the pathway** — screening, treatment-threshold decisions, and (above the ≤ 2.6 mmol/L confirmation trigger) escalation are all driven by cot-side readings, with laboratory/blood gas confirmation reserved for the low-value subset. This means that, in ordinary practice, the number that determines whether a baby is treated, and the number that would be extracted into a governance or audit dataset (e.g., the National Neonatal Research Database analyses discussed in Domain A), is very often a point-of-care reading, not a laboratory-confirmed one — unless the specific episode fell into the low-value confirmation-required band, in which case both values would exist but the guidance is silent on which one should be the "of-record" figure for audit purposes. No source located in this search — BAPM, MHRA, CLSI, or the peer-reviewed accuracy literature — makes an explicit, freestanding statement of the form "governance/audit diagnosis of neonatal hypoglycaemia must be made on laboratory-confirmed values." The nearest thing to a contrary, minority position comes from a very recent (2026) Canadian retrospective study: Hamama et al. (J Pediatr Endocrinol Metab, PMID 42137950, DOI 10.1515/jpem-2025-0736; erratum at PMID 42387304, DOI 10.1515/jpem-2026-0438), comparing 108 NICU infants' hypoglycaemic Nova StatStrip readings against confirmatory central-laboratory serum glucose, found strong correlation ($r=0.825$) with the point-of-care meter reading systematically higher than the laboratory value (Deming regression slope 1.223), and concludes — provocatively, and against the mainstream of this literature — that this "questions the need for confirmatory serum samples in the presence of bedside hypoglycemic results," since a systematic positive bias in the meter means a low POC reading is, if anything, more likely to be a true or worse hypoglycaemic value than a falsely low one. This is a single small retrospective single-centre study using one specific device (StatStrip, already established above as one of the better-performing chemistries) and should not be read as displacing BAPM's confirmation requirement; but it is a genuine, recent, peer-reviewed voice arguing the opposite direction to the conventional "always confirm low values" position, and per the brief's rule on giving contested findings fair prominence it is reported here rather than omitted.

(b) Why — physiological and methodological rationale

The physiological basis for reduced device accuracy in neonates rests on three converging factors, all well established in the biochemistry of these assays rather than being a mere

empirical curiosity. First, **haematocrit interference**: most POC chemistries (glucose oxidase and several glucose dehydrogenase formulations) partition glucose between plasma and red cells during the brief reaction time of a strip-based assay, so at neonatal-typical haematocrit (commonly 45–70%, well above the healthy adult range) the true whole-blood-to-plasma glucose ratio is altered in ways the device's calibration — built for adult haematocrit — does not correctly compensate for (Dietzen et al. 2013; Schlebusch et al. 1998). Second, **the low end of the measurement range is inherently the hardest part of any colorimetric or electrochemical assay to resolve precisely**, because the signal-to-noise ratio of the enzymatic reaction degrades as substrate concentration falls, which is why device manufacturers — whose primary market and validation population is adult diabetic hyperglycaemia management — have historically had little commercial incentive to optimise performance in the 1.5–3.0 mmol/L (27–54 mg/dL) range that is central to neonatal practice but peripheral to adult diabetes care (Demers & Smith 1999; Lang & Hussain 2014, Adv Clin Chem, PMID 24783355, DOI 10.1016/b978-0-12-800094-6.00006-6). Third, **capillary heel-stick sampling itself introduces additional variance beyond the device chemistry** — peripheral vasoconstriction, sample dilution with interstitial fluid, and clotting/haemolysis artefacts from a slow, squeezed capillary draw are all more prevalent in sick, cold, or hypoperfused neonates, which is the mechanistic reason Brooks et al. 2023 found capillary readings materially less reliable than arterial readings using the same meter and the same infants — the discrepancy is in the sample, not only the device. Electrochemical hexokinase- and glucose-dehydrogenase-based methods (StatStrip-class devices) outperform photometric glucose-oxidase methods in the St Clair 2024 meta-analysis at least in part because hexokinase is intrinsically less sensitive to the maltose/galactose cross-reactivity and oxygen-tension dependence that affect older photometric and some electrochemical GO chemistries (Warner et al. 2011, Diabetes Technol Ther, PMID 21175272, DOI 10.1089/dia.2010.0129, explicitly flagging Accu-Chek's older maltose-sensitive strips as unsafe for neonatal use for this reason).

(c) GRADE certainty summary

Diagnostic-accuracy evidence is graded here using the same GRADE framework the brief specifies for interventional claims, since the practical clinical action (confirm vs act on a POC reading) flows directly from these diagnostic-accuracy findings.

Claim	Certainty	Downgrade domain(s)
POC glucometers using electrochemical hexokinase or glucose-dehydrogenase chemistry (StatStrip-class devices) perform better than older photometric glucose-oxidase devices for neonatal glucose measurement	Moderate	Imprecision (St Clair 2024's HK+electrochemistry stratum contained only 2 studies); indirectness (pooled across a very heterogeneous 71-study evidence base spanning multiple neonatal populations and reference standards)
POC glucometers systematically under-perform at glucose	Moderate	Inconsistency (direction and magnitude of bias varies materially across studies — Roth-Kleiner 2010 found

Claim	Certainty	Downgrade domain(s)
concentrations below approximately 60 mg/dL (3.3 mmol/L), with lower correlation to laboratory reference values than at higher concentrations		overestimation, Parikh 2025 found underestimation, using different devices and eras); risk of bias (most individual studies are small, single-centre, non-blinded operator assessments)
Capillary (heel-stick) POC glucose readings are less reliable than arterial POC readings from the same device in preterm/ill neonates	Moderate	Imprecision (single study, n=40, Brooks et al. 2023); indirectness (all-comers preterm/ill NICU population, not specifically the healthy-late-preterm/term population the BAPM screening pathway primarily targets)
Continuous glucose monitoring (CGM) has poor sensitivity for detecting discrete hypoglycaemic episodes in preterm infants despite good specificity and good hyperglycaemia detection, and should not be used as a substitute for point-of-care/laboratory confirmation of hypoglycaemia	Moderate	Risk of bias (Nava 2020 meta-analysis notes lack of representative population/reference-standard/device standardisation across included studies); imprecision (individual patient data available for only 5 of 11 included studies)
A modern, well-validated point-of-care device (electrochemical HK/GDH chemistry) can achieve accuracy comparable to a blood gas analyser across the full gestational-age range including extremely preterm infants	Low-Moderate	Indirectness (single multicentre study, one device, one country — Kuboi 2025, Japan; not yet independently replicated); imprecision for the extremely preterm subgroup specifically (subgroup analysis, not the primary powered comparison)
The BAPM ≤ 2.6 mmol/L (47 mg/dL) point-of-care confirmation threshold is the correct numeric boundary at which laboratory/blood gas confirmation should be required, as opposed to a higher (e.g. 2.8–3.5 mmol/L / 50–63 mg/dL) or lower cutoff	Very low	Risk of bias and inconsistency (the underlying device-accuracy literature derives materially different confirmation cutoffs by device and by study — Ngercham 2011 ~63 mg/dL, Dietzen 2013 ~50 mg/dL — and BAPM's own ≤ 2.6 mmol/L figure is a consensus operational choice rather than a value derived from a single definitive accuracy study); indirectness (no single study tested the BAPM cutoff itself as a diagnostic-accuracy question)

(d) Where the evidence is weak, contested, or extrapolated

Four genuine tensions and gaps stand out, none resolved by the current literature:

1. **No single, universally evidence-derived "confirm below X" number exists, despite BAPM stating one clearly.** BAPM's ≤ 2.6 mmol/L (47 mg/dL) is the clearest and most operationally important figure in this domain — it is stated plainly, it is the figure a UK unit should use, and it usefully coincides with the CHYLD/Sugar Babies treatment threshold (Domain C) — but it is a consensus/operational choice, not a value derived from a device-accuracy study testing that specific cutoff. The peer-reviewed device literature independently derives higher cutoffs (2.8–3.5 mmol/L / 50–63 mg/dL, depending on device and study) using formal sensitivity/NPV analysis against a laboratory reference, and the UK's own MHRA guidance implies an even more

conservative >3 mmol/L (54 mg/dL) reliability floor. This should be stated plainly to a UK Level 3 unit rather than papered over: BAPM's number is the one to follow operationally, but it is not the same number the device-accuracy literature would derive if asked "at what point-of-care reading is the negative predictive value for true hypoglycaemia adequate."

2. **The direction of point-of-care bias relative to the laboratory reference is inconsistent across studies and eras**, which undermines any simple "POC reads X higher/lower than lab, so correct by Y" heuristic. Older studies (Roth-Kleiner 2010) found systematic overestimation; a recent study (Parikh 2025) found systematic underestimation with a different device in a different population; Hamama 2026 found overestimation again with StatStrip specifically. This is most plausibly explained by genuine device-generation and population heterogeneity rather than by a single underlying truth the literature has been converging on — it has not converged, and that should be stated rather than resolved by picking the most recent study as if it were definitive.
3. **The Hamama 2026 contrarian finding sits in real tension with the BAPM confirmation requirement and with the great majority of the accuracy literature**, and is not resolved here. It is a single, small (n=108), retrospective, single-device, single-centre study; the great weight of evidence (St Clair 2024's 71-study meta-analysis; the low-end-accuracy-degradation literature; Brooks 2023's capillary-specific caution) supports continuing to require confirmation of low point-of-care readings before treatment escalation, and BAPM's guidance should not be read as under active peer-reviewed challenge on the strength of this one paper. But per the brief's instruction to give contested and minority findings fair prominence, it is flagged rather than silently dismissed: it is a genuine, recent, peer-reviewed voice, from a specialist neonatal endocrinology/laboratory-medicine collaboration, arguing that routine confirmatory serum sampling may be unnecessary overhead given a well-performing modern device — a position that, if replicated at scale, would have material workflow and infant-comfort implications (fewer repeat blood draws) for UK practice.
4. **The governance/audit-diagnosis question is a genuine evidence gap, not merely an underexplored one**. No guideline or accuracy study located in this search directly addresses whether a governance dataset (e.g., NNRD-derived audit of admissions for hypoglycaemia, as used in the Domain A real-world evaluation of the BAPM Framework) should, or in practice does, distinguish point-of-care from laboratory-confirmed values when recording an episode of hypoglycaemia. This matters because the practical answer — that most recorded episodes are point-of-care values by default, with laboratory confirmation only for the subset that fell in the low-value confirmation band — has real implications for how audit incidence figures should be interpreted (a device with a positive bias, as several studies above found, would tend to under-record hypoglycaemia in audit data; a device with a negative bias would over-record it), but this implication does not appear to have been tested empirically anywhere in this literature.

Directness caveat specific to this domain: unlike most other domains in this review, the underlying device-accuracy evidence base here is not concentrated in a single late-

preterm/term-at-risk trial programme — it spans healthy term infants, at-risk NICU populations, and (via Kuboi 2025, Tendl 2013, Hwang 2015, Beardsall 2012, Bonet 2024, Nava 2020) preterm and very preterm infants directly. This is a genuine strength of this domain relative to Domains D and E. The caveat that remains is that CLSI POCT12-A3 and ISO 15197:2013 are themselves general (adult/paediatric, not neonate-specific) standards being applied, sometimes awkwardly, to a neonatal population — the standards were not derived from neonatal data, even where individual studies test neonatal devices against them.

(e) Practical implication for a UK Level 3 unit

Three points should govern local practice and any local point-of-care policy document. First, **the BAPM ≤ 2.6 mmol/L (47 mg/dL) confirmation threshold is the number to operationalise** — a point-of-care reading above this level is actionable on its own for feeding/first-line management without waiting for a laboratory or blood gas result (avoiding a treatment delay that would itself be harmful), while a reading at or below it should trigger blood gas or laboratory confirmation before further escalation, consistent with both BAPM guidance and the weight of the device-accuracy literature (even though that literature would, on its own diagnostic-accuracy terms, sometimes derive a slightly higher cutoff). Second, **device selection matters materially and should not be assumed generic**: the evidence favours electrochemical hexokinase/glucose-dehydrogenase chemistry (StatStrip-class devices) over older photometric glucose-oxidase devices, particularly for the extremes of haematocrit and the low glucose range relevant to a Level 3 unit's preterm caseload, and a unit should confirm its own in-use device has been validated against a neonatal (not only adult) reference population — the 2025 Shikoku multicentre study (Kuboi et al.) is the most reassuring single piece of evidence that a well-chosen modern device can perform adequately even in extremely preterm infants, but this has not yet been independently replicated outside Japan. Third, **capillary sampling technique is a source of error independent of the device itself**: given Brooks et al.'s finding that over a quarter of capillary readings fell outside tolerance criteria in preterm/ill neonates even when the same device performed reliably on arterial samples, technique (adequate perfusion, avoiding excessive squeezing, prompt analysis) deserves as much governance attention as device choice. Finally, **CGM should not be adopted, and should not be assumed by staff, to substitute for point-of-care or laboratory confirmation of a hypoglycaemic episode** — its evidence base supports a trend-monitoring and screening-adjunct role (flagging when a confirmatory blood glucose check is needed) but not a diagnostic one, given its documented poor sensitivity for discrete hypoglycaemic events.

Domain C — BAPM core management pathway for term infants (first 48 hours), its evidence base, the permissive-hypoglycaemia controversy, and dextrose gel Cochrane status

Directness: This domain concerns the **term and late-preterm (from ~34-35 weeks) infant, birth to 72 hours** — the exact population the BAPM Framework targets and the exact population enrolled in the CHYLD/Sugar Babies programme (see Domain E). Every RCT-grade element below is direct evidence for this population; every extrapolation to preterm infants (Lucas 1988, and the general question of whether BAPM's numeric thresholds apply below 34 weeks) is flagged explicitly. Grey literature (BAPM, NICE) is labelled as such throughout and never folded into a peer-reviewed evidence count.

(a) What the pathway is and how it was derived

The BAPM Framework itself — grey literature, labelled

The British Association of Perinatal Medicine's "**Identification and Management of Neonatal Hypoglycaemia in the Full Term Infant**" was convened as a joint BAPM/NHS Improvement working group in 2016 in response to wide, unexplained national variation in the identification, management, and admission thresholds for term hypoglycaemia, and to reduce unnecessary mother-baby separation. It was first published in **April 2017** and has since been revised, with the current version — retitled "...(**Birth - 72 hours**)" — dated **24 January 2024** (a public consultation on the revision ran in June 2023). **This session could not directly retrieve either bapm.org document** (WebFetch to bapm.org returned `EGRESS_BLOCKED`); the pathway detail below is triangulated from BAPM's own page as indexed by web search, cross-checked against five independent NHS trust clinical guidelines that explicitly cite BAPM as their source (Royal Cornwall, Ashford & St Peter's, North West Neonatal ODN, NHSGGC, Milton Keynes) — used here as convergent secondary confirmation, not as independent primary sources, and the numeric values below should be treated as triangulated-but-not-independently-document-verified. This limitation is stated plainly rather than papered over.

Element by element, as triangulated:

1. **Risk-factor identification.** Infant of a diabetic mother (any type, including gestational), small-for-gestational-age (SGA), large-for-gestational-age (LGA), late preterm (34–36+6 weeks), and other clinical risk factors (e.g., perinatal stress/acute illness, maternal beta-blocker use). Feeding difficulty and clinical instability are additional flagged concerns.

2. **First-line management: feeding.** Early feeding (within the first hour) and frequent (3-hourly) feeding for all at-risk infants, with capillary blood glucose checked immediately before each feed.
3. **Screening timing/frequency.** First check before the second feed or by 4 hours of age (whichever is first), then pre-feed 3-hourly, continuing until stable normoglycaemia is demonstrated on serial readings.
4. **Treatment threshold(s), tiered by severity.** A target/screening floor of blood glucose ≥ 2.0 mmol/L (36 mg/dL) with active management triggered at < 2.6 mmol/L (47 mg/dL) in most transcribing guidelines: asymptomatic blood glucose 1.0–1.9 mmol/L (18–34 mg/dL) → buccal 40% dextrose gel (200 mg/kg) as an adjunct to continued feeding, up to two doses.
5. **Escalation to IV dextrose.** Blood glucose < 1.0 mmol/L (18 mg/dL), and/or any clinical signs of hypoglycaemia at any glucose level, and/or persistent hypoglycaemia despite two doses of dextrose gel → IV 10% dextrose bolus (approximately 2.5 mL/kg) followed by a maintenance infusion (approximately 60 mL/kg/day), with admission to a neonatal unit.
6. **Discharge criteria.** Not before 24 hours of age, and not before at least three consecutive pre-feed glucose readings ≥ 2.0 mmol/L (36 mg/dL) with effective feeding established and a completed breastfeeding assessment.

NICE cross-reference — also grey literature

NICE's operative numeric infant-glucose recommendation lives in **NG3** ("Diabetes in pregnancy," most recently updated as a living guideline), not in the general postnatal care guideline **NG194** (published 2021, most recently updated June 2026 per NICE's own site), which this session's web search did not find to carry its own numeric hypoglycaemia threshold — NG194 appears to defer operational detail to BAPM/local guidance. NG3's recommendation, for infants of diabetic mothers specifically: blood glucose testing every 2–3 hours until pre-feed levels are maintained ≥ 2.0 mmol/L (36 mg/dL); escalation to tube feeding or IV dextrose only if < 2.0 mmol/L (36 mg/dL) on two consecutive readings despite maximal feeding support, or immediately if the infant is symptomatic; minimum 24-hour postnatal stay. **This is worth flagging as a genuine cross-guidance divergence:** NICE NG3's single-threshold, two-consecutive-readings rule for infants of diabetic mothers is simpler than BAPM's tiered 1.0–1.9/ < 1.0 /symptomatic structure for the broader at-risk population, and a unit following both documents needs to reconcile them explicitly for infants of diabetic mothers (who are covered by both) rather than assume they say the same thing. Neither document was independently document-verified this session (WebFetch blocked); both are cited as grey literature per web-search triangulation.

(b) Why — physiological and methodological rationale

The Framework's design directly implements the screening/diagnostic threshold distinction established by Cornblath et al. (2000, PMID 10790476, DOI 10.1542/peds.105.5.1141):

"the definition of clinically significant hypoglycemia remains one of the most confused and contentious issues in contemporary neonatology," and the paper's explicit purpose is to offer "pragmatic recommendations for operational thresholds, i.e., blood glucose levels at which clinical interventions should be considered" — deliberately not a claim about the exact level at which neuroglycopenic injury occurs. BAPM's tiered structure (a lower floor for asymptomatic infants, a higher and immediate trigger for symptomatic infants or very low readings) operationalises this Cornblath principle directly: a genuinely diagnostic threshold for injury does not exist (there is no bedside test for neuroglycopenia — see Alsweiler et al. 2022, PMID 36568425), so the Framework instead sets an intervention threshold with a margin of safety above it, by the same logic used for phototherapy thresholds in neonatal jaundice. The rationale for the specific 2.6 mmol/L (47 mg/dL) figure traces to neurophysiological evidence, not RCT evidence: Koh et al. (1988, PMID 3202642, DOI 10.1136/ad.63.11.1353) found abnormal sensory evoked potentials in 10 of 11 children (including 5 who were clinically asymptomatic) whose blood glucose fell below 2.6 mmol/L, and no evoked-potential abnormality in the 6 who stayed above it — this is the primary neurophysiological evidence for the number itself, predating any RCT outcome data by nearly three decades.

(c) What evidence supports each pathway element, graded

Pathway element	Primary evidence	GRADE certainty	Downgrade domain(s)
Treatment threshold (≥ 2.6 mmol/L/47 mg/dL as the treat-to target)	CHYLD: treating hypoglycaemia to maintain glucose ≥ 2.6 mmol/L was not associated with excess combined neurosensory impairment at 2y (McKinlay et al. 2015, PMID 26465984, RR 0.95, 95% CI 0.75–1.20) or 4.5y (McKinlay et al. 2017, PMID 28783802, composite RR 0.96, 95% CI 0.77–1.21) — see Domain E for full detail, cited directly from that verified source	Moderate	Imprecision; no untreated comparator exists or could ethically be created — the RCT-grade evidence tests a treat-to-threshold policy, not the threshold's exact numeric optimality
Dextrose gel as first-line/ adjunct treatment of confirmed hypoglycaemia	Sugar Babies RCT (Harris et al. 2015 2y, PMID 26613985; Harris et al. 2022 4.5y, PMID 35940872; St Clair et al. 2022 9–10y, PMID 36516806 — cited directly from Domain E); current Cochrane treatment review (Edwards et al. 2022, PMID 35302645, DOI 10.1002/14651858.CD011027.pub3): gel probably increases correction of hypoglycaemia (rate ratio 1.08, 95% CI 0.98–1.20), probably reduces mother-infant separation (RR 0.54, 95% CI 0.31–0.93), increases exclusive breastfeeding (RR 1.10, 95% CI 1.01–1.18)	Moderate (correction, separation, breastfeeding) / Low (long-term neurological disability, RR 0.46, 95% CI 0.09–2.47 — spans large benefit and large harm) / Low (the replicated visual-processing adverse secondary signal at 4.5y and 9–10y — see Domain E)	Imprecision (only 2 trials pooled in the Cochrane review; one included study is abstract-only with unclear-to-high risk of bias); the long-term-outcome CI is uninformatively wide

Pathway element	Primary evidence	GRADE certainty	Downgrade domain(s)
Prophylactic dextrose gel (contrast case — NOT part of the BAPM treatment pathway, included for completeness)	Current Cochrane prophylaxis review (Roberts et al. 2023, PMID 38014716, DOI 10.1002/14651858.CD012152.pub4): high-certainty reduction in hypoglycaemia incidence (RR 0.87, 95% CI 0.79–0.95) but the neurological-disability estimate has reversed direction between editions as more follow-up data accrued (pub3, PMID 33998668: RR 0.21 favouring gel; pub4, current: RR 1.00, 95% CI 0.59–1.68, no clear effect)	High (incidence) / Low (neurological disability)	Imprecision; the review authors' own explicit conclusion is that prophylactic gel "should not be incorporated into routine practice until additional information is available about the balance of risks and harms"
Risk-factor-based screening (IDM/SGA/LGA/late preterm)	Consensus/operational only — Cornblath 2000 (PMID 10790476) established the conceptual approach; O'Brien et al. 2023 (PMID 37782488, DOI 10.1001/jamapediatrics.2023.3957) found 18 international guidelines it appraised showed "variability in the assessed quality of guidelines and a lack of consensus between screening criteria"	Very low (as a specific risk-factor list, not as the general screening concept)	Risk of bias (no guideline, BAPM included, has been validated against a randomised screened-vs-unscreened comparison — Alsweiler et al. 2022 confirms none exists); indirectness (O'Brien's appraisal is international, not BAPM-specific)
Feeding-based first-line management (early/frequent feeding as the initial response)	Physiologically and operationally embedded in Cornblath 2000 and every appraised guideline in O'Brien 2023; no dedicated RCT of feeding-frequency-as-intervention was located in this session's searches	Very low (consensus/physiological reasoning, not RCT-tested as a standalone intervention)	Risk of bias (no comparator arm exists in any of the sources located); this element sits closest to "unsupported by direct RCT evidence, extrapolated from physiological reasoning" of any element in the pathway
Escalation to IV dextrose when gel/feeding fail	Roberts et al. 2024 (PMID 39536722, DOI 10.1159/000541471): systematic review found only 6 studies (2 RCTs, 4 cohort, 711 infants total) addressing IV dextrose for neonatal hypoglycaemia treatment at all; one RCT comparison (IV dextrose vs oral sucrose bolus, 80 infants) suggested reduced repeated hypoglycaemia (RR 0.67, 95% CI 0.20–2.18) but possibly increased hyperglycaemia risk (RR 2.33, 95% CI 0.65–8.39)	Very low	Imprecision (n=80 or n=60 per comparison); risk of bias (cohort designs predominate); this is the single weakest-evidenced step in the entire pathway — a treatment used routinely worldwide for decades has

Pathway element	Primary evidence	GRADE certainty	Downgrade domain(s)
			almost no dedicated trial evidence behind its specific use for confirmed hypoglycaemia
Discharge criteria (24h minimum, 3 consecutive normal readings)	Consensus/operational only; no dedicated outcome study of discharge-timing-as-intervention was located	Very low	Risk of bias (no comparator); extrapolated from general newborn observation practice, not glucose-specific trial evidence
Framework's real-world effect on term admissions	Nezafat Maldonado et al. 2026 (PMID 41338963, DOI 10.1136/archdischild-2025-328872), NNRD population study: term admissions for hypoglycaemia fell from 4.9 (95% CI 4.8–5.1) to 3.3 (95% CI 3.1–3.4) per 1000 term live births comparing 36 months before vs after the April 2017 Framework, with unchanged time-to-admission and length of stay	Moderate (as evidence of the Framework's practical effect on admission behaviour, explicitly not as evidence of the threshold's neurodevelopmental safety, which is CHYLD's separate contribution)	Risk of bias (before/after design, no contemporaneous control, cannot fully exclude secular trend as an alternative explanation) — and note 46.5% of admitted infants had ≥ 1 BAPM risk factor, meaning the majority (53.5%) did not , an important caveat on how cleanly the risk-factor list predicts who is actually admitted

(d) Where the evidence is weak, contested, or extrapolated — the permissive-hypoglycaemia controversy

On locating the specific "cause for concern" commentary named in the brief: a piece titled almost exactly "Thresholds for hypoglycaemic screening — a cause for concern?" was confirmed to exist via a CORE.ac.uk repository record, and its indexed content matches the brief's description precisely — but despite PubMed title searches, multiple author-guess web searches, and direct attempts to fetch the CORE.ac.uk, Infant journal, Europe PMC, CrossRef, and Semantic Scholar records (all blocked by this session's network egress proxy), **no PMID, DOI, author name, journal name, or year could be recovered**. Per the citation-integrity rule, it is **not cited here** and is logged in full in `05-references/could-not-verify-domain-C.md`. The controversy it represents is genuine and

important, so it is presented below using verified primary sources that carry the same substantive arguments.

The critique (the "permissive hypoglycaemia" position), with its verified evidentiary basis: The argument is that BAPM's threshold structure — an asymptomatic floor as low as 1.0–1.9 mmol/L (18–34 mg/dL) before gel is even offered, and treatment aimed at ≥ 2.0 –2.6 mmol/L (36–47 mg/dL) rather than a more conservative historical target — undertreats risk relative to older UK practice, and that the Framework does not adequately distinguish a screening threshold (deliberately set to catch a margin of at-risk infants) from a diagnostic threshold (the level at which injury is believed to occur), by direct analogy with jaundice/phototherapy thresholds. Two verified historical papers supply the evidentiary weight behind this position: **Koh et al. 1988** (PMID 3202642) showed neurophysiological (evoked-potential) abnormality below 2.6 mmol/L even in asymptomatic children, arguing against relaxing the floor; **Lucas et al. 1988** (PMID 2462455, DOI 10.1136/bmj.297.6659.1304) — a multicentre UK cohort of 661 **preterm** infants — found that ≥ 5 days of "moderate" hypoglycaemia (< 2.6 mmol/L) was associated with a 3.5-fold increase (95% CI 1.3–9.4) in neurodevelopmental impairment and 13–14 point reductions in Bayley mental/motor scores at 18 months, concluding "reappraisal of current management is urgently required." **Directness flag: Lucas 1988 is a preterm cohort**, not term, and is nonetheless the single most historically influential paper behind the conservative pre-BAPM UK practice the critique is defending — using a preterm-derived alarm to argue for more caution in term management is itself an extrapolation, one the critique's proponents do not typically make explicit.

The reassurance (the position BAPM/CHYLD evidence supports): the randomised, prospective CHYLD evidence (Domain E, McKinlay et al. 2015/2017) directly tests a treat-to- ≥ 2.6 -mmol/L policy in the population BAPM's Framework targets and finds **no excess combined neurosensory impairment** at 2 or 4.5 years — a materially stronger evidence base (a masked, prospective, treat-to-threshold cohort with continuous interstitial monitoring) than either 1988 paper had available. Alsweiler et al. 2022 (PMID 36568425) supplies the modern, more nuanced middle position: screening for neonatal hypoglycaemia cannot cleanly satisfy classical screening-test principles because neuroglycopenia — the condition that actually matters — has no bedside diagnostic test, so blood glucose is an imperfect proxy by construction, not because BAPM chose the wrong number. On this reading, the two positions are answering different questions (the Koh/Lucas evidence describes what happens near and below the threshold; CHYLD describes what happens when a treat-to-threshold policy is actually followed), and the disagreement is not fully resolved by either evidence base — the domain-specific executive-function and visual-motor dose-response signal in CHYLD's own 4.5-year data (Domain E) is, if anything, a point of partial convergence: hypoglycaemia is not neurologically inert even under BAPM's current threshold, which is exactly what the "cause for concern" critique would predict, even though the global impairment measures the critique implies should be markedly worse were not.

This is presented as a genuine, unresolved two-sided disagreement, per the brief's explicit instruction — it is not adjudicated here.

A second, distinct tension — treatment vs prophylaxis dextrose gel, with a directly opposite Cochrane conclusion: the current treatment Cochrane review (Edwards et al. 2022, PMID 35302645) concludes oral dextrose gel is "probably an effective and safe first-line treatment"; the current prophylaxis Cochrane review (Roberts et al. 2023, PMID 38014716) concludes prophylactic gel "should not be incorporated into routine practice" pending more safety data, after its own neurological-disability point estimate reversed direction between its pub3 and pub4 editions (RR 0.21 favouring gel → RR 1.00, no clear effect) as more follow-up data accrued. **These are not the same clinical question and must not be conflated:** BAPM's pathway uses gel to treat infants who are already hypoglycaemic (the higher-certainty review), not to prevent hypoglycaemia in at-risk-but-currently-normoglycaemic infants (the more cautious review, corresponding to the hPOD programme in Domain E, which itself showed an adverse motor/cognitive secondary signal at 2 years).

A third gap, reported plainly rather than papered over: the escalation step to IV dextrose — arguably the single highest-stakes step in the pathway, since it marks admission and separation from the mother — rests on only 6 studies totalling 711 infants (Roberts et al. 2024, PMID 39536722), with very-low-certainty evidence throughout. This is a genuinely thin evidence base for a universally-practised intervention, and it is the least-evidenced element of the entire BAPM pathway.

Generalisability caveat: every dextrose-gel RCT located in any Cochrane review here was conducted in a high-income country; a 2021 Campbell Systematic Review of neonatal nutrition interventions in low- and middle-income countries (PMID 37133295) explicitly found **zero** eligible dextrose-gel trials from LMIC settings — the entire evidence base for this now-standard UK first-line treatment is high-income-country-derived.

(e) Practical implication for a UK Level 3 unit

The BAPM pathway's individual elements sit on markedly uneven evidence footing, and a governance document should say so plainly rather than treat the Framework as a uniformly evidence-based document: the **treatment threshold and dextrose-gel treatment step are the two elements with genuine RCT support** (CHYLD and Sugar Babies respectively, both direct evidence in the target population); **risk-factor screening, feeding-based first-line management, and discharge criteria are consensus/operational**, extrapolated from Cornblath-style pragmatic reasoning and physiological plausibility rather than tested against a randomised comparator; and **escalation to IV dextrose is the weakest-evidenced single element**, resting on a handful of small, mostly non-randomised studies despite being the step that separates mother and baby and triggers neonatal unit admission. The real-world NNRD evidence (Nezafat Maldonado et al. 2026) shows the Framework has reduced term admissions by roughly a third without detectably increasing time-to-admission or length of stay, which is reassuring at the population level but is evidence about admission behaviour, not about neurodevelopmental safety — that reassurance comes only from CHYLD. The permissive-hypoglycaemia critique should be taken seriously as a live, evidence-literate position rather than dismissed, precisely because CHYLD's own dose-response data (Domain E) shows hypoglycaemia

under the current threshold is not neurologically inert in the executive-function and visual-motor domains — the disagreement is really about how much residual, domain-specific risk is acceptable at a threshold that reliably prevents global impairment, not about whether BAPM "got the number wrong" in any simple sense. Finally, units following both BAPM and NICE NG3 for infants of diabetic mothers should reconcile the two documents' differently-structured thresholds explicitly in local protocol, rather than assume they are identical.

Domain D — Preterm-specific glucose thresholds and long-term outcome evidence

Directness: This domain concerns infants born preterm (<37 weeks), with particular attention to very preterm (<32 weeks) and extremely preterm (<28 weeks) infants — the population that dominates a Level 3 NICU caseload. This is explicitly **not** the population enrolled in the CHYLD/Sugar Babies/hPOD programme (Domain E), which recruited from ~34–35 weeks upward. Nothing in this domain should be read as validated by, or interchangeable with, the Domain E evidence base; every point below states its own population and flags extrapolation where it occurs.

(a) What the evidence shows

The landmark positive finding: Duvanel et al. 1999

Duvanel et al. (J Pediatr 1999, PMID 10190926, DOI 10.1016/s0022-3476(99)70209-x) remains the single most-cited preterm-specific hypoglycaemia–outcome study and, on current evidence, the most influential positive finding in this entire domain. The cohort was 85 **small-for-gestational-age (SGA) preterm** neonates (Centre Hospitalier Universitaire Vaudois, Lausanne) undergoing systematic glucose detection (hypoglycaemia defined as <2.6 mmol/L, 47 mg/dL), with prospective serial evaluation of physical growth and psychomotor development, infants retrospectively grouped by glycaemic status.

- Incidence of hypoglycaemia in this SGA preterm cohort: **72.9%**.
- Infants with **repeated episodes** of hypoglycaemia had significantly **reduced head circumference** and **lower scores on specific psychometric tests at 3.5 years**.
- Hypoglycaemia was also associated with reduced head circumference at **18 months** and lower psychometric scores at **5 years**.
- Infants with **moderate recurrent** hypoglycaemia had **lower scores at 3.5 and 5 years** than infants who experienced a **single severe** hypoglycaemic episode — i.e., recurrence, not peak severity, was the stronger predictor.
- The authors' own conclusion: recurrent hypoglycaemia was strongly correlated with persistent neurodevelopmental and growth deficits to 5 years, was a more reliable predictor of long-term effect than single-episode severity, and on this basis they recommended repetitive glucose monitoring and rapid treatment of even mild hypoglycaemia specifically in SGA infants.

This finding — recurrence outweighing single-episode severity — is structurally the same dose-response pattern later described in the term/late-preterm CHYLD cohort (Domain E), which lends it some cross-population coherence, but Duvanel 1999 is a **26-year-old, single-centre, retrospective-grouping cohort of 85 infants with no contemporaneous untreated comparator arm and no CGM/interstitial data** — it

predates modern neonatal intensive care practice (including current parenteral nutrition, thermoregulation, and glucose-monitoring standards) by nearly three decades. It is a genuine, oft-repeated finding, not a myth — but it is old, and, as covered in (d) below, most of the modern literature testing the same question in the same population (SGA preterm) has since returned null.

The modern nulls, given equal prominence: Palazzo 2024 and Palazzo/Correani 2025

Two companion papers from the same Italian group (Polytechnic University of Marche/G. Salesi Children's Hospital, Ancona) directly re-test the Duvanel question in a much larger, contemporary <32-week cohort, splitting by SGA/AGA status, and **both return null results**. These are reported here with equal prominence to Duvanel, not subordinated to it, per the brief's non-negotiable rule on nulls.

Palazzo et al. 2024 (Eur J Pediatr, PMID 39707054, DOI 10.1007/s00431-024-05936-2) — **SGA preterm infants <32 weeks**: retrospective review of 1364 preterm infants (gestational age 24–31 weeks/days), classified by blood glucose in the first 6 hours of life as <40 mg/dL (2.2 mmol/L) or ≥40 mg/dL (2.2 mmol/L), and by birthweight z-score as SGA or AGA, with propensity-score matching. Of 747 infants meeting inclusion criteria, 173 (23.2%) had early hypoglycaemia (<40 mg/dL) and 574 (76.8%) did not. SGA infants had significantly higher 2-year cognitive impairment (Bayley-III <85) than matched AGA infants regardless of glycaemic group (+20 percentage points, $p=0.040$, in the hypoglycaemic group; +17 percentage points, $p=0.029$, in the non-hypoglycaemic group) — i.e., **SGA status itself, not hypoglycaemia, was the driver of excess risk**. Critically, early hypoglycaemia was **not** independently associated with 2-year cognitive impairment either in the whole cohort or restricted to SGA infants specifically, after adjustment for gestational age, sex, 5-minute Apgar <7, SGA status, prematurity complications, prolonged (>7 day) ventilation, cumulative energy intake to 36 weeks, and maternal education (adjusted OR 1.077, $p=0.768$, whole cohort; adjusted OR 0.993, $p=0.935$, SGA subgroup). The authors' conclusion, stated plainly: in SGA preterm infants 24–31 weeks, early hypoglycaemia (first 6 hours) was **not an independent risk factor** for 2-year cognitive impairment.

Palazzo, Bonanni, Correani et al. 2025 (Brain Dev, PMID 40609275, DOI 10.1016/j.braindev.2025.104391) — the **AGA companion**, newly identified in this session's search (not in the original brief seed list): retrospective analysis of 598 **appropriate-for-gestational-age** preterm infants, same gestational age band (24–31 weeks/days), same <40 mg/dL (2.2 mmol/L) hypoglycaemia threshold within the first 6 hours of life. 129/598 (21.6%) were classified as hypoglycaemic, 469/598 (78.4%) were not; time from birth to intravenous glucose infusion did not differ significantly between groups. After propensity-score matching (79 matched pairs) and multiple logistic regression, early hypoglycaemia was **not** associated with a significantly higher risk of either 2-year cognitive (ExpB 1.150, $p=0.656$) or motor (ExpB 0.982, $p=0.834$) impairment. The authors' conclusion: in a clinical setting with strict glycaemic monitoring and standardised glucose management,

early hypoglycaemia in the first 6 hours was not an independent risk factor for 2-year cognitive or motor impairment in AGA preterm infants 24–31 weeks.

Read together, these two 2024–2025 companion papers constitute the **most direct, most contemporary, and largest** evidence available specifically for the <32-week population on the question Duvanel 1999 first raised, and both are null for early (first-6-hour) hypoglycaemia as an independent predictor — in both SGA and AGA infants. This does not overturn Duvanel's finding (a different exposure window — first 6 hours only, versus Duvanel's serial/recurrent monitoring approach across the neonatal course) but it materially narrows how far the Duvanel finding can be generalised to a modern, strictly glucose-monitored NICU population, and both papers deserve equal billing with Duvanel in any account of this domain, not a footnote status.

The historical preterm null: Goode et al. 2016

Goode et al. (Pediatrics 2016, PMID 27940690, DOI 10.1542/peds.2016-1424) is a secondary analysis of the Infant Health and Development Program (a US multisite cohort recruited in the **1980s**), 745/985 preterm infants with recorded glucose, stratified into four glucose-level strata, assessed at 3, 8, and 18 years. No significant differences in cognitive or academic outcomes at any age between hypoglycaemia-exposed and unexposed preterm children; more severe hypoglycaemia was associated with fewer reported behaviour problems at 18 years, a finding the authors themselves did not judge clinically meaningful. Full extraction detail is carried in the shared evidence matrix as row **E17** (02-extraction/extraction.csv) and is not duplicated here; it is discussed in this domain because it bears directly on whether the term/late-preterm null-and-dose-response pattern (Domain E) generalises to a preterm population.

Double directness caveat, flagged explicitly: Goode 2016 carries two independent directness problems that must both be stated, not just the more obvious one. First, it is genuinely preterm-population evidence (unlike most of Domain E), which is a strength. Second, and easily overlooked, the cohort was recruited in the **1980s** — glucose-monitoring frequency, treatment thresholds, and general neonatal intensive care practice at that time differ substantially from current standards (compare: Palazzo 2024/2025's "strict glycaemic monitoring and standardized glucose management" in a modern NICU). A null finding in a 1980s-era, non-standardised-monitoring cohort is not equivalent in strength to a null finding in a 2020s cohort using contemporary protocolised glucose management — the historical null could reflect either true absence of effect, or noisier/less complete glucose ascertainment obscuring a real effect. Both caveats apply simultaneously and neither substitutes for the other.

Glycaemic variability/lability as a distinct preterm-relevant construct: the CGM/interstitial monitoring literature

A structurally different evidence base addresses preterm glycaemic instability as a variability problem rather than a single-threshold-breach problem — directly responsive to the brief's Domain D instruction to treat this as a distinct concept. This literature is a live,

active RCT research area in preterm infants, in contrast to the absence of threshold-outcome RCTs (see (d)):

- **Galderisi et al. 2017** (Pediatrics, PMID 28916591, DOI 10.1542/peds.2017-1162): RCT of 50 infants ≤ 32 weeks or ≤ 1500 g, randomised within 48 hours of birth to computer-guided glucose infusion rate with real-time continuous glucose monitoring (CGM) versus standard-of-care blinded-CGM glucometer-guided titration. CGM-guided infants spent more time in the euglycaemic range (72–144 mg/dL; median 84% vs 68%, $p < .001$), less time in mild ($p = .04$) and severe ($p = .007$) hypoglycaemia and severe hyperglycaemia ($p = .04$), and had lower glycaemic variability (SD 21.6 ± 5.4 vs 27 ± 7.2 mg/dL, $p = .01$; coefficient of variation $22.8\% \pm 4.2\%$ vs $27.9\% \pm 5.0\%$, $p < .001$). This is a **process/monitoring-modality RCT**, not a threshold-outcome RCT — it establishes that CGM changes glycaemic exposure patterns, not that any particular threshold or variability metric changes neurodevelopmental outcome.
- **REACT (Beardsall et al. 2021)** (Lancet Child Adolesc Health, PMID 33577770, DOI 10.1016/S2352-4642(20)30367-9): international, open-label RCT across 13 NICUs (UK, Spain, Netherlands), 180 infants randomised, birthweight ≤ 1200 g, gestational age up to 33+6 weeks, to real-time CGM versus standard care (masked CGM for comparison). Primary efficacy outcome (proportion of time sensor glucose 2.6–10 mmol/L in week 1) favoured real-time CGM (mean 84% vs 94% — reported as such in the abstract; adjusted mean difference 8.9%, 95% CI 3.4–14.4, equivalent to 13 hours). Fewer infants in the real-time CGM group had a prolonged (>1 h) episode of sensor glucose < 2.6 mmol/L (47 mg/dL) (4/70, 6%, vs 13/85, 15%, standard care). No serious device-related adverse events. The paper's own conclusion states explicitly that "further studies using CGM are required to determine optimal glucose targets, strategies to obtain them, and the potential effect on long-term health outcomes" — i.e., REACT is silent on long-term neurodevelopmental outcome by design, and its authors say so.
- **Battaglini et al. 2026** (Neonatology, PMID 41984740, DOI 10.1159/000550797): single-centre RCT, 53 infants, real-time CGM ($n = 26$) versus blinded-CGM standard care ($n = 27$), randomised within 6 hours of life. Significantly fewer hypoglycaemic and hyperglycaemic episodes in the CGM group in week 1 (hypoglycaemia 339 vs 934 episodes, $p = 0.0001$; hyperglycaemia 145 vs 830, $p = 0.0001$) and at 32 weeks postmenstrual age (hypoglycaemia 260 vs 785, $p = 0.0001$; hyperglycaemia 1 vs 38, $p = 0.0001$). This is the **only one of the three CGM RCTs that reports a long-term neurodevelopmental outcome at all** — assessed at 2 years corrected age with the Griffiths Scales — and it found **no significant difference in neurodevelopmental outcome** between groups, a null result reported here with equal prominence to the positive glycaemic-control finding. At $n = 53$ (26 vs 27), the trial is self-evidently underpowered to detect anything but a large neurodevelopmental effect, and should be read as hypothesis-generating on the long-term question, not as evidence of safety-equivalence.
- **Cochrane review (Galderisi, Trevisanuto, Russo, Hall, Bruschettini)**, current edition 2021 (Cochrane Database Syst Rev, PMID 34931697, DOI 10.1002/14651858.CD013309.pub3 — supersedes the 2020 pub2 edition, PMID 33348448): pooled four RCTs, 300 preterm infants at risk of hypoglycaemia or

hyperglycaemia. The review's own primary outcome was neurodevelopmental impairment — **none of the four included trials reported it**, nor did any report seizures. Effect on in-hospital mortality was uncertain (RR 0.59, 95% CI 0.16–2.13; very low certainty). The review's explicit conclusion: "there is insufficient evidence to determine if CGM affects preterm infant mortality or morbidities," certainty very low across all outcomes for risk of bias and imprecision, and **no studies at all were identified in preterm infants with proven (rather than at-risk-of) hypoglycaemia or hyperglycaemia**. This is a Cochrane-level confirmation, independent of this domain's own primary search, that the preterm CGM literature has not yet closed the gap to long-term outcome evidence.

Two further preterm CGM/variability papers were identified and are noted for completeness rather than given full extraction: **Gutierrez-Rosa et al.** (PMID 41894434) established glycaemic-variability reference percentiles in VLBW infants via CGM (a descriptive/normative contribution, not an outcome trial), and **Jagla et al. 2019** (Dev Period Med, PMID 30954975) found CGM-derived variability metrics (SD/CV/MAGE) associated with severe IVH and treated ROP on univariate analysis in 74 VLBW infants, but the association did not survive adjustment in logistic regression — a further null-on-adjustment finding consistent with the pattern that preterm glycaemic-variability signals seen in unadjusted analyses tend to attenuate once confounders (largely, degree of prematurity and illness severity) are accounted for.

The near-miss: May et al. 2026 addresses "variability" but not the preterm population

May et al. 2026 (Pediatr Res, PMID 42050117, DOI 10.1038/s41390-026-04958-w) — fully detailed in Domain E — is worth flagging here specifically because its research question (does glycaemic variability, as opposed to simple threshold-breach, better predict neurodevelopmental outcome?) is precisely the Domain D-relevant question the CGM trials above are unable to answer with long-term outcome data. May et al. pooled 18,947 blood glucose readings from 2,894 infants across three Auckland-programme cohorts and found dichotomous threshold measures (severe/recurrent hypoglycaemia) more strongly predictive of 2-year Bayley motor scores than glycaemic gradient variability. **This is directly relevant in method and question to the Domain D variability construct, but its population is the same late-preterm/term at-risk cohort as the rest of the CHYLD/hPOD programme (from ~34–35 weeks) — not a preterm or very preterm population.** It cannot be used as preterm-specific evidence on the variability-versus-threshold question, however tempting the topical overlap; it is flagged here precisely because a reader assembling the "variability" literature across domains could easily and wrongly treat it as filling this domain's gap.

(b) Why — physiological and methodological rationale

Preterm infants, and SGA preterm infants specifically, are physiologically distinct from the term/late-preterm at-risk population in ways that are directly relevant to why this domain

cannot simply borrow Domain E's conclusions. Reduced hepatic glycogen stores, immature counter-regulatory hormonal responses (catecholamine, glucagon, cortisol), a higher brain-to-body mass ratio increasing relative glucose demand, and — in SGA infants — additional intrauterine growth restriction affecting substrate reserve, together plausibly increase both the frequency and the potential neurological consequence of hypoglycaemic episodes relative to an appropriately grown late-preterm/term infant. This is the physiological rationale historically cited for extrapolating a lower tolerance threshold downward in gestation. However, the same immaturity that increases physiological vulnerability also means preterm infants in a modern NICU are subject to far more intensive glucose surveillance and more standardised, protocolised correction (as the Palazzo 2024/2025 papers explicitly note) than either the 1980s IHDP cohort (Goode) or, arguably, than a bedside-glucometer-monitored late-preterm/term infant on a postnatal ward. This may partly explain why the two most contemporary, well-monitored preterm cohorts (Palazzo 2024, 2025) return null findings for early hypoglycaemia specifically: aggressive, protocolised correction may be preventing the exposure gradient (recurrent/prolonged hypoglycaemia) that Duvanel's 1999 cohort — managed under 1990s standards — was able to demonstrate.

(c) GRADE certainty summary

Claim	Certainty	Downgrade domain(s)
Recurrent hypoglycaemia is associated with reduced head circumference and lower psychometric scores to age 5 in SGA preterm infants (Duvanel 1999)	Low	Risk of bias (retrospective glycaemic-status grouping, no untreated comparator, single centre); indirectness (26-year-old cohort, pre-dates current NICU glucose-management standards)
Early (first-6-hour) hypoglycaemia is not an independent risk factor for 2-year cognitive/motor impairment in SGA or AGA preterm infants <32 weeks under modern, protocolised glucose management (Palazzo 2024, Palazzo/Correani 2025)	Low	Risk of bias (retrospective, single-region Italian cohort, propensity-matched not randomised); imprecision (matched subgroup analyses relatively small, e.g. 79 matched pairs in the 2025 paper); indirectness (single first-6-hour exposure window only — does not test recurrent/prolonged hypoglycaemia across the full NICU stay, so cannot be read as contradicting Duvanel's recurrence finding directly)
Hypoglycaemia is not associated with cognitive/academic impairment in preterm infants to age 18 (Goode 2016)	Low	Risk of bias and indirectness (1980s-era recruitment and glucose-management practice; see double directness caveat above)
Real-time CGM increases time-in-euglycaemic-range and reduces hypo-/hyperglycaemic exposure in very preterm infants (Galderisi 2017, REACT 2021, Battaglini 2026)	Moderate (for the glycaemic-exposure outcome)	Imprecision (small trials, 50–182 participants); the outcome is a surrogate (glycaemic exposure), not a patient-important outcome
CGM use in preterm infants improves neurodevelopmental outcome, mortality, or major morbidity	Very low / evidence absent	Per Cochrane 2021 (PMID 34931697): none of the pooled trials reported neurodevelopmental outcome; mortality estimate very low certainty (imprecision,

Claim	Certainty	Downgrade domain(s)
		risk of bias); Battaglini 2026's single 2-year neurodevelopmental readout is null but grossly underpowered (n=53)
A specific glucose treatment threshold, tested against a long-term neurodevelopmental outcome by RCT, applies to infants <32 weeks (or <28 weeks) gestation	No evidence identified — not gradeable	See (d)

(d) Where the evidence is weak — the central finding of this domain

This is the most important section of this domain, and the brief's prediction is confirmed, not merely suspected: **across an exhaustive search (five PubMed search strings plus two Consensus queries, approximately 247 raw PubMed hits and 40 Consensus results, all screened — full log in 01-search-log/domain-D-searches.md), no randomised controlled trial testing a specific glucose treatment threshold against a long-term neurodevelopmental outcome in infants <32 weeks gestation was identified. None exists for <28 weeks either.** This is stated plainly, as the brief requires, as a genuine and clinically important evidence gap — not a search failure, and not something to be papered over with extrapolated term/late-preterm data presented as if it were preterm-specific.

What does exist, and what does not, should be kept sharply distinct:

1. **What exists:** a landmark but 26-year-old single-centre cohort (Duvanel 1999) showing a positive association between recurrent hypoglycaemia and adverse outcome in SGA preterm infants; two large, contemporary, well-conducted retrospective cohorts from the same modern Italian NICU network (Palazzo 2024, Palazzo/Correani 2025) testing the same population (<32 weeks, SGA and AGA respectively) and finding no independent association for early (first 6 hours) hypoglycaemia; a 1980s historical US cohort (Goode 2016) also null but doubly confounded by era; and a small but active RCT literature (Galderisi 2017, REACT 2021, Battaglini 2026, Cochrane review) testing whether a monitoring modality (CGM) rather than a threshold improves glycaemic exposure — which it does — without yet being able to show, by the Cochrane review's own explicit statement, whether this translates into improved neurodevelopmental outcome.
2. **What does not exist:** an RCT (or even a large prospective cohort using a pre-specified, standardised threshold definition tested against blinded long-term follow-up) that establishes what glucose threshold, sustained for what duration, at what gestational age within the <32-week or <28-week bands, causes or predicts long-term neurodevelopmental harm. The two Palazzo papers come closest methodologically (large n, propensity matching, standardised modern protocol) but are retrospective, cover only the first 6 hours of life (not recurrent/prolonged exposure across the NICU

stay, which is precisely the exposure pattern Duvanel's positive finding turned on), and are single-region (one Italian network), so their null result — while the best currently available evidence for the <32-week population — cannot be treated as definitively closing the question Duvanel raised.

3. **The <28-week gap is total.** Search D1 (specifically targeting "28 weeks"/"extremely preterm"/ELBW combined with hypoglycaemia) returned 55 hits, screened in full, with no threshold-outcome paper surfacing at all — almost entirely gestational-diabetes, growth-chart, thermoregulation, transport, and vaccine-safety literature with hypoglycaemia as an incidental listed outcome. The Consensus cross-check (search D4) independently confirmed a striking asymmetry: at <28 weeks, the glucose-and-outcome literature that does exist is predominantly about **hyperglycaemia and insulin therapy** (Domain G territory), not hypoglycaemia — extremely preterm infants appear in the literature more often as subjects of hyperglycaemia/insulin studies than of hypoglycaemia-threshold studies. This asymmetry is itself a finding: a clinician searching for "what glucose threshold matters for my 25-week infant's hypoglycaemia" will find comparatively little, while a search on hyperglycaemia in the same population is comparatively richer.
4. **Practical consequence for extrapolation.** Every glucose threshold currently applied to a <32-week or <28-week infant in routine UK practice — including any threshold derived by downward extrapolation from BAPM's term framework (Domain A/C) or from the CHYLD/Sugar Babies late-preterm/term RCT evidence (Domain E) — is, on this evidence, an **extrapolation without direct RCT support**, not a directly evidenced, gestation-specific standard. This should be stated to trainees and in any governance document as a plain fact about the evidence base, not softened.

(e) Practical implication for a UK Level 3 unit

For infants <32 weeks, and especially <28 weeks, current practice (protocolised glucose monitoring, correction thresholds broadly consistent with the general neonatal hypoglycaemia treatment threshold of 2.6 mmol/L/47 mg/dL, adapted locally) rests on physiological plausibility, an aggregation of a small and heterogeneous historical/modern cohort literature, and downward extrapolation from term/late-preterm RCT evidence — **not** on a preterm-specific RCT evidence base, because none exists. Three points should inform unit practice and how this gap is communicated: (i) the strongest single positive signal (Duvanel 1999) implicates recurrence, not single-episode severity, as the more clinically important exposure pattern — consistent with, and reinforced by, the analogous recurrence-driven dose-response later shown in the term/late-preterm CHYLD cohort (Domain E), which is at least indirect corroboration of the underlying biological plausibility even though the populations differ; (ii) the most contemporary, best-conducted preterm-specific cohorts (Palazzo 2024, 2025) are reassuringly null for isolated early (first-6-hour) hypoglycaemia under current standardised, closely monitored management — which supports continued confidence in existing protocolised early correction, without extending that reassurance to recurrent or prolonged hypoglycaemia later in the NICU stay, which those papers did not test; (iii) continuous/interstitial glucose monitoring is an active and

promising preterm-specific research area that demonstrably improves glycaemic exposure metrics, but the field itself (via its own Cochrane review) states it cannot yet answer whether this improves long-term neurodevelopmental outcome — CGM should be understood, and presented to families and trainees, as a monitoring-quality improvement of unproven long-term neuroprotective benefit, not yet as an evidence-based means of preventing neurodevelopmental harm. The absence of any <32-week or <28-week RCT-based threshold-outcome evidence should be stated explicitly in any unit guideline derived from this synthesis, rather than left implicit behind a threshold number presented as if it carried the same evidentiary weight as the term BAPM/CHYLD threshold.

Domain E — Term and late-preterm neurodevelopmental outcomes after neonatal hypoglycaemia

Directness: All primary evidence in this domain is from the New Zealand Liggins Institute/Auckland programme (CHYLD, Sugar Babies, hPOD/pre-hPOD) and a handful of independent European cohorts (Sweden, Denmark, Germany). Populations are **late preterm (from ~34-35 weeks) and term infants born at risk of hypoglycaemia** (infant of diabetic mother, small/large for gestational age, late preterm, acute illness) — not unselected term infants, and not the very/extremely preterm population that dominates a Level 3 NICU. Every extrapolation beyond this population is flagged explicitly below (see Domain D for the preterm-specific evidence, which is materially weaker).

(a) What the evidence shows

The core cohort: CHYLD

The Children with Hypoglycaemia and Their Later Development (CHYLD) study is a prospective cohort of 614 neonates ≥ 35 weeks' gestation with ≥ 1 risk factor for hypoglycaemia, recruited at Waikato Hospital, Hamilton, New Zealand (2006–2010). Clinicians were masked to interstitial glucose; all infants with hypoglycaemia (blood glucose < 2.6 mmol/L, 47 mg/dL) were treated to maintain glucose ≥ 2.6 mmol/L (47 mg/dL) — i.e., the cohort tests outcomes **under a treat-to-threshold protocol**, not outcomes of untreated hypoglycaemia. Severe hypoglycaemia was defined as < 2.0 mmol/L (36 mg/dL); recurrent as ≥ 3 episodes; an "interstitial" (occult/clinically undetected) episode as continuous glucose monitoring reading < 2.6 mmol/L (47 mg/dL) for ≥ 10 minutes despite normal contemporaneous blood glucose.

- **2 years** (McKinlay et al. 2015, NEJM, PMID 26465984, DOI 10.1056/NEJMoa1504909): 404/528 eligible children assessed (77%). 216/404 (53%) had neonatal hypoglycaemia. Treated hypoglycaemia was **not** associated with increased risk of the primary outcomes: neurosensory impairment (risk ratio 0.95, 95% CI 0.75–1.20, $p=0.67$) or processing difficulty (risk ratio 0.92, 95% CI 0.56–1.51, $p=0.74$). Unrecognised (interstitial-only) hypoglycaemia, lowest glucose, number of episodes, and negative interstitial increment (area under 2.6 mmol/L) also did not predict outcome. **This is a null primary result and is reported with the same weight as the positive findings below** — it is the basis for the current BAPM/CHYLD-derived position that treated hypoglycaemia to a 2.6 mmol/L (47 mg/dL) floor is not associated with early neurosensory harm.
- **4.5 years** (McKinlay et al. 2017, JAMA Pediatr, PMID 28783802, DOI 10.1001/jamapediatrics.2017.1579): 477/604 eligible children assessed (79.0%); 280/477

(58.7%) exposed to hypoglycaemia. The **composite** primary outcome (combined neurosensory impairment) remained null (risk difference 0.01, 95% CI -0.07 to 0.10; risk ratio 0.96, 95% CI 0.77-1.21). However, hypoglycaemia was associated with a **dose-dependent** increase in two specific domains: low executive function (risk difference 0.05, 95% CI 0.01-0.10; risk ratio 2.32, 95% CI 1.17-4.59) and visual-motor function (risk difference 0.03, 95% CI 0.01-0.06; risk ratio 3.67, 95% CI 1.15-11.69), with the highest risk in children with **severe, recurrent** (≥ 3 episodes), or **clinically undetected (interstitial-only)** hypoglycaemia. This is the central dose-response finding in the literature: severity and recurrence gradient, and detectability, all independently associated with domain-specific (not global) impairment. The "occult hypoglycaemia is still hypoglycaemia" finding — that interstitial-only episodes carried comparable risk to clinically detected ones — is one of the most practically important and most easily overlooked results in this evidence base.

- **9-10 years, academic performance** (Shah et al. 2022, JAMA, PMID 35315886, DOI 10.1001/jama.2022.0992): 480/587 eligible children assessed (82%), mean age 9.4 years. The primary outcome, low educational achievement (reading or mathematics below curriculum-normative level), was **not** significantly different between exposed and unexposed children (47% vs 48%; adjusted risk difference -2%, 95% CI -11% to 8%; adjusted relative risk 0.95, 95% CI 0.78-1.15) — a second null primary result at a later timepoint. Of 47 pre-specified secondary outcomes, one reached significance in a **counter-intuitive direction**: children exposed to neonatal hypoglycaemia were teacher-rated as less likely to be below curriculum level for reading (24% vs 31%; adjusted risk difference -9%, 95% CI -17% to -1%; adjusted relative risk 0.72, 95% CI 0.53-0.99, $p=0.04$). With 47 secondary comparisons tested, this single nominally significant result in the direction opposite to the hypothesised harm is most parsimoniously explained by multiplicity/chance rather than a genuine protective effect of hypoglycaemia, and the paper itself does not claim causal protection. **This finding is reported here specifically because the brief requires nulls and counter-intuitive secondary signals to be given equal billing, not because it is believed to be real.**
- **Structural correlate — caudate volume** (Kennedy et al. 2022, Pediatr Res, PMID 36513807, DOI 10.1038/s41390-022-02410-3): in a subset of 101 CHYLD children with MRI and neurodevelopmental data at 9-10 years (70 with prior neonatal hypoglycaemia), smaller caudate volume was associated with emotional/behavioural difficulties and poorer prosocial behaviour, and (in children **not** exposed to neonatal hypoglycaemia only) with visual memory. There was no significant caudate-volume-by-hypoglycaemia interaction on the main outcomes. **This is explicitly a parallel, not-yet-mechanistically-linked finding**: the paper's own conclusion is that neonatal hypoglycaemia is associated with smaller caudate volume (from earlier work) but that this "does not appear to contribute to clinically relevant neurodevelopmental deficits" in this cohort — i.e., a structural correlate exists but has not been shown to explain the behavioural/cognitive associations reported elsewhere in CHYLD.
- **Glycaemic variability vs dichotomous threshold measures** (May et al. 2026, Pediatr Res, PMID 42050117, DOI 10.1038/s41390-026-04958-w): pooled analysis of

18,947 blood glucose concentrations from 2,894 infants across three of the Auckland-programme cohorts. Higher glycaemic gradient variability was associated with poorer neurodevelopment at 2 years, but **dichotomous threshold-based measures remained more strongly predictive**: a 1 SD increase in gradient variability was associated with a -0.9-point adjusted mean Bayley motor score (95% CI -1.4 to -0.3), versus -3.1 (95% CI -4.8 to -1.5) for severe hypoglycaemia (<2.0 mmol/L, 36 mg/dL) and -2.1 (95% CI -3.6 to -0.6) for recurrent hypoglycaemia (≥ 3 episodes). This is a very recent paper (2026) and is the first head-to-head test, in this evidence base, of whether glycaemic variability adds predictive information beyond simple threshold-breach counting; on this analysis it does not clearly outperform the established dichotomous severe/recurrent categories, though the authors state variability "warrants further investigation" rather than dismissing it — treat as an open rather than closed question.

Dextrose gel treatment follow-up: Sugar Babies

The Sugar Babies trial randomised infants ≥ 35 weeks with confirmed hypoglycaemia (<2.6 mmol/L, 47 mg/dL) in the first 48 hours to 40% dextrose gel or placebo gel (both alongside feeding), with the primary acute outcome (published separately, not re-verified here as it is outside this domain's ND-outcome scope) being reversal of hypoglycaemia.

- **2 years** (Harris et al. 2015, J Pediatr, PMID 26613985, DOI 10.1016/j.jpeds.2015.10.066): 184/237 randomised children assessed. Neurosensory impairment: 38% (dextrose) vs 34% (placebo), relative risk 1.11 (95% CI 0.75–1.63) — no significant difference, but a **notably high absolute rate of impairment in both arms (36% overall: 1 severe, 6 moderate, 59 mild)**, which the paper itself flags as showing "neurosensory impairment is common among these children" regardless of gel allocation — i.e., this cohort's baseline risk (from the underlying risk factors for hypoglycaemia, not from gel treatment) is substantial. Processing difficulty also did not differ (10% vs 18%, relative risk 0.52, 95% CI 0.23–1.15).
- **4.5 years** (Harris et al. 2022, Arch Dis Child Fetal Neonatal Ed, PMID 35940872, DOI 10.1136/archdischild-2022-324148): 185/237 assessed (78%). Neurosensory impairment remained similar (dextrose 38% vs placebo 39%, relative risk 0.96, 95% CI 0.66–1.34, $p=0.83$). **Secondary signal, reported here with equal prominence to the null primary result per the brief's non-negotiable rule**: children randomised to dextrose gel had worse visual processing scores (mean 94.5 vs 99.8, $p=0.02$), though the proportion scoring in the impaired range (<85) did not differ. Dextrose-gel children were also taller and heavier at follow-up (z-scores 0.18 vs -0.17, $p=0.001$; 0.57 vs 0.29, $p=0.01$) — an anthropometric signal of uncertain long-term significance.
- **9-10 years, mid-childhood** (St Clair et al. 2022, Neonatology, PMID 36516806, DOI 10.1159/000527715): 184/227 eligible assessed (81%). Low educational achievement was similar between groups (42% dextrose vs 45% placebo, relative risk 1.04, 95% CI 0.76–1.44, $p=0.79$). **The visual-processing signal persisted and sharpened at this timepoint**: children allocated to dextrose gel had lower visual perception standard scores (95.2 vs 100.6, mean difference -5.68, 95% CI -9.79 to -1.57, $p=0.006$) and a greater proportion scoring below 85 (23% vs 11%, relative risk 2.23, 95% CI 1.13–4.37,

p=0.02). This is now a **replicated adverse secondary signal across two follow-up timepoints (4.5 and 9-10 years) in the same trial**, and per the brief's explicit instruction is reported with the same prominence as the reassuring primary-outcome nulls, not subordinated to them. The trial's own interpretation is that dextrose gel "does not appear to result in any clinically significant differences in educational achievement or other neurodevelopmental outcomes" but that the visual-processing finding is real enough to warrant continued scrutiny, not dismissal.

Prophylactic dextrose gel: hPOD / pre-hPOD

- **hPOD, 2 years, intention-to-treat prophylaxis effect** (Edwards et al. 2022, JAMA, PMID 35315885, DOI 10.1001/jama.2022.2363): 1197/1321 (91%) of infants randomised in the 9 New Zealand hPOD centres to prophylactic 40% dextrose gel (n=681) vs placebo (n=678) at 1 hour of age. The primary outcome, neurosensory impairment, did not differ significantly (20.8% vs 18.7%; adjusted risk ratio 1.13, 95% CI 0.90-1.41). **Adverse secondary signal, given equal prominence per the brief's rule:** the dextrose-gel group had a significantly higher risk of motor delay (2.5% vs 0.7%; adjusted risk ratio 3.79, 95% CI 1.27-11.32) and significantly lower composite Bayley-III scores for cognitive (adjusted mean difference -1.30, 95% CI -2.55 to -0.05), language (-2.16, 95% CI -3.86 to -0.46), and motor (-1.40, 95% CI -2.60 to -0.20) domains, with no differences on the other 27 secondary outcomes. The authors explicitly flag that the trial "may have been underpowered to detect a small but potentially clinically important increase in risk" — this is not a reassuring null on the secondary outcomes and should not be read as one.
- **hPOD, 2 years, observational analysis by hypoglycaemia exposure (not treatment allocation)** (Edwards et al. 2022, JAMA Netw Open, PMID 36219444, DOI 10.1001/jamanetworkopen.2022.35989): a distinct, cohort-style (non-randomised-comparison) analysis of the same hPOD population, comparing children who did versus did not experience neonatal hypoglycaemia regardless of prophylaxis allocation. Here hypoglycaemia **was** associated with higher neurosensory impairment (23% vs 18%; adjusted risk ratio 1.28, 95% CI 1.01-1.60), with the highest risk after severe episodes (28% vs 18%; adjusted risk ratio 1.68, 95% CI 1.20-2.36) but not after recurrent episodes (19% vs 18%; adjusted risk ratio 1.06, 95% CI 0.63-1.80). This finding sits in apparent tension with the null CHYLD primary result above; the key methodological difference is that this is an **observational, confounded-by-indication comparison** (children who became hypoglycaemic despite prophylaxis vs those who did not) rather than a randomised comparison, so it is appropriately graded lower (see GRADE table) and should not be read as contradicting the CHYLD treat-to-threshold trial evidence.
- **pre-hPOD dosage trial, early school age (6-7 years), prophylaxis dosing** (Wei et al. 2024, Arch Dis Child Fetal Neonatal Ed, PMID 38307710, DOI 10.1136/archdischild-2023-326452): 309/392 eligible children assessed. No significant difference in the primary outcome (neurocognitive impairment) between placebo (56%) and any dextrose-gel dose (46-50% across four dosing regimens). Combined dextrose-gel exposure was associated with **lower** risk of motor impairment (3% vs 14%, adjusted risk difference -11%, 95% CI -19% to -3%) and **higher** mean cognitive scores (106.0

vs 101.1, adjusted mean difference 5.4, 95% CI 1.8–8.9) — a positive secondary signal for prophylaxis at school age, in contrast to the adverse motor/cognitive secondary signal seen in the hPOD 2-year ITT analysis above. The two results are not directly comparable (different outcome batteries, different ages, dose-ranging vs single-dose design) and the tension between them is not resolved in the current literature — flagged here rather than adjudicated.

- **pre-hPOD, early school age, hypoglycaemia exposure (observational)** (Wei et al. 2024, J Pediatr, PMID 38815750, DOI 10.1016/j.jpeds.2024.114119): 315/392 (80%) assessed; 47% experienced hypoglycaemia. Neurocognitive impairment was similar by exposure (51% vs 50%; adjusted risk difference –4%, 95% CI –15% to 7%) — another null primary result — but children with **severe or recurrent** hypoglycaemia specifically had worse visual motion perception and higher risk of emotional-behavioural difficulty, consistent with the CHYLD dose-response pattern (severity/recurrence, not simple exposure, driving risk).

Independent (non-Auckland) cohorts

- **Shah et al. 2018 systematic review and meta-analysis** (Neonatology, PMID 30408811, DOI 10.1159/000492859): 11 studies (12 publications) pooled with ROBINS-I risk-of-bias assessment and GRADE. In early childhood, hypoglycaemia exposure was not associated with pooled neurodevelopmental impairment (n=1657, OR 1.16, 95% CI 0.86–1.57) but was associated with visual-motor impairment (n=508, OR 3.46, 95% CI 1.13–10.57) and executive dysfunction (n=463, OR 2.50, 95% CI 1.20–5.22) — the same domain-specific pattern as CHYLD, now cross-validated across a broader evidence base. In mid-childhood, pooled data (much sparser, n=54 for the impairment outcome) showed an association with neurodevelopmental impairment (OR 3.62, 95% CI 1.05–12.42) and with low literacy (n=1395, OR 2.04, 95% CI 1.20–3.47) and numeracy (OR 2.04, 95% CI 1.21–3.44) — in tension with the later, larger, single-cohort CHYLD academic-achievement null (Shah 2022 JAMA, above); the meta-analysis predates that null result, illustrating how a single large well-conducted cohort study can shift the picture that a smaller pooled evidence base suggested.
- **Wickström et al. 2018** (Eur J Epidemiol, PMID 30030683, DOI 10.1007/s10654-018-0425-5): Swedish population-based register cohort, 101,060 singletons, moderate hypoglycaemia in otherwise healthy term infants (diabetic mothers and NICU admissions excluded). Any neurological/neurodevelopmental outcome: adjusted OR 1.48 (95% CI 1.17–1.88); any developmental delay OR 2.53 (95% CI 1.71–3.73); motor delay OR 1.91 (95% CI 1.06–3.44); cognitive delay OR 2.85 (95% CI 1.70–4.76). Early hypoglycaemia (<6h) carried the highest risk (any outcome OR 1.94, 95% CI 1.30–2.89; cognitive delay OR 3.17, 95% CI 1.35–7.43). This is a **large, register-based, unselected-population** study (not an at-risk-only cohort like CHYLD) and is one of the few sources of population-attributable evidence rather than at-risk-cohort evidence; it is retrospective/register-based (not a masked prospective cohort with continuous glucose monitoring) and cannot establish the same dose-response granularity as CHYLD, but its scale is a genuine strength.

- **Roeper et al. 2024** (JAMA Netw Open, PMID 38530314, DOI 10.1001/jamanetworkopen.2024.243683 [pii 10.1001/jamanetworkopen.2024.3683]): German single-centre, retrospectively recruited, frequency-matched cohort, 140 children assessed at mean 9.1 years. Severe transitional hypoglycaemia (≥ 1 reading ≤ 1.7 mmol/L, 30 mg/dL) was associated with a 4.8-point lower mean full-scale IQ (107.0 vs 111.8), a 4.9-fold increased odds of abnormal fine motor function (95% CI 1.5–15.5), a 5.3-fold increased odds of abnormal visual-motor integration (95% CI 2.1–13.3), and higher parent-reported attention-problem and ADHD-symptom T-scores. This is the **lowest glucose threshold used to define the exposure group** of any study in this domain (≤ 1.7 mmol/L/30 mg/dL, well below the 2.6 mmol/L/47 mg/dL BAPM/CHYLD treatment threshold) and its findings — a clear, consistent IQ and motor deficit — are most directly relevant to the question of what happens when glucose is allowed to fall well below the treatment threshold, rather than to the treat-to-2.6-mmol/L policy question CHYLD addresses.
- **Rasmussen et al. 2020** (Eur J Pediatr, PMID 32666280, DOI 10.1007/s00431-020-03729-x): Danish cohort, 71 children with neonatal hypoglycaemia (< 1.7 mmol/L, 30 mg/dL, treated to > 2.5 mmol/L, 45 mg/dL) versus 32 healthy sibling controls, assessed at mean 7.75–9.17 years. No difference in cognitive function (Wechsler IV) or behaviour (Child Behavior Checklist). Fine motor scores were lower in the hypoglycaemia group (age-adjusted percentile 42.6 vs 57.2, $p=0.03$; sibling-paired analysis $p=0.008$), particularly in boys ($\beta=-16.4$, $p=0.048$). A **mixed** result — null on cognition/behaviour, positive (adverse) on fine motor function — reported in full rather than reduced to a single headline direction.
- **Goode et al. 2016** (Pediatrics, PMID 27940690, DOI 10.1542/peds.2016-1424) — **preterm-relevant, cross-referenced from Domain D**: secondary analysis of the Infant Health and Development Program (745/985 infants with recorded glucose), stratified by glucose level, assessed at 3, 8, and 18 years. No significant differences in cognitive or academic skills at any age between hypoglycaemia-exposed and unexposed preterm children; children with more severe hypoglycaemia reported fewer problem behaviours at 18 years (not judged clinically meaningful by the authors). This is a **null finding in a preterm population** using an old (1980s-recruited) cohort with historical glucose-monitoring and treatment practices, and is discussed in full in Domain D; it is cross-referenced here because it is one of the few sources bearing directly on whether the term/late-preterm null and dose-response pattern generalises to preterm infants — inconsistent in this instance, since Goode found no dose-response at all.

(b) Why — physiological and methodological rationale

The domain-specific pattern (impairment concentrated in executive function and visual-motor function, sparing global cognitive/neurosensory composites in most studies) is consistent with the known vulnerability of parieto-occipital white matter and watershed cortical regions to neuroglycopenic injury (see Domain F, Burns et al. 2008), which subserve visuomotor integration and are anatomically and functionally linked to fronto-executive circuits. The consistent dose-response signal by severity and recurrence (not by

simple exposure) across CHYLD, the Auckland pre-hPOD cohort, and (for IQ/motor outcomes) Roeper et al. is mechanistically coherent with a cumulative-insult model of neuroglycopenic injury rather than a single-threshold-crossing model — reinforcing the conceptual point established in the protocol (§4) that "hypoglycaemia" is a proxy for neuroglycopenia, and the degree and duration of the proxy signal's abnormality tracks injury risk better than a binary threshold crossing.

(c) GRADE certainty summary

Claim	Certainty	Downgrade domain(s)
Treated hypoglycaemia (to ≥ 2.6 mmol/L/47 mg/dL) is not associated with combined neurosensory impairment at 2 or 4.5 years in late-preterm/term infants at risk (CHYLD)	Moderate	Imprecision (wide CIs on several component outcomes); single-centre cohort, not randomised (no untreated comparison arm exists or could ethically exist)
Severe, recurrent, or clinically undetected (interstitial-only) hypoglycaemia is associated with dose-dependent risk of low executive function and visual-motor impairment at 4.5 years	Moderate	Imprecision (executive-function risk ratio 95% CI 1.17–4.59 is wide); indirectness (single cohort; population is at-risk late-preterm/term, not unselected)
Dextrose gel treatment of confirmed hypoglycaemia does not alter neurosensory impairment at 2, 4.5, or 9–10 years, but is associated with a replicated adverse visual-processing signal at 4.5 and 9–10 years (Sugar Babies)	Low (for the visual-processing signal) / Moderate (for no effect on primary neurosensory outcome)	Imprecision (secondary outcome, not powered as primary); single trial population; risk of bias low (randomised, though gel allocation could not be blinded to families administering it, only outcome assessors were masked)
Prophylactic dextrose gel does not reduce neurosensory impairment at 2 years and is associated with an adverse secondary signal (motor delay, lower cognitive/language/motor composite scores) in the hPOD ITT population	Low	Imprecision (trial explicitly underpowered for this signal per its own authors); risk of bias low (randomised, masked assessors)
Moderate hypoglycaemia in an unselected term population is associated with increased odds of developmental delay (Wickström, Swedish register cohort)	Low	Risk of bias (retrospective register data, residual confounding by indication for glucose testing not fully excludable); indirectness (population differs from CHYLD's at-risk cohort)
Severe transitional hypoglycaemia (≤ 1.7 mmol/L/30 mg/dL) is associated with lower IQ and motor/visual-motor deficits at school age (Roeper)	Low	Risk of bias (retrospective single-centre recruitment, frequency-matched not randomised); imprecision (n=140)

(d) Where the evidence is weak, contested, or extrapolated

Three genuine tensions exist within this domain, and none is resolved by the current literature — all are reported as open, per the brief's rule against artificial resolution:

1. **CHYLD's reassuring treat-to-threshold null versus the hPOD observational hypoglycaemia-exposure signal.** The randomised CHYLD/Sugar Babies evidence (treat hypoglycaemia to ≥ 2.6 mmol/L, no excess neurosensory impairment) and the observational, non-randomised hPOD exposure analysis (hypoglycaemia associated with 28% excess neurosensory impairment after severe episodes) are not strictly contradictory — they answer different questions (effect of a treatment policy vs. association of experiencing hypoglycaemia despite that policy, which is confounded by whatever made prophylaxis fail) — but a reader skimming both papers could reasonably come away with opposite headline impressions. This synthesis treats the randomised/treat-to-threshold evidence as the higher-certainty answer to the clinical policy question ("does treating to 2.6 mmol/L prevent harm?") and the observational exposure analyses as hypothesis-generating for which infants remain at residual risk despite treatment.
2. **The Shah 2018 meta-analysis's mid-childhood signal versus the larger, later CHYLD academic-achievement null.** The 2018 pooled estimate (OR 3.62 for mid-childhood impairment, based on only 54 children pooled across small studies) preceded the CHYLD cohort's own 9–10 year academic-achievement result (Shah 2022 JAMA, $n=480$, null). This is a case where accumulating evidence in a single well-powered cohort moved the estimate toward the null relative to an earlier, much smaller pooled estimate — worth stating explicitly rather than citing the 2018 meta-analysis's mid-childhood OR as if it were still the best current estimate.
3. **Dextrose gel's secondary visual-processing/motor signal, replicated but unexplained.** Both the treatment (Sugar Babies, confirmed hypoglycaemia) and prophylaxis (hPOD, at-risk-only) dextrose gel trials show an adverse secondary signal — worse visual processing in Sugar Babies at 4.5 and 9–10 years, worse motor/cognitive/language composite scores in hPOD at 2 years — that neither trial's authors are able to explain mechanistically, and that sits in tension with the positive motor/cognitive secondary signal in the dose-ranging pre-hPOD school-age follow-up. Whether this reflects a genuine biological effect of buccal dextrose gel (osmotic, glycaemic-swing, or other mechanism), chance findings from underpowered secondary-outcome batteries, or population/design differences between the trials cannot be adjudicated from the current literature. **This is flagged, not resolved** — it is one of the more clinically important open questions given how widely dextrose gel has been adopted as first-line UK practice (see Domain C).

Directness caveat applying to this entire domain: every trial and cohort above enrolled infants ≥ 32 –35 weeks' gestation (late preterm and term). None of this evidence is generated in, and none of it should be silently assumed to apply to, very preterm (<32 weeks) or extremely preterm (<28 weeks) infants — the population that dominates a Level

3 NICU caseload. See Domain D for the separate, and much thinner, preterm-specific evidence base, and the cross-cutting preterm evidence-gap statement in the final synthesis.

(e) Practical implication for a UK Level 3 unit

For the late-preterm/term at-risk population this domain actually covers, the CHYLD/Sugar Babies programme provides reassuring randomised evidence that a treat-to-2.6 mmol/L (47 mg/dL) policy does not produce excess global neurosensory impairment by mid-childhood, which underpins confidence in the BAPM treatment threshold (Domain C). But three points should inform how this evidence is used in practice and in parent counselling: (i) the absence of global impairment does not mean the absence of any effect — executive function and visual-motor function are consistently, if modestly, affected in a severity/recurrence-dependent way, including when hypoglycaemia is clinically undetected (interstitial-only); (ii) dextrose gel, now first-line UK practice, carries a replicated but mechanistically unexplained adverse visual-processing secondary signal at two follow-up timepoints in the trial that established its use, which is a legitimate basis for continued outcome surveillance rather than for changing current practice; (iii) none of this reassurance transfers to the very/extremely preterm infant without an explicit, acknowledged extrapolation.

Domain F — Hypoglycaemia + seizure: is MRI indicated?

Directness: The central evidence base is a single UK cohort (Burns et al. 2008, n=35, Hammersmith Hospital — the same trust as this synthesis). Everything else is small, retrospective, non-UK case-series material (China, India, Turkey, Australia, Canada, Japan) describing the injury-pattern phenotype and, more recently, MRI timing. There is no RCT and cannot ethically be one (imaging cannot be randomised against a "no scan" arm in a child who has had a hypoglycaemic seizure). No BAPM- or NICE-authored guidance on MRI in this specific scenario was found. One 2025 multi-society practice guideline (Spanish Neonatal Brain Group, first author Hammersmith/Imperial College Healthcare NHS Trust-affiliated) addresses MRI timing across neonatal pathologies including hypoglycaemia-associated brain damage, but its condition-specific recommendation could not be extracted beyond the abstract this session. Every recommendation below is flagged accordingly.

(a) What the evidence shows

The central paper: Burns et al. 2008

Burns, Rutherford, Boardman & Cowan (Pediatrics 2008, PMID 18595988, DOI 10.1542/peds.2007-2822) studied 35 term infants with early brain MRI after symptomatic neonatal hypoglycaemia (median glucose 1 mmol/L, 18 mg/dL) without evidence of hypoxic-ischaemic encephalopathy, at Hammersmith Hospital, Imperial College Healthcare NHS Trust — the same trust this synthesis is written for, which is worth stating plainly rather than treating as incidental. Findings: white matter abnormalities in 94% (severe in 43%, a posterior-predominant pattern in 29%); cortical abnormalities in 51%; white matter haemorrhage in 30%; basal ganglia/thalamic lesions in 40%; abnormal posterior limb of the internal capsule (PLIC) in 11%; and — the finding most relevant to differential diagnosis, below — **3 of 35 infants had middle cerebral artery (MCA) territory infarctions**. At a minimum of 18 months, 65% (23/35) had neurodevelopmental impairment, related to the severity of white matter injury and PLIC involvement. The paper's own conclusion, stated verbatim because it is the single most important sentence in this domain: **"Early MRI findings were more instructive than the severity or duration of hypoglycemia for predicting neurodevelopmental outcomes."** This directly answers the brief's central question — in this cohort, MRI pattern outperformed biochemical severity/duration as a prognostic tool.

Injury pattern: broader and more heterogeneous than "posterior-only"

Burns 2008 itself corrects an oversimplified prior teaching point: injury is not confined to the parieto-occipital region. It described white matter injury (not confined posteriorly),

haemorrhage, basal ganglia/thalamic lesions, and even arterial infarction. Independent case-series material — none of it UK, all screened and verified this session — is broadly consistent with a posterior-predominant predilection while confirming this heterogeneity:

- **Gu et al. 2019** (Clin Med Insights Pediatr, PMID 31447599, DOI 10.1177/1179556519867953): retrospective cohort of 66 hypoglycaemic neonates undergoing MRI (China). Abnormal MRI in 54.5% (36/66), most commonly parietal/occipital. **Key finding, verified against the abstract text as instructed:** the number of days with hypoglycaemia — not the lowest glucose value — was significantly associated with abnormal MRI ($P<.001$), and prolonged/recurrent hypoglycaemia was likewise associated with abnormal MRI ($P<.001$); lowest glucose did **not** differ between infants with and without abnormal MRI ($P>.05$). However, lowest glucose was significantly lower specifically in the subgroup who had seizures ($P<.01$) — i.e., glucose nadir tracks with seizure occurrence, but duration/recurrence, not nadir, tracks with imaging abnormality. This is a genuinely nuanced (not simply "duration beats severity") finding and is reported as such.
- **Mao et al. 2008** (Chinese, Zhongguo Dang Dai Er Ke Za Zhi, PMID 18433524 — **[abstract only]**, Chinese language, English abstract retrieved via PubMed metadata, full text not accessed) and **Mao et al. 2007** (Chinese, Zhonghua Er Ke Za Zhi, PMID 17953809 — **[abstract only]**, same limitation): two small serial-MRI case series ($n=16$ and $n=6$) from the same Chinese group, both showing occipital/parietal cortex as most vulnerable, both showing basal ganglia/thalamic and white matter involvement in the more severe cases, and both showing DWI hyperintensity as the earliest detectable sign, evolving to T1/T2 changes later. The 2008 series found minimum glucose was significantly lower in infants with widespread cortical injury than those with occipital-parietal injury alone (0.71 ± 0.35 vs 1.19 ± 0.42 mmol/L, $P<0.05$) — in this small, non-UK series, severity of the glucose nadir did track with the extent of injury, which sits in tension with Gu 2019's larger finding above about duration vs nadir; both are reported rather than adjudicated, since neither is a UK-comparable, adequately powered source.
- **Hu et al. 2014** (Acta Paediatr, PMID 24798709, DOI 10.1111/apa.12673): case-control study (15 hypoglycaemia cases vs 11 controls, China) using flash visual evoked potentials (FVEP) alongside MRI. 80% of the hypoglycaemia group had occipital-lobe injury on MRI; FVEP was 100% sensitive but only 25% specific for parieto-occipital injury, so cannot substitute for MRI. Notably, the infants who went on to have seizures were detected as hypoglycaemic later (median 51 hours of age) than those who did not (median 22 hours, $P=0.003$) — consistent with a picture in which delayed recognition, not simply glucose depth, contributes to the seizure phenotype.
- **Kurahashi et al. 2023** (World J Clin Cases, PMID 37383118, DOI 10.12998/wjcc.v11.i16.3899): a single case report of an 8-year-old with parieto-temporo-occipital injury from neonatal hypoglycaemia/seizures, notable for demonstrating a **non-visual** long-term deficit (slowed reading and impaired phonological awareness, without cortical blindness or visual field defect) — a useful reminder that the outcome phenotype from this injury topography extends beyond the commonly emphasised cortical visual impairment.

Later epilepsy phenotype: variable, and occipital-predominant

- **Fong & Harvey 2014** (Dev Med Child Neurol, PMID 24861161, DOI 10.1111/dmcn.12496): retrospective case series of 11 children (Melbourne) with seizures beyond infancy after neonatal hypoglycaemia (perinatal asphyxia excluded). All 11 had MRI evidence of occipital $\pm$ parietal gliosis/atrophy. Seizure phenotype was genuinely variable: focal occipital seizures in 9 (infrequent or remitted in 6 of these 9), generalised tonic seizures in 2 (both frequent and refractory). The paper's own framing — mild occipital, refractory occipital, and symptomatic generalised epilepsy all recognised from the same antecedent injury — is the headline finding and is reported as a genuine null-to-mixed result on the question of whether injury pattern alone predicts epilepsy severity.
- **Kapoor et al. 2020** (Seizure, PMID 32446209, DOI 10.1016/j.seizure.2020.05.010): the largest case series in this domain by a wide margin — 170 children with epilepsy secondary to documented neonatal hypoglycaemic brain injury, over a 10-year period at a single Indian tertiary epilepsy clinic explicitly described by the authors as a **low-resource setting**. MRI showed occipital $\pm$ parietal gliosis/encephalomalacia in 96.5%. Epileptic spasms were the dominant seizure type (76.5%), and 68.2% (116/170) had drug-resistant epilepsy. Comorbidity was substantial: global developmental delay 91.2%, cerebral palsy 48.7%, vision impairment 48.2%, microcephaly 38.2%. **This PMID was flagged as unresolvable in the parent scoping session's could-not-verify list; it resolved cleanly this session both as a solo `get_article_metadata` call and via an independent title search — see §"Kapoor PMID" below.** This series sits at the most severe end of the spectrum (LMIC referral-clinic population with established, often refractory epilepsy) and its very high comorbidity rates should not be read as representative of the outcome distribution in a UK cohort with earlier detection and treatment.
- **Yalçın et al. 2022** (Noro Psikiyatri Arsivi, PMID 36514522, DOI 10.29399/npa.27997): independent Turkish replication (n=21 with documented hypoglycaemia and typical parieto-occipital injury) showing the same outcome cluster — 85.7% developed epilepsy, 38.1% refractory, 33.3% cerebral palsy, 100% of those tested (18/18) had abnormal VEP. Prolonged hypoglycaemia (recurrent episodes >12h) was associated with significantly greater parieto-occipital involvement and spasticity (P<0.01). This is now a **third independent non-UK cohort** (after Fong 2014 and Kapoor 2020) cross-validating the injury-pattern-to-epilepsy/CP/visual-impairment linkage, which strengthens confidence in the phenotype description even though none of the underlying cohorts is UK-based or population-representative.

MRI timing: when does the scan need to happen, and what does it show at that point?

This is the sub-question the brief specifically asked to be chased for 2015+ literature, since Burns 2008 (2008) does not itself resolve it.

- **Tam et al. 2008** (Pediatrics, PMID 18762519, DOI 10.1542/peds.2007-2002) — pre-2015 but foundational and retained per the brief's rule that foundational imaging papers are not date-filtered. In 45 neonates (term and preterm) with diffusion-weighted MRI (DWI) after hypoglycaemia: among 25 infants scanned **within 6 days** of hypoglycaemia onset, occipital diffusion restriction was seen in 8/16 (50%) **term** infants but in **zero** preterm infants; among 20 infants scanned **beyond 6 days**, no occipital diffusion restriction was seen at all, even where hypoglycaemia was ongoing. This is a directly quantified timing-window finding: DWI is sensitive for the classic injury signature only within roughly the first week, and — critically for directness — **this sensitivity does not transfer to preterm infants at all** in this dataset. Diffusion restriction correlated with abnormal visual evoked potentials and later cortical visual deficit.
- **Zhang et al. 2022** (Eur J Pediatr, PMID 36166098, DOI 10.1007/s00431-022-04637-y) — the most direct 2015+ MRI-timing paper found. Retrospective analysis of 139 DWI scans in 86 Chinese neonates with hypoglycaemic brain injury: 84 scans showed injury-site DWI hyperintensity **within 7 days** of hypoglycaemia; 39 scans showed the signal evolving to abnormal T1/T2-weighted changes (with DWI often pseudonormalising) at **11-23 days**; repeat scans in 13 infants at 9-20 days showed no abnormal signal at all. Vulnerable sites: occipital lobe 98%, splenium of the corpus callosum 60%, parietal lobe 30%. Extensive injury (involving internal capsule, basal ganglia, thalamus; n=20) carried significantly higher rates of recurrent convulsions, respiratory failure, coma, and death/severe neurodevelopmental impairment than focal injury (n=66) — the authors explicitly note this extensive pattern may partly reflect coexisting hypoxia-ischaemia rather than hypoglycaemia in isolation, a caveat directly relevant to the differential-diagnosis point below.
- Taken together, Tam 2008 and Zhang 2022 — a decade and a different country apart, using the same DWI methodology — converge on a consistent early-DWI-positive, later-pseudonormalising timing pattern, with the injury becoming harder to detect on DWI after roughly one to two weeks even as conventional T1/T2 sequences may still show or newly show abnormality. This is the empirical basis for the "scan early" recommendation in §(e), and it is corroborated (though not independently timed) by both Mao 2007 and Mao 2008 reporting DWI as the earliest-detectable abnormality in their own small serial-imaging series.

Differential diagnosis and coexistence: the stroke-mimic point

Burns 2008's finding that **3 of 35 infants (8.6%) undergoing MRI for hypoglycaemic seizure workup had MCA territory infarctions** is the direct evidential basis for treating MRI as having a differential-diagnosis role that is separable from — and does not depend

on — its prognostic role. A dedicated PubMed search for literature specifically pairing "hypoglycaemic seizure" with "stroke mimic" terminology returned no hits: this point rests on the internal finding within the Burns cohort itself, not on a dedicated literature, and should be presented as such rather than implying a larger evidence base exists. The practical implication is straightforward regardless: a clinical picture of "hypoglycaemia plus seizure" cannot be assumed on biochemistry alone to be caused by hypoglycaemia — MRI (not glucose values, not EEG alone) is the modality that would detect a coexisting or confounding arterial infarction, and by extension would also be the modality to detect coexisting HIE, a structural lesion, or an atypical/multifocal pattern that might prompt a metabolic or genetic work-up rather than (or in addition to) a hypoglycaemia-only diagnosis. This is a qualitatively different indication for MRI than the prognostic one, and the two should not be conflated: even where prognosis for a straightforward hypoglycaemic injury tracks the imaging pattern well (per Burns 2008's headline finding), MRI retains a separate, undiminished role in excluding or detecting a coexisting alternative diagnosis.

A 2025 guideline addressing MRI timing across neonatal pathologies, including hypoglycaemia

Martínez-Biarge et al. 2025 (An Pediatr [Engl Ed], PMID 41022518, DOI 10.1016/j.anpede.2025.503935): a practice-guideline document from the Spanish Neonatal Brain Group and paediatric neuroradiologists, covering MRI logistics, condition-specific protocols, and — of direct relevance here — "clinical indications and optimal timing of MRI... to maximize its diagnostic and prognostic value in different pathologies such as... brain damage associated with hypoglycemia." The first author is affiliated with **Hammersmith Hospital, Imperial College Healthcare NHS Trust** — the same trust as Burns 2008 and as this synthesis's intended use, which is a genuinely useful practice link even though the guideline itself is Spanish multi-society output, not a UK/BAPM/NICE document. No PMC full text was available (confirmed via `convert_article_ids` this session), so the specific timing value recommended for the hypoglycaemia section could not be extracted beyond the abstract — this is marked **[abstract only]** and should be re-obtained in full text before being used to write a specific timing rule into unit guidance. This is nonetheless the only guideline-tier (as opposed to case-series-tier) source located anywhere in this domain that names hypoglycaemia-associated brain damage as a condition with its own MRI timing considerations, and is treated throughout as **grey literature**, not folded into the peer-reviewed evidence count, per the project's non-negotiable rules.

Kapoor PMID: resolved, not unresolvable

The brief flagged PMID 32446209 (Kapoor et al., Seizure 2020) as having failed verification in the parent scoping session, with the tool silently dropping it from a 20-PMID batch call. This session re-attempted it twice: once as a solo `get_article_metadata(["32446209"])` call, and once via an independent PubMed title search — both resolved cleanly with full, consistent metadata (title, DOI 10.1016/j.seizure.2020.05.010, abstract, journal, author affiliations). **The PMID is valid and the paper is real and verifiable**; the earlier failure

was a batch-call artefact of the tool, not a bad citation. This is not re-added to `could-not-verify-domain-F.md` since it now verifies — see the note at the end of this document for how the parent's master could-not-verify list should be reconciled.

(b) Why — physiological and methodological rationale

The posterior (parieto-occipital) predilection described across this literature is consistent with the developmental metabolic-demand gradient of the neonatal brain: the occipital and parietal cortices are undergoing particularly active synaptogenesis and have high regional glucose utilisation in the perinatal period, making them selectively vulnerable to substrate (glucose) deprivation — the same "neuroglycopenia as the true injury mechanism, glucose as an imperfect proxy" framing established in the project's conceptual framing (RESEARCH-BRIEF.md §4) applies directly here. Burns 2008's finding that injury is not confined to this region — extending to white matter, basal ganglia/thalamus, haemorrhage, and even arterial infarction — indicates that severe or prolonged neuroglycopenic insult can produce a watershed-type or more globally distributed pattern once the posterior predilection is overwhelmed, analogous to the way basal ganglia/thalamic and white matter/watershed patterns are described in hypoxic-ischaemic injury; several of the case series here explicitly flag that their most severe, most extensive-injury cases may reflect coexisting hypoxia-ischaemia rather than hypoglycaemia in isolation (Zhang 2022), which is itself an argument for MRI's differential-diagnosis role rather than an argument against it.

The MRI-timing pattern (Tam 2008, Zhang 2022) is consistent with the known evolution of cytotoxic oedema on diffusion-weighted imaging generally: DWI hyperintensity from restricted diffusion appears early (in this literature, within roughly 6–7 days) and pseudonormalises over one to three weeks as the oedema resolves or the tissue becomes established/gliotic, at which point conventional T1/T2-weighted sequences — not DWI — carry the diagnostic signal. This is the same DWI evolution principle seen in other perinatal cerebral injury (e.g. arterial ischaemic stroke, HIE), and is the physiological basis for recommending imaging in the first week rather than at an arbitrarily later, more "convenient" time.

(c) GRADE certainty summary

Claim	Certainty	Downgrade domain(s)
Early MRI pattern (white matter injury severity, PLIC involvement) predicts neurodevelopmental outcome better than glucose severity/duration after symptomatic neonatal hypoglycaemia	Moderate	Imprecision (single-centre, n=35, no untreated/comparator arm — none could ethically exist); this is nonetheless the most direct, UK-based source in the domain
	Low	

Claim	Certainty	Downgrade domain(s)
Duration/recurrence of hypoglycaemia (not the lowest glucose value) is associated with abnormal MRI findings		Risk of bias (retrospective, single-centre, non-UK); imprecision; inconsistency (smaller Chinese case series — Mao 2008 — found the opposite pattern, glucose nadir tracking with injury extent, in a much smaller sample)
DWI-MRI is sensitive for the classic occipital injury signature only within approximately the first week after hypoglycaemia onset, pseudonormalising thereafter	Low	Risk of bias (retrospective, non-UK, single-centre in both supporting studies); imprecision (small samples scanned within the sensitive window — 16 term infants in Tam 2008); consistency is a strength (two independent cohorts, 14 years apart, concordant findings)
DWI-MRI timing sensitivity established in term infants does not extend to preterm infants (zero diffusion restriction detected in preterm infants scanned within 6 days in Tam 2008)	Low	Very serious imprecision (small preterm subgroup); this is a directness-limiting negative finding, not a well-powered preterm-specific study — treated as a caution against extrapolation, not as proof of absence of preterm injury
MRI performed for hypoglycaemic seizure workup can detect coexisting arterial ischaemic stroke (MCA territory infarction in 3/35 infants, Burns 2008)	Low (as a rate estimate) but the underlying principle (MRI's differential-diagnosis role is independent of its prognostic role) is not seriously contested	Imprecision (single cohort, small numerator); no dedicated literature found to corroborate or refute the specific coexistence rate — the qualitative point (MRI can find things glucose values and EEG cannot) is not in doubt, but the 8.6% figure should not be quoted as a stable, generalisable coexistence rate
The later epilepsy phenotype after neonatal hypoglycaemic brain injury is occipital-predominant but clinically variable (from mild focal to refractory generalised/drug-resistant)	Low-Moderate (consistent direction across four independent cohorts — Fong 2014, Kapoor 2020, Yalçın 2022, plus corroborating case-report material — increases confidence in the qualitative pattern; certainty for any specific proportion remains low)	Risk of bias (all cohorts are retrospective and referral/clinic-selected, not population-based); indirectness (none UK-based; Kapoor 2020 explicitly a low-resource LMIC setting at the severe end of the spectrum)

(d) Where the evidence is weak, contested, or extrapolated

1. **The entire evidence base outside Burns 2008 is small, retrospective, and non-UK.** No study in this domain other than Burns 2008 was conducted in a UK population or setting comparable to a modern UK Level 3 unit's glycaemic monitoring and treatment practice. The largest series (Kapoor 2020, n=170) is explicitly from a low-

resource LMIC referral clinic with established, often drug-resistant epilepsy — its very high comorbidity rates (91.2% global developmental delay, 48.7% cerebral palsy) almost certainly overstate the risk that would be seen in a UK cohort with earlier glucose detection, treatment, and imaging access, and should not be quoted to parents or in governance documents as if representative of UK-expected outcomes.

2. **Severity vs duration as the driver of injury is not settled.** Gu 2019 (n=66, the larger and more recent of the two duration-focused series) found duration/recurrence, not glucose nadir, predicted abnormal MRI; Mao 2008 (n=16, smaller, earlier) found glucose nadir did track with injury extent in the more severe subgroup. Both are non-UK, retrospective, and modestly sized. This is reported as an open tension, not resolved — the honest summary is that both duration and severity plausibly matter, and no single study in this domain is powered or designed to adjudicate between them.
3. **The MRI-timing evidence (Tam 2008, Zhang 2022), while internally consistent, rests on only two studies, 14 years apart, both from single non-UK tertiary centres,** using retrospective and likely non-protocolised (clinically-driven) scan timing. The recommendation to scan within the first week is well-supported qualitatively by DWI physiology and by these two studies' concordance, but the exact optimal day (day 3? day 5? day 7?) cannot be pinned down more precisely than "within roughly a week, and preferably not later" from the current literature.
4. **The stroke-mimic/coexistence point has essentially one data point.** The 3/35 MCA infarction finding in Burns 2008 is the entire evidentiary basis for this argument in the peer-reviewed literature identified this session — no dedicated study of stroke-hypoglycaemia coexistence rates was found. This does not make the underlying clinical logic (MRI, not biochemistry, is the tool that detects a coexisting stroke) any less sound, but the specific 8.6% figure should be treated as illustrative of the phenomenon existing, not as a stable estimate of its population frequency.
5. **The 2025 Spanish guideline (Martínez-Biarge et al.) could not be fully interrogated this session** — its abstract confirms that hypoglycaemia-associated brain damage is one of the conditions for which the document gives specific timing/indication recommendations, but the actual recommended timing window was not extractable without full text (no PMC copy available). This is the single most promising near-term source for upgrading the "when to scan" recommendation from case-series-derived to guideline-derived, and should be a priority follow-up (obtain via the journal directly, or via the first author's NHS trust) before this domain's recommendation is finalised for governance use.
6. **The preterm population is essentially unaddressed.** Tam 2008 is the only study in this domain reporting on preterm infants specifically, and its finding is negative (no preterm infant showed the expected DWI signature even within the sensitive window) — this could mean preterm infants are genuinely less susceptible to this injury pattern, or (more likely, given the small preterm subgroup and DWI's known technical limitations at preterm gestations) that DWI is simply less sensitive in the preterm brain for reasons unrelated to whether injury occurred. Neither explanation can be distinguished from the current literature, and no other study in this domain reports

preterm-specific data at all. This should be stated as an evidence gap, not silently extrapolated over.

(e) Practical implication for a UK Level 3 unit

Working recommendation: In an infant who has had a hypoglycaemic seizure, brain MRI is indicated for two distinct reasons that should both be named explicitly when requesting it: (i) **differential diagnosis** — to detect or exclude a coexisting or alternative cause (arterial ischaemic stroke, HIE, structural lesion, or a pattern atypical enough to prompt metabolic/genetic work-up), a role that holds regardless of how straightforward the biochemical picture seems, given Burns 2008's own cohort included unexpected MCA infarctions; and (ii) **prognostication**, where the evidence (principally Burns 2008, the only UK/directly-relevant source) indicates that the MRI pattern — specifically white matter injury severity and PLIC involvement — is a better predictor of 18-month neurodevelopmental outcome than the glucose value or duration reached during the acute episode. **Timing:** on the balance of the two DWI-timing studies located (Tam 2008, Zhang 2022), MRI should be obtained as early as practicable and, if at all possible, within the first week of the hypoglycaemic seizure — DWI sensitivity for the characteristic occipital/parieto-occipital signature declines sharply beyond roughly 6–7 days, after which conventional T1/T2-weighted sequences rather than DWI carry the diagnostic information, and a scan performed purely for logistical convenience at 2–3 weeks risks a falsely reassuring or uninterpretable DWI even where earlier injury occurred.

Which infants should be scanned: the literature located this session does not support restricting imaging only to the most severe or prolonged cases as a validated triage rule — no study in this domain tested or validated a severity/duration threshold below which MRI can safely be omitted. Given that (a) the differential-diagnosis rationale applies to essentially any infant with a seizure regardless of biochemical severity, and (b) Burns 2008 found imaging outperformed biochemical severity for prognosis specifically because biochemical severity was a poor discriminator, the working position taken here is that **any infant with a clinical seizure attributed to (or occurring alongside) neonatal hypoglycaemia warrants brain MRI**, not only those with the most severe or longest biochemical derangement — but this is a reasoned extrapolation from the available evidence's logic, not a validated triage threshold from a trial or guideline.

Directness caveat — state this plainly, not as a footnote: this entire recommendation, other than the central Burns 2008 finding, is extrapolated from small, heterogeneous, largely non-UK case series (China, India, Turkey, Canada, Japan, Australia) rather than from a guideline-level source. No BAPM or NICE document addressing MRI in hypoglycaemic seizure specifically was found. The one guideline-tier document located (Martínez-Biarge et al. 2025, Spanish Neonatal Brain Group) is promising — not least for its first author's Hammersmith/Imperial College Healthcare NHS Trust affiliation, the same trust as Burns 2008 and as this synthesis — but its specific hypoglycaemia-timing recommendation could not be extracted from the abstract alone this session and should be obtained in full text before this recommendation is written into unit guidance as anything

more authoritative than "an evidence-informed working position, extrapolated from case-series-level data, pending guideline confirmation."

Note on the Kapoor PMID for the parent session

PMID 32446209 (Kapoor et al., Seizure 2020) was logged in the master `05-references/could-not-verify.md` as unresolvable (silently dropped from a 20-PMID batch `get_article_metadata` call). This session resolved it cleanly, twice (solo metadata call and independent title search) — see `01-search-log/domain-F-searches.md` for full detail. **This PMID is valid and should be removed from the master could-not-verify list on merge**, with the full citation (DOI 10.1016/j.seizure.2020.05.010) moved into the verified reference set instead. This agent has not edited the master file directly per instructions; flagging here for the parent to action.

Domain G — Hyperglycaemia and long-term follow-up

Directness: This domain is fundamentally more heterogeneous in population than Domain E. Three distinct clinical scenarios are conflated in casual use of the word "neonatal hyperglycaemia" and are kept separate throughout: **(i) primary/stress hyperglycaemia in extremely and very preterm infants** (relative insulin deficiency, high parenteral glucose load — the dominant Level 3 NICU scenario), **(ii) parenteral-nutrition-related hyperglycaemia in preterm infants** (a related but mechanistically distinguishable nutritional-management phenomenon), and **(iii) secondary hyperglycaemia in term/near-term infants with neonatal encephalopathy** (a stress-response phenomenon in a sick-term population, not a primary metabolic disorder of prematurity). Unlike Domain E, there is **no equivalent of CHYLD or Sugar Babies** for hyperglycaemia — no adequately powered randomised trial has evaluated a hyperglycaemia treatment threshold against long-term neurodevelopmental outcome in any population. Every claim below states which of the three scenarios it applies to.

(a) What the evidence shows

Operational definitions — and the confirmed absence of UK/BAPM or NICE guidance

No single numeric threshold for neonatal hyperglycaemia has consensus or guideline status. The most current attempt at a synthesised operational definition comes from the secondary review at the centre of this domain:

- **Lagacé & Tam 2024** (Pediatr Res, PMID 38972961, DOI 10.1038/s41390-024-03411-0): drawing on the accumulated clinical/physiological/epidemiological literature, propose ≤ 47 mg/dL (2.6 mmol/L) for hypoglycaemia and **>150-180 mg/dL (>8.3-10.0 mmol/L) for hyperglycaemia** as "safe operational definitions... to delineate" initiation of treatment, and describe a **"u-shape" relationship between dysglycaemia and adverse neurological outcomes** — i.e., both ends of the glucose distribution, not hyperglycaemia or hypoglycaemia alone, carry risk. The review explicitly states that **glycaemic lability has been recognised as a key factor in adverse neurodevelopmental outcomes**, over and above either threshold considered in isolation. This is a review's synthesis, not new primary data — it is cited directly here, and its constituent claims are traced back below.

A dedicated search of BAPM and NICE grey literature this session (see [01-search-log/domain-G-searches.md](#) §4) found **no BAPM Framework for Practice addressing hyperglycaemia**. Every version of BAPM's glycaemia-related Framework located, including the January 2024 update, is titled "Identification and Management of Neonatal Hypoglycaemia in the Full Term Infant (Birth-72 hours)" — hyperglycaemia is outside its

scope entirely. No NICE guideline sets a neonatal hyperglycaemia treatment threshold; NICE NG3 addresses maternal glycaemic control in diabetic pregnancy and its neonatal hypoglycaemia risk, not a neonatal hyperglycaemia cut-off. **This is a confirmed absence, stated plainly rather than papered over:** UK national guidance on neonatal hyperglycaemia does not exist, and unit-level practice (e.g., the individual NHS regional guidelines identified in web search, typically using treatment triggers in the 8-10 mmol/L/144-180 mg/dL range) represents local consensus, not national policy.

Population 1 — primary/stress hyperglycaemia in extremely/very preterm infants: the core long-term outcome evidence

- **Zamir et al. 2021** (Arch Dis Child Fetal Neonatal Ed, PMID 33863775, DOI 10.1136/archdischild-2020-319926, PMC8394751): the EXPRESS national cohort, 533 Swedish infants born <27 completed gestational weeks (2004-2007), 436 (82%) assessed at 6.5 years. **Duration** of hyperglycaemia >8 mmol/L (144 mg/dL) — not a single threshold breach — was associated with WISC-IV FSIQ: each additional day carried a **0.33-point decrease** (95% CI 0.03-0.62). Hyperglycaemia >8 mmol/L on **3 consecutive days** was associated with lower MABC-2 motor scores (adjusted mean difference **-4.90**, 95% CI -8.90 to -0.89), and each day with hyperglycaemia carried a further **0.55-point** MABC-2 decrease (95% CI 0.17-0.93). **Insulin treatment was not associated with either worsened or improved neurodevelopmental outcomes on any measure** — a null finding given equal prominence here, and one the authors themselves use to call explicitly for RCTs to clarify insulin's role. This is the single most directly relevant primary study for the extremely preterm population and is the clearest trace-back for Lagacé & Tam's U-shape claim in this gestational band: it demonstrates a genuine dose-response (duration-dependent) relationship between hyperglycaemia and adverse cognitive/motor outcome in extremely preterm infants specifically.
- **Boscarino et al. 2021** (Nutrients, PMID 34199741, DOI 10.3390/nu13061930, PMC8227040): prospective cohort, Rome, infants <32 weeks GA or <1500 g birth weight, n=108 (32 exposed/76 unexposed) assessed with Bayley-III at 24 months corrected age. **Parenteral-nutrition-related** hyperglycaemia (glucose >180 mg/dL, 10.0 mmol/L, in the first week) was associated with cognitive delay (44% vs 22%, p=0.024) and motor delay (38% vs 8%, p<0.001); hyperglycaemia **remained an independent risk factor for motor delay after adjustment** for background characteristics (the cognitive-delay association's adjusted result is not separately reported in the abstract). This is the clearest evidence specifically implicating the **nutritional-management route** to hyperglycaemia (high early macronutrient/PN intake, as distinct from illness-severity-driven stress hyperglycaemia), with a plausible mechanistic story: excess early PN load produces glucose intolerance in an immature endocrine-pancreatic system, and the resulting hyperglycaemia (or a shared upstream driver) tracks with motor and cognitive delay.
- **van der Lugt et al. 2010** (BMC Pediatr, PMID 20646308, DOI 10.1186/1471-2431-10-52, PMC2915976): retrospective cohort, Leiden, 859 infants ≤32 weeks screened over 2002-2006; 66/859 (8%) developed hyperglycaemia (glucose

≥10 mmol/L, 180 mg/dL, treated with insulin ≥12 hours). **Mortality was strikingly higher in the hyperglycaemia group** — 41% (27/66) versus 8% (62/793) in those without hyperglycaemia ($p<0.001$), concentrated in infants ≤1000 g or 24–28 weeks. Among the 33 hyperglycaemia survivors assessed at 2 years against 63 matched controls, growth was similar, but **neurological ($p=0.036$) and behavioural ($p=0.021$) development were more frequently abnormal**. This is a genuinely important companion finding to the neurodevelopmental signal: the mortality difference is so large that survivor-bias (the sickest infants died, so surviving "hyperglycaemia" infants may be a residually higher-risk group even after matching on the variables the authors could match on) is a real threat to interpreting the neurodevelopmental result as a hyperglycaemia-specific effect rather than an illness-severity marker — flagged explicitly, not resolved.

Population 1b — the pivotal interventional trial, and why the brief's "COIN trial" does not exist

A dedicated search this session for the "COIN trial" named in the research brief found **no trial by that name in the peer-reviewed literature** (see search log §4 for the specific negative search). The trial that actually matches the brief's description — a multicentre RCT of early insulin therapy for preterm hyperglycaemia — is:

- **Beardsall et al. 2008, NIRTURE trial** ("Neonatal Insulin Replacement Therapy in Europe"; N Engl J Med, PMID 18971490, DOI 10.1056/NEJMoa0803725): multicentre RCT, 10 centres UK/Europe, 389 VLBW infants (<1500 g) randomised to continuous insulin infusion (0.05 U/kg/h with 20% dextrose support, days 1–7) versus standard care. **The trial was stopped early for futility on its primary outcome (mortality at expected date of delivery) and concern about potential harm**. Early insulin did reduce hyperglycaemia (mean glucose 6.2 vs 6.7 mmol/L, $p=0.007$; hyperglycaemia >10% of week 1 in 21% vs 33%, $p=0.008$) and improved short-term nutritional/growth markers (more carbohydrate infused, less weight loss). But it **significantly increased hypoglycaemia** (<2.6 mmol/L for >1 hour: 29% vs 17%, $p=0.005$), and **28-day mortality was higher in the early-insulin group** ($p=0.04$) in intention-to-treat analysis. **No published long-term neurodevelopmental follow-up of the NIRTURE cohort was located this session** — a targeted author-based search returned nothing, and this is reported as a genuine evidence gap rather than a search failure: the single largest RCT of a hyperglycaemia intervention in the population most relevant to a Level 3 NICU has no long-term outcome data, and was in any case stopped before it could answer its own primary question. **This null/harm finding is given equal prominence here to the positive observational associations above** — the observational studies show hyperglycaemia is associated with worse outcome, but the one adequately randomised attempt to correct it by lowering glucose with insulin found no mortality benefit and a signal of harm, with hypoglycaemia traded for hyperglycaemia rather than either being solved. Zamir et al.'s own null finding for insulin treatment (above) is consistent with this — insulin exposure itself does not appear to be the answer, in either direction.

Glycaemic variability/lability for hyperglycaemia — a distinct construct, tested directly, and generally not confirmed as an independent predictor

The task brief and Lagacé & Tam both flag glycaemic lability as potentially more clinically important than either threshold in isolation. Three primary studies test this directly for hyperglycaemia/variability specifically (as distinct from Domain D/E's hypoglycaemia-focused variability literature):

- **Tottman et al. 2017** (J Pediatr, PMID 28647271, DOI 10.1016/j.jpeds.2017.05.052): retrospective cohort, Auckland, 443 infants <1500 g or <30 weeks, classified into normoglycaemic/hypoglycaemic/hyperglycaemic/"unstable" groups by first-week glucose profile, with glucose variability separately quantified as the SD of log-transformed glucose values. **Unadjusted**, hyperglycaemic and unstable infants were less likely to survive without neonatal morbidity or without neurodevelopmental impairment at 2 years, and higher mean glucose tracked with worse outcomes at both timepoints. **Critically, none of these associations — including glucose variability itself — remained significant after adjustment for gestation, birth-weight z-score, and socioeconomic status.** The authors' own conclusion is that measures of neonatal glycaemia in this population are **markers of gestational age and intrauterine growth restriction, not independent predictors** of illness or 2-year outcome. This is the domain's most direct test of the variability-as-independent-predictor hypothesis, and it returns a **null after adjustment** — reported with full prominence, not subordinated to the more striking unadjusted numbers.
- **Battaglini et al. 2026** (Neonatology, PMID 41984740, DOI 10.1159/000550797): single-centre RCT, Genoa, 53 preterm infants randomised within 6 hours of life to real-time CGM-guided versus blinded-CGM/intermittent-glucose-guided management. CGM guidance **massively and significantly reduced both hyperglycaemic and hypoglycaemic episode counts** (e.g., hyperglycaemia episodes in week 1: 145 vs 830, $p=0.0001$) — i.e., it is an effective way of reducing glycaemic instability. But **neurodevelopmental outcome at 2 years (Griffiths Scales) did not differ significantly between groups.** This is the most direct interventional test available of whether reducing glycaemic lability improves neurodevelopmental outcome, and in this small trial the answer is no — though the trial's sample size ($n=53$), adequate for the glycaemic-control primary outcome, was very likely underpowered for a neurodevelopmental secondary outcome, so this null should not be read as a definitive refutation of the lability hypothesis, only as failing to confirm it.
- **Lasagni et al. 2026** (Sci Rep, PMID 41702956, DOI 10.1038/s41598-026-37286-4): small prospective study, Padova, 60 very preterm infants (43 completed 12-month follow-up) with CGM in the first postnatal week versus 31 full-term controls, using both standardised developmental scales and lab-based visual attention/memory tasks at 12 months corrected age. VPT infants performed worse than term infants across cognitive, motor, and language domains (an expected, non-specific prematurity effect). The one glycaemia-specific finding: **tighter glycaemic control at birth was associated with better memory performance at 12 months** — a positive but small, single-outcome,

non-replicated signal from a small sample, reported here as suggestive rather than confirmatory.

Taken together, this three-study set does not confirm glycaemic variability as an independent driver of long-term outcome distinct from gestational immaturity itself — the best-adjusted analysis (Tottman) attributes the unadjusted association to confounding by gestation/growth, and the one RCT that successfully reduced variability (Battaglini) did not detect a neurodevelopmental benefit. This tempers, without discarding, the "lability matters more than threshold" framing found in Lagacé & Tam and elsewhere in the review literature — it is a genuinely open question, not a settled one in either direction.

Population 2 — neonatal encephalopathy with secondary hyperglycaemia (a distinct population)

- **Puzone et al. 2023** (Arch Dis Child Fetal Neonatal Ed, PMID 37316160, DOI 10.1136/archdischild-2023-325592, PROSPERO CRD42022368870): systematic review and meta-analysis, 12 studies included after screening 82. In infants with **neonatal encephalopathy**, exposure to hyperglycaemia was associated with death or neurodisability at ≥ 18 months in 7 studies/807 infants (46.1% vs 28.0%; **OR 3.07, 95% CI 2.17-4.35, $p < 0.00001$**) — the largest pooled effect size for hyperglycaemia and adverse outcome anywhere in this domain. Hypoglycaemia in the same NE population showed a comparable, slightly smaller association (6 studies/685 infants, OR 2.17, 95% CI 1.46-3.25, $p = 0.0001$). Both findings were confirmed in the subgroup restricted to infants who underwent therapeutic hypothermia. Risk of bias was assessed with ROBINS-I (pooling observational studies) and GRADE-rated per included study.

This is flagged as a distinct population, per the task's explicit instruction, and must not be read alongside the extremely-preterm evidence above as though it were the same phenomenon. In NE, hyperglycaemia is most plausibly a **secondary stress response** — a marker of the severity of the underlying hypoxic-ischaemic insult and its counter-regulatory hormonal response — rather than the primary nutritional/endocrine-immaturity phenomenon seen in preterm infants. The strong OR here may in substantial part reflect confounding by encephalopathy severity itself (sicker brains produce more stress hyperglycaemia and also have worse intrinsic prognosis) rather than an independent glucotoxic effect of the hyperglycaemia. The review's own conclusion is appropriately cautious ("these data suggest... may be associated with"), and the directness caveat here is as important as the effect size: this is term/near-term NE evidence, not preterm evidence, and not evidence about primary/isolated hyperglycaemia.

Background/context (reviews, not independently extracted)

- **Ramel & Rao 2020** (NeoReviews, PMID 32005719, DOI 10.1542/neo.21-2-e89): narrative review of hyperglycaemia in extremely preterm infants (< 28 weeks). Confirms the epidemiology (lower GA/birth weight, illness severity, and higher parenteral glucose intake all increase risk; higher amino acid/lipid provision and earlier full enteral feeding reduce it) and states explicitly that data on long-term

neurodevelopmental, growth, and cardiovascular/metabolic effects are **"limited"** — a statement this domain's search substantially confirms rather than overturns. It also names the core management tension explored above: lowering glucose infusion rate risks caloric deficit and its own adverse neurodevelopmental consequences, while insulin use (per NIRTURE/Beardsall) increases hypoglycaemia risk — there is no clearly superior strategy in the current evidence base, a point the review makes and this synthesis's primary-source tracing confirms.

(b) Why — physiological and methodological rationale

Three distinct mechanisms are plausible and not mutually exclusive, mapping onto the three populations above. In extremely/very preterm infants, relative pancreatic β -cell immaturity and peripheral insulin resistance limit glucose disposal in the face of standard parenteral glucose loads, producing hyperglycaemia as a marker of physiological immaturity itself (consistent with Tottman's finding that glycaemia measures are markers of gestation/growth rather than independent predictors) — but hyperglycaemia occurring on top of this may also cause harm directly through osmotic diuresis, intraventricular haemorrhage risk, and possibly direct glucotoxic/oxidative injury to an immature, still-myelinating brain, which is the mechanism implicit in Zamir's duration-dependent dose-response. In PN-related hyperglycaemia specifically (Boscarino), excess early macronutrient delivery is the proximate driver, distinguishing this route from illness-severity-driven stress hyperglycaemia even though both converge on the same biochemical endpoint. In NE (Puzone), hyperglycaemia is best understood as a marker/mediator of the catecholamine and cortisol stress response to hypoxic-ischaemic injury, plausibly compounding (through additional osmotic/oxidative stress on already-injured tissue) rather than solely marking the primary insult — the review literature does not yet distinguish these possibilities. The consistent finding across CGM studies (Galderisi 2017 RCT, cross-referenced from Domain D; Battaglini 2026) that CGM-guided management can substantially reduce both hyper- and hypoglycaemic episode burden, without yet demonstrating a neurodevelopmental benefit from doing so, is consistent with either (i) glycaemia being predominantly a marker rather than a modifiable causal pathway in this population, or (ii) current trials being underpowered to detect a real but modest neurodevelopmental effect of tighter control — the evidence does not yet distinguish between these.

(c) GRADE certainty summary

Claim	Certainty	Downgrade domain(s)
Duration of hyperglycaemia >8 mmol/L (144 mg/dL) is associated with dose-dependent lower FSIQ and motor scores at 6.5 years in extremely preterm (<27 week)	Moderate	Imprecision (wide CIs, e.g. FSIQ 0.03–0.62 per day); observational design (confounding by illness severity/ indication for glucose monitoring not excluded)

Claim	Certainty	Downgrade domain(s)
infants; insulin treatment itself is not associated with outcome (Zamir)		
PN-related hyperglycaemia (>180 mg/dL/10.0 mmol/L) in preterm infants <32 weeks/<1500g is associated with cognitive and motor delay at 24 months, with motor delay remaining significant after adjustment (Boscarino)	Moderate	Risk of bias (non-randomised exposure, single-centre); imprecision (n=108)
Insulin-treated hyperglycaemia (≥ 10 mmol/L/180 mg/dL) in infants ≤ 32 weeks is associated with markedly higher in-hospital mortality and, among survivors, more frequent neurological/behavioural abnormality at 2 years (van der Lugt)	Low	Risk of bias (retrospective, small matched comparison, substantial survivor-bias risk given the scale of the mortality difference); imprecision (n=33 exposed survivors)
Early insulin therapy in VLBW infants reduces hyperglycaemia but increases hypoglycaemia and shows a signal toward higher 28-day mortality, with no long-term neurodevelopmental follow-up published (Beardsall/NIRTURE)	Low (for the mortality/hypoglycaemia signal)	Risk of bias (trial stopped early for futility/harm — early stopping systematically inflates apparent harm and truncates power); imprecision (reduced sample from early stopping)
Glycaemic variability (as a construct distinct from threshold breach) is an independent predictor of 2-year outcome in very preterm infants (Tottman)	Low , and the direction of the finding is null — variability did not survive adjustment	Indirectness/confounding (unadjusted association attributable to gestation/growth restriction); single-centre imprecision
Reducing glycaemic instability (both hyper- and hypoglycaemic episodes) via real-time CGM improves neurodevelopmental outcome at 2 years in preterm infants (Battaglini)	Low , and the direction of the finding is null for the neurodevelopmental outcome (though strongly positive for the glycaemic-control outcome)	Imprecision (n=53, very likely underpowered for the neurodevelopmental secondary outcome)
Hyperglycaemia in neonatal encephalopathy is associated with death/neurodisability at ≥ 18 months (Puzone SR/MA)	Low-Moderate	Risk of bias (ROBINS-I moderate-to-serious across pooled observational studies); indirectness (population is NE-with-secondary-hyperglycaemia, not primary/isolated hyperglycaemia — do not extrapolate this effect size to the preterm populations above)

(d) Where the evidence is weak, contested, or extrapolated

1. **There is no RCT evidence, in any population, that treating neonatal hyperglycaemia to any specific threshold improves long-term neurodevelopmental outcome.** This is the single most important gap in this domain and mirrors, in an even starker form, the preterm evidence gap already flagged in Domain D for hypoglycaemia. The one large RCT that attempted an interventional answer (NIRTURE/Beardsall) was stopped early for futility/harm on its primary mortality outcome before it could inform a neurodevelopmental question, and no follow-up of that cohort exists. Every other piece of evidence in this domain is observational, and every observational association between hyperglycaemia and adverse outcome in preterm infants carries the same structural threat to causal interpretation: hyperglycaemia is itself substantially a marker of gestational immaturity, growth restriction, and illness severity (as Tottman's adjusted analysis demonstrates directly), so associations with adverse outcome may in unknown proportion reflect confounding by the underlying vulnerability rather than an independent glucotoxic effect. **State this plainly: the field does not currently know whether treating preterm hyperglycaemia (by any means — GIR reduction or insulin) improves neurodevelopmental outcome, and the one trial that tried to answer this found a signal of harm from the insulin approach specifically.**
2. **The "COIN trial" named in the original research brief does not exist under that name.** This synthesis traced the brief's description (a multicentre RCT of insulin therapy for preterm hyperglycaemia) to the correctly-named NIRTURE trial (Beardsall et al. 2008). This is reported transparently as a correction rather than silently substituted, since a governance document that repeats an unverifiable trial name would itself become an unverifiable claim.
3. **The glycaemic-lability/variability hypothesis is plausible, explicitly endorsed by the secondary review literature (Lagacé & Tam), but not currently confirmed by direct primary evidence for hyperglycaemia specifically.** The best-designed adjustment analysis (Tottman) attributes apparent variability associations to confounding by gestation and growth restriction, and the one randomised attempt to reduce variability and measure neurodevelopmental outcome (Battaglini) found no benefit, albeit in a trial almost certainly underpowered for that specific question. This should be presented as a genuinely open question — worth continued investigation, particularly with adequately powered trials, but not yet as an established clinical construct that changes management.
4. **PN-related, illness/stress-related, and NE-secondary hyperglycaemia are mechanistically distinct phenomena that the literature does not cleanly separate.** Boscarino's PN-specific finding, Zamir's/van der Lugt's illness-severity-adjacent extremely/very preterm cohorts, and Puzone's NE population all report broadly similar directions of association (hyperglycaemia with worse outcome), which is reassuring for external validity in one sense but also risks the reader collapsing three

different clinical scenarios, with different plausible causal structures and different management implications, into a single "hyperglycaemia is bad" heuristic. This synthesis has deliberately kept them separate; a Level 3 unit's clinical response to PN-related hyperglycaemia in a stable growing preterm infant should not be assumed identical to its response to stress hyperglycaemia in a sick ventilated infant or secondary hyperglycaemia during therapeutic hypothermia for NE.

5. **The U-shaped dysglycaemia curve (Lagacé & Tam)** is a reasonable synthesis of the fact that both hypoglycaemia (Domains D/E) and hyperglycaemia (this domain) are independently associated with adverse outcome in various populations, but **no single study in this domain's search directly plots outcome against the full glucose range within one cohort to demonstrate a U-shape empirically** — the U-shape is an inference from separately-conducted hypoglycaemia and hyperglycaemia literatures, not a directly observed within-study curve. This is stated because the brief specifically asked for the claim to be traced to primary evidence rather than accepted from the review; the primary evidence supports "both tails are bad" but does not, in the studies located this session, supply a single dose-response curve spanning both directions in the same population.

Directness caveat applying to this entire domain: the extremely/very preterm evidence (Zamir, Boscarino, van der Lugt, Beardsall, Tottman, Battaglini) is direct for the population that dominates a Level 3 NICU caseload, which is a meaningfully better position than Domain D's hypoglycaemia evidence gap for the same population — but it remains entirely observational (bar one stopped-early, harm-signalling RCT with no long-term outcome data) and none of it establishes causality or a treatment threshold with any confidence. The NE evidence (Puzone) is direct for that specific population and must not be extrapolated to primary/isolated preterm hyperglycaemia, or vice versa.

(e) Practical implication for a UK Level 3 unit

Four points follow for practice and for parent counselling, stated with the same care about certainty as above. First, **no BAPM or NICE guidance exists to defer to** for a hyperglycaemia treatment threshold — a unit using a numeric trigger (commonly in the 8–10 mmol/L/144–180 mg/dL range, consistent with Lagacé & Tam's proposed >150–180 mg/dL) is following reasonable extrapolation from the observational outcome literature and physiological reasoning, not a national standard, and this should be stated as such in local guidelines rather than presented as guideline-derived. Second, **the observational evidence that hyperglycaemia (however caused) tracks with worse motor and cognitive outcome in extremely/very preterm infants is reasonably consistent across independent cohorts (Zamir, Boscarino, van der Lugt)** and is enough to justify continued clinical attention to hyperglycaemia as a marker worth minimising where practical — but not enough, on its own, to justify aggressive insulin-based correction, given NIRTURE's harm signal. Third, **populations that the literature identifies as needing long-term neurodevelopmental follow-up specifically because of neonatal hyperglycaemia** are: (i) extremely and very preterm infants with sustained or recurrent hyperglycaemia in the neonatal period, particularly where PN-related or prolonged (Zamir,

Boscarino, van der Lugt); (ii) infants of any gestation who received insulin therapy for neonatal hyperglycaemia, given the absence of reassuring long-term data and the presence of a harm signal in the pivotal RCT (Beardsall/NIRTURE; also the population in whom Zamir found no insulin-outcome association either way, which does not resolve the question); and (iii) infants with neonatal encephalopathy who develop secondary hyperglycaemia, who already warrant follow-up for their primary diagnosis but in whom hyperglycaemia is an independently-associated marker of higher risk (Puzone) that should sharpen surveillance intensity. Fourth, **glycaemic variability/lability should be tracked as a research question, not yet treated as an established independent target for intervention** — the current direct evidence (Tottman, Battaglini) does not confirm that reducing variability, over and above treating frank hyperglycaemia/hypoglycaemia, changes neurodevelopmental outcome, notwithstanding its prominent billing in the secondary review literature.